

Foundational Reasoning and Feedback Protocol (FRFP): A Formal Architecture for Long-Horizon Human–AI Scientific Collaboration

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Abstract

We propose the Foundational Reasoning and Feedback Protocol (FRFP), a formal architecture for long-horizon scientific collaboration between human agents and AI systems. FRFP enforces a sharp separation between explicit representations, which may be manipulated by machines, and tacit epistemic states, which are exclusive to human agents and bear correctness authority. On top of this separation, FRFP builds a free algebra of explicit task decompositions and navigation steps, realized as a Grothendieck-style semidirect product $N \ltimes E$ of explicit pipelines with a navigation groupoid. A semantic functor $J\text{--}K : N \ltimes E \rightarrow T$ into a preorder-enriched tacit category, together with a correctness quotient $\gamma : \text{Ob}(T) \rightarrow J$, yields observable judgments that are invariant under admissible refactorings and normalization in $N \ltimes E$.

Using this architecture, we develop a dynamic theory of long-run interaction. Tacit context degrades under use, requirement maps encode what is needed to safely endorse explicit artifacts in a given context, and hallucination is defined structurally as a mismatch between tacit state and requirements at the moment of acceptance. Under qualitative but pessimistic assumptions on degradation, feedback, and recurring nontrivial tasks, we prove inevitability theorems: along sufficiently long admissible runs, hallucination occurs almost surely, and protocol-level “safe horizons” are necessarily finite. We then lift the framework to epistemic populations, institutions, and governance operators, showing that explicit-only consensus and monitoring cannot in general eliminate hallucination or recover tacit correctness. Finally, we extract protocol and interface rules—together with FRFP-compatible system instructions—for building explicit-only AI tools that support powerful exploration while preserving human-exclusive grounding and auditability at the level of whole scientific and regulatory workflows.

Appendix ?? provides a practitioner-facing “business version” of FRFP: a non-technical overview of the architecture, examples in scientific and regulated workflows, an implementation roadmap, common pitfalls, and a short design checklist for teams deploying AI systems.

Contents

1 Interaction Protocols

This layer specifies interface-level constraints on human–AI interaction traces: who may say what, in which modes, and in which order. We distinguish content-producing messages (explicit messages, which both human and AI may send) from correctness-bearing messages (meta-level messages, which only the human may send); Section ?? makes this distinction precise as explicit versus tacit epistemic spaces.

1.1 Interaction Traces

We model an interaction session as a finite sequence of messages tagged by origin.

Definition 1.1 (Interaction trace). A *human–AI interaction trace* is a finite sequence

$$S = (m_1, m_2, \dots, m_n),$$

where each message m_i is tagged as either human-generated (H) or AI-generated (A) and carries explicit content (e.g., text, code, mathematical expressions).

We write $S_k = (m_1, \dots, m_k)$ for the k -prefix of S . We impose a family of interaction constraints Φ_0 on such prefixes.

1.2 Interaction Constraints

The constraints are operational: they enforce mode discipline and prevent the AI from taking tacit roles.

Intent locking. Early in the trace, the human states a coarse-grained intent (e.g., “analyse this conjecture”, “design an experiment”). The AI treats this intent as fixed unless the human explicitly revises it. This prevents uncontrolled drift in task framing.

Ambiguity resolution. If an AI message admits more than one task-relevant interpretation, the next AI message must be a clarification query, not a content-producing response. This prohibits silent resolution of deep ambiguity.

Mode discipline. Each message is typed as either explicit (facts, derivations, questions) or meta-level (requests for tacit judgment). Both human and AI may send explicit messages. Only the human may send meta-level, correctness-bearing messages; the AI is confined to explicit mode.

No tacit emulation. The AI may not simulate tacit operations such as evaluating overall correctness or deciding whether a result should be accepted. It may propose checks, but it is not a source of correctness.

Correctness stability. Once the human issues a tacit correctness judgment, the AI may not reinterpret or override it via further explicit manipulation. Tacit states are fixed points with respect to AI operations.

Definition 1.2 (Interaction admissibility). An interaction trace S is *admissible* if every prefix S_k satisfies all interaction constraints. We write $S \models \Phi_0$ for this property.

These constraints are intentionally strict and normative. FRFP commits to: (i) an explicit/tacit distinction, in the sense of Polanyi and subsequent work on organisational knowledge creation; and (ii) human-exclusive correctness in high-stakes settings, as reflected in human-oversight requirements for high-risk AI systems. Following Polanyi’s account of tacit knowledge as embodied, context-dependent, and not fully articulable ?, and Nonaka’s treatment of explicit and tacit knowledge as distinct but interacting modalities in organisational learning ??, we model explicit artefacts and tacit human states as separate epistemic spaces (Section ??) linked by a one-way boundary. In parallel, contemporary governance frameworks require human oversight of high-risk AI systems, with humans able to monitor, intervene, and override AI outputs ?. FRFP packages these commitments as an architectural constraint for disciplined human–AI workflows and studies their formal consequences. It is not intended as a description of current ad hoc practice, but as a target architecture: in this volume, explicit/tacit separation and human-exclusive correctness are design principles, not optional knobs; later work may study carefully delimited departures.

In the running airfare example introduced in Section ??, the interaction trace $S = (m_1, \dots, m_{10})$ consists of alternating human and AI messages H_1 – H_5 and A_1 – A_5 . We write the human messages as $H_1, \dots, H_5$ and the AI messages as $A_1, \dots, A_5$, in order of appearance. The interaction constraints require, in particular, that the AI remain in explicit mode: it may elicit preferences, propose itineraries, and analyse trade-offs, but it may not itself issue the final “Option A is best for you” verdict. The human’s final choice at H_5 is a tacit correctness judgment, which the AI can at most record explicitly. Any trace in which the AI directly declares an option to be “best” or overwrites the human’s recorded decision is operationally inadmissible.

The interaction protocols constrain the observable structure of human–AI conversations: who may say what, in which modes, and in which order. The next layer introduces the epistemic spaces that such conversations act on: explicit artefacts and tacit human states that carry knowledge- and correctness-relevant information.

2 Epistemic Spaces

The interaction protocols in Section ?? constrain which conversational messages are allowed; this layer specifies what those messages act on: an explicit space E of machine-manipulable artefacts and a tacit space T of human-only correctness-bearing states, linked by a one-way boundary.

2.1 Ambient Category and Explicit–Tacit Separation

We work inside an ambient category C_{full} with distinguished objects $\emptyset$, E , and T , as detailed in Appendix A. The Explicit–Tacit Separation axiom (ETS) asserts:

- explicit morphisms end at E ;
- tacit morphisms end at T ;
- there are no morphisms $T \rightarrow E$;
- there is a unique boundary morphism $\text{RB} : E \rightarrow T$;
- RB is not invertible.

Intuitively, explicit and tacit modes are *disjoint*: there is a one-way flow from explicit artefacts into tacit evaluation, and no way back.

2.2 Explicit and Tacit Primitives

Within C_{full} , we identify six primitive morphisms that play irreducible epistemic roles:

- $\text{RI} : \emptyset \rightarrow E$ (Representation Initiation),
- $\text{EC} : E \rightarrow E$ (Explicit Computation),
- $\text{ED} : E \rightarrow E$ (Explicit Diagnostics),
- $\text{RB} : E \rightarrow T$ (Representational Backflow),
- $\text{TE} : T \rightarrow T$ (Tacit Evaluation),
- $\text{HFD} : T \rightarrow T$ (Human Final Decision).

The first three live entirely in explicit space; RB crosses the boundary into tacit space; and TE, HFD live in tacit space. They compose into the core correctness architecture:

$$\text{RI} \rightarrow (\text{EC/ED})^* \rightarrow \text{RB} \rightarrow \text{TE} \rightarrow \text{HFD}.$$

Here $(\text{EC/ED})^*$ denotes an arbitrary (possibly empty) finite sequence of explicit computations and diagnostics. The pipeline ends with tacit evaluation and final decision: no correctness-bearing steps occur in explicit space.

2.3 Correctness in Tacit Space

Tacit states form a preorder-enriched one-object category $\mathcal{T}$ with primitives TE, HFD and a correctness quotient $\gamma : \text{Ob}(\mathcal{T}) \rightarrow J$ into a finite set of judgments. The key fact for the main text is:

Proposition 2.1 (Tacit correctness quotient). *Tacit primitives $\text{TE}, \text{HFD} : T \rightarrow T$ are idempotent and commute, and the correctness quotient γ satisfies $\gamma \circ \text{TE} = \gamma = \gamma \circ \text{HFD}$. Thus tacit transformations do not change the observable correctness class.*

The proof and full structure appear in Appendix A. For the main text, it suffices to know that correctness is localised in tacit space and is observable only via γ .

Interpretation and scope of tacit correctness. Treating correctness as living in tacit space T rather than in explicit space E is a substantive architectural commitment, not a definitional firewall. It targets domains in which correctness-bearing judgment does not reduce to a fixed, finite explicit specification or benchmark suite: long-horizon workflows in which tasks evolve, standards and constraints shift, and background understanding and norms determine whether an explicit artefact is adequate. In such settings, explicit artefacts, tests, and formal checks do not exhaust correctness-bearing judgment; Appendix A.11 shows that, under a nontrivial tacit structure, a broad class of explicit-only oracles and governance mechanisms are structurally impossible. By contrast, some regimes admit fully explicit correctness—for example, proof-assistant settings or closed-world tasks with sound and complete test oracles. FRFP does not treat these regimes as primary instances of its epistemic dynamics; rather, they appear as tacit-insensitive inner layers embedded in a broader tacit governance context (Appendix D.4). The results in this section are therefore conditional on a genuine tacit layer in the sense above, and on explicit/tacit separation and human-exclusive correctness as non-negotiable design principles in that domain.

Hallucination as a protocol-level phenomenon. Within this architecture, hallucination is a tacit-level notion. We do not apply “hallucination” to the mere production of incorrect strings in E ; at that stage outputs are explicit candidates, and, when they violate a trusted explicit oracle (e.g. a test harness or proof checker), explicit errors. Hallucination arises at the protocol level, when explicit artefacts cross the boundary $RB : E \rightarrow T$ and are incorporated into tacit correctness. Informally, a hallucination occurs along a run when tacit operations such as TE and HFD end up treating an explicit artefact as acceptable even though, relative to the domain’s standards, it ought to have been judged inadequate or incorrect. Because correctness is localised in T and observable only via γ , hallucination cannot be read off from explicit artefacts in isolation or from the behaviour of the generator that produced them: the same explicit string may be harmless when rejected and hallucinated when taken up into tacit acceptance. Later sections refine this picture dynamically; the core point is that, once explicit and tacit spaces are separated, hallucination in FRFP is a property of endorsement by tacit states, not of explicit generation alone.

Explicit correctness engines. In many workflows, parts of correctness can be discharged by *explicit correctness engines*: unit-test harnesses, static analysers, and interactive theorem provers (e.g., Coq, Isabelle, Lean) that, relative to a fixed formal specification, can certify whether an explicit artefact satisfies a given property. Within FRFP these engines live entirely in explicit space E : they are instances of explicit computation or diagnostics (EC/ED), possibly composed into larger explicit pipelines. They can reduce the space of explicit errors tacit agents must consider, but they do not collapse the tacit layer. Tacit correctness still governs which specifications are adequate, when they must be revised, how to interpret formally verified artefacts in context, and what residual risk is acceptable. Explicit correctness engines are therefore treated as powerful inner modules inside E , coupled to but not replacing the outer tacit governance layer.

2.4 Uniqueness of the FRFP Epistemic Ontology

Appendix A shows that, under reasonable axioms, the FRFP Epistemic Ontology is essentially unique.

Theorem 2.2 (Initiality of the FRFP ontology, informal). *Any epistemic ontology that:*

- *distinguishes explicit and tacit modalities,*
- *enforces a one-way boundary $E \rightarrow T$,*
- *localises correctness in tacit operators TE, HFD, and*
- *represents AI systems as acting only in explicit space,*

admits a unique structure-preserving mapping from the FRFP ontology. In this precise sense, the FRFP ontology is minimal and essentially unique.

We refer the reader to Appendix A for the formal category of Epistemic Spaces frameworks and the proof that the FRFP ontology is initial in that category.

Operational significance of the FRFP Epistemic Ontology. The initiality result is not a formal restatement of “we built a free category”. It says that any ontology satisfying (i) explicit/tacit modality, (ii) a one-way boundary $E \rightarrow T$, (iii) tacit localisation of correctness, and (iv) explicit-only AI action must factor uniquely through the FRFP Epistemic Ontology. In particular, any framework

that preserves the same primitive epistemic roles and axioms, but proposes a different wiring of RI, EC, ED, RB, TE, HFD, still receives a unique structure-preserving functor from FRFP. This makes later non-automation and no-go results robust under local redesigns: one cannot bypass them by relabelling primitives or rearranging explicit/tacit roles while keeping the Epistemic Spaces axioms intact.

Given an admissible interaction trace S , the epistemic-ontology layer associates to each prefix S_k a pair of states (e_k, t_k) , where $e_k \in E$ is the explicit state and $t_k \in T$ the tacit state.

In the airfare example introduced in Section ??, after H_2 the explicit state e_2 contains at least the travel dates and route, the human’s stated priority weights, and the assistant’s preference template. The tacit state t_2 reflects the human’s internal attitudes toward those trade-offs, which may not match any explicit labels. After A_3 , the explicit state e_3 additionally contains candidate itineraries and their attributes; the tacit state t_3 reflects the human’s (possibly tentative) sense of which options are promising.

The boundary operator $RB : E \rightarrow T$ (representational backflow) is applied when the human reviews an explicit artefact. At H_5 , the human evaluates the explicit summaries and options, applies RB to the relevant explicit configuration, and updates the tacit state t_5 to encode acceptance of Option A as “best” in context. The correctness quotient $\gamma : \text{Ob}(\mathcal{T}) \rightarrow J$ then records this tacit acceptance as an observable judgment (e.g., “Option A accepted”).

Our treatment of tacit space is deliberately spare. We assume only that tacit states carry correctness-relevant information, are updated via a one-way boundary from explicit artefacts, and admit a finite observable judgment quotient. We do not model the internal structure of human belief or institutional reasoning; instead, substantive structure is pushed into explicit workflows that we can represent and analyse compositionally.

3 Epistemic Algebra

The previous section introduced the FRFP epistemic spaces: an explicit space E of machine-manipulable artefacts and a tacit space T of human-only correctness-bearing states, linked by a one-way boundary. This section equips *explicit* workflows with algebraic structure over that ontology. Together with the interaction protocols of Section ??, these constructions form the FRFP human–AI conversational calculus. All algebra in this section lives on explicit workflows; tacit state appears only as the semantics of those workflows, not as an object we rewrite.

Given the explicit/tacit ontology, FRFP’s technical structure sits on the explicit side. We define an epistemic algebra of explicit workflows: a task decomposition grammar, navigation and refactoring operations, and a semantics into tacit states. Tacit evaluation enters only through the boundary and correctness quotient; the algebraic invariance results are entirely explicit.

This layer has two ingredients:

- a Task Decomposition Grammar (TDG) that generates explicit pipelines from the primitives RI, EC, ED; and
- a navigation groupoid ($\mathcal{N}$) that captures reversible changes of perspective or representation.

They are combined via a Grothendieck-style semidirect product $\mathcal{N} \ltimes \mathcal{E}$, a standard construction in categorical semantics ???, yielding an algebra of reasoning trajectories.

3.1 Task Decomposition Grammar

TDG presents the free algebra of explicit pipelines. Informally, its grammar is:

$$\begin{aligned} Q &::= \text{RI} \mid Q; \text{EC} \mid Q; \text{ED} \mid \text{BRANCH}(Q_1, \dots, Q_k) \mid \text{PARALLEL}(Q_1, \dots, Q_k) \mid \text{LOOP}(Q_{\text{cond}}, Q_{\text{body}}), \\ P &::= Q; \text{RB}; \text{TE}; \text{HFD}. \end{aligned}$$

Here Q ranges over explicit pipelines and P over complete FRFP pipelines.

The explicit category E_{sem} generated by TDG satisfies the usual universal property: any interpretation of the generators in an explicit category extends uniquely to a functor from E_{sem} .

3.2 Navigation Groupoid

Navigation steps—trying alternative models, adding or dropping assumptions, restarting from a new shape—are represented as morphisms in a groupoid $\mathcal{N}$. Objects of $\mathcal{N}$ are *shapes* of explicit reasoning; morphisms are structural rewritings between shapes, freely inverted to form a groupoid.

Each navigation morphism $n \in \mathcal{N}$ acts on explicit pipelines via a fiberwise functor $n^* : E_S \rightarrow E_{S'}$ between fibers of a fibration $p : \mathcal{E} \rightarrow \mathcal{S}$ over shapes. This makes E an indexed category over $\mathcal{N}$.

3.3 Semidirect Product $\mathcal{N} \ltimes \mathcal{E}$

The combined reasoning algebra is the Grothendieck construction $\mathcal{N} \ltimes E$. A morphism in $\mathcal{N} \ltimes E$ is a pair (n, e) consisting of a navigation morphism $n \in \mathcal{N}$ and an explicit arrow $e \in \mathcal{E}$ in the appropriate fiber. Composition interleaves navigation and explicit reasoning according to:

$$(n_2, e_2) \circ (n_1, e_1) = (n_2 n_1, e_2 \circ n_2^*(e_1)).$$

Intuitively:

- $\mathcal{N}$ says *which view* of the explicit reasoning we are in (e.g., which model, which set of assumptions);
- $\mathcal{E}$ records *what* explicit computation and diagnostics have been applied;
- $\mathcal{N} \ltimes \mathcal{E}$ packages these into a single object representing a reasoning trajectory.

3.4 Confluence, Normal Forms, and Decidability

Appendix A equips $\mathcal{N} \ltimes E$ with a reduction system that combines explicit rewrites and navigation moves. Provided that:

- explicit reduction is terminating and locally confluent, and
- navigation reduction normalizes within each component of $\mathcal{N}$,

we obtain:

Theorem 3.1 (Confluence and normal forms (informal)). *Every reasoning trajectory in $\mathcal{N} \ltimes \mathcal{E}$ admits a canonical normal form, unique up to navigation isomorphism. In particular, explicit steps and navigation steps may be permuted and simplified without changing the canonical form.*

Moreover, given a well-specified reduction system and typing rules, the question “*Is this composite a valid FRFP pipeline?*” is decidable.

Theorem 3.2 (Decidability of admissibility (informal)). *Given a composite term in $\mathcal{N} \ltimes \mathcal{E}$, it is decidable whether it corresponds to an admissible pipeline of the form $\text{RI} \rightarrow (\text{EC}/\text{ED})^* \rightarrow \text{RB} \rightarrow \text{TE} \rightarrow \text{HFD}$.*

These properties make $\mathcal{N} \ltimes \mathcal{E}$ a suitable intermediate language for representing and analyzing complex reasoning trajectories.

Operational significance of Epistemic Algebra canonicity. Appendix A strengthens this picture by showing that the Epistemic Algebra architecture of FRFP is canonical among all architectures with the same primitive structure: a free explicit category generated by TDG, a navigation groupoid acting fiberwise via a fibration $p : E \rightarrow S$, a Grothendieck-style semidirect product $N \ltimes E$, and preservation of the RB–TE–HFD boundary under navigation. Any other architecture that satisfies these axioms admits a unique structure-preserving morphism from the FRFP Epistemic Algebra construction. Operationally, this means that “small local changes” to task decomposition or navigation—while still respecting the same roles and constraints—cannot evade the correctness-preservation, non-automation, and inevitability results: they are inherited along the unique morphism from FRFP. To genuinely escape those consequences, one must step outside the Epistemic Algebra axioms themselves (for example by allowing tacit-to-explicit morphisms or by breaking boundary preservation), rather than modifying TDG or navigation in a cosmetic way.

At the explicit algebra layer, the airfare interaction (introduced in Section ??), is represented as an explicit pipeline $P_{\text{air}} \in N \ltimes E$ that composes AI-assisted steps such as preference elicitation, option generation, and trade-off analysis. Different prompt sequences that are operationally admissible but differ only by refactoring—for example, asking for trade-offs before seeing full itineraries, or regrouping constraints into two separate messages—are related by navigation morphisms in N and rewrite rules in the explicit reduction system.

By construction, our rewrite system on $N \ltimes E$ is terminating and confluent, so each such pipeline admits a canonical normal form. Correctness is defined at the level of the canonical pipeline via the semantics functor $\llbracket - \rrbracket : N \ltimes E \rightarrow T$ and the quotient γ . Informally: if one way of using the airfare assistant is correct (the human, after reviewing the explicit artefacts, makes an acceptable tacit choice), then all ways of using the assistant that are rewritable to that pipeline by admissible refactorings are equally correct.

3.5 Semantic Correctness and Invariance

The combination of Epistemic Space and Epistemic Algebra yields a semantic map from composite trajectories to tacit states.

3.5.1 Semantic Functor and Observable Correctness

The semantic functor

$$\llbracket - \rrbracket : \mathcal{N} \ltimes E_{\text{sem}} \rightarrow \mathcal{T}$$

interprets a trajectory by running its explicit part, applying the boundary RB, and then applying tacit evaluation and final decision. Observable correctness is given by $\gamma(\llbracket - \rrbracket)$.

Theorem 3.3 (Correctness preservation (informal)). *Correctness judgments are invariant under explicit reasoning, navigation, and semidirect-product composition: for any admissible trajectory τ and any navigation morphism $n \in \mathcal{N}$,*

$$\gamma(\llbracket \tau \rrbracket) = \gamma(\llbracket n \cdot \tau \rrbracket).$$

In other words, changing the route by which we arrive at a result—so long as we respect explicit typing and Epistemic Spaces/Algebra constraints—does not change the tacit correctness judgment.

3.5.2 Canonical Correctness Form

Every admissible pipeline admits a canonical factorization:

Theorem 3.4 (Canonical correctness form (informal)). *Any trajectory in $\mathcal{N} \ltimes E_{\text{sem}}$ that is admissible in the FRFP sense is equivalent, up to navigation, to a unique normal form in which:*

- *all explicit operations (including navigation) occur before the boundary morphism RB;*
- *the tacit suffix is $\text{TE} \rightarrow \text{HFD}$, with no interleaving explicit operations;*
- *no additional boundary or tacit operators appear.*

This expresses, at the algebraic level, the design principle that AI systems may explore and manipulate explicit representations in many ways, but correctness judgments always arise from a single passage through RB followed by tacit evaluation and decision.

4 Epistemic Dynamics

The conversational calculus described so far is static: it assigns tacit judgments to whole explicit workflows but does not yet model how those judgments evolve over time. This section adds a dynamic layer on top of the conversational calculus, introducing tacit degradation, requirement schedules, hallucination, and safe horizons.

The structures introduced in Epistemic Spaces and Epistemic Algebra characterise *single* FRFP pipelines: how explicit and tacit operations are separated, how explicit reasoning and navigation combine, and how correctness is preserved under admissible transformations. In typical long-horizon use, however, a human agent engages with an AI system repeatedly. The tacit context from one episode influences the next, and errors or misplaced trust may accumulate.

Epistemic Dynamics extends the framework to this setting by equipping FRFP pipelines with a dynamic semantics on tacit states. The aim is to capture how tacit grounding, task requirements, and confidence interact over time, and to make precise in what sense hallucinations become structurally difficult to avoid. The formal development of this dynamic semantics is given in Appendix ??; the present section outlines the main ingredients and results.

4.1 Tacit Context, Degradation, and Requirements

We view the human agent’s epistemic background as a tacit context space T equipped with a preorder $\preceq$:

$$t_1 \preceq t_2 \quad \text{“}t_1 \text{ is epistemically no stronger than } t_2\text{”}.$$

Strict inequality $t_1 \prec t_2$ denotes $t_1 \preceq t_2$ and $t_2 \not\preceq t_1$. No further algebraic structure is imposed on T at this stage; the categorical structure of tacit space from Appendix A is still present but the dynamic analysis works at the level of the underlying preordered set.

A tacit state $t \in T$ is not merely isolated propositional knowledge, but a compressed record of prior interaction, aggregating experience accumulated over years through performing manual checks, encountering edge cases, debugging failures, and observing where formal specifications cease to align with practice. Tacit state is path-dependent: one only reaches a given t by walking through particular

sequences of work, not by reading a finite specification once. In this picture, the degradation operator $\delta : T \rightarrow T$ and the requirement map $\text{Req} : E \times \mathbb{R}_{\geq 0} \rightarrow T$ express how this compressed history evolves over time: δ erodes available grounding through repeated use, while $\text{Req}(e, c)$ encodes the tacit floor needed to treat an explicit artefact e at credence c as epistemically adequate. Our later inevitability results will rely precisely on this interaction between path-dependent tacit history, degradation, and growing or recurring requirements.

4.2 Epistemic degradation under long-run AI assistance

A central premise of this paper is that sustained reliance on AI assistants can induce *epistemic degradation* in human users. We use this term in a local, task-relative sense, not as a global claim that “humans become stupid.” Let D be a family of tasks (e.g., reading and evaluating papers, designing experiments, checking proofs). For a given user, let $t_t \in T$ denote their tacit epistemic state at time t : a compressed record of prior interactional experience—performing manual checks, encountering edge cases, debugging failures, and seeing where formal specifications cease to match practice—that enables them to solve tasks in D and to notice when something is wrong.

Definition 4.1 (Epistemic degradation). Fix a task family D and a baseline learning process in which a user develops by performing tasks in D without an AI assistant (or with only minimal, non-substitutive tools). We say that the user is *epistemically degraded* on D at time t (relative to this baseline) if, when evaluated *without* access to an assistant, their independent performance on D and their ability to detect and correct errors on D are strictly worse than under the baseline, due to differences in the evolution of their tacit state t_t .

Our claim is that, under conditions that already hold in many research workflows, long-run use of AI assistance makes epistemic degradation on relevant task families D the default outcome. We cannot prove this as a theorem about all possible humans and all possible AI systems, but we can justify it as a modeling assumption based on three well-established mechanisms.

Automation, complacency, and deskilling. A large human-factors literature shows that when automation and decision aids become reliable, humans tend to become *monitors* rather than *primary actors*, with resulting automation bias (over-trust in automated outputs) and automation-induced complacency (insufficient checking) (e.g. ???). In aviation, for example, increased reliance on autopilot and flight management systems has been linked to loss of basic manual flying skills, mode confusion, and degraded situation awareness, especially among less experienced pilots (?). Similar dynamics are now documented for AI systems in medicine: a mixed-method review of AI in clinical practice identifies *deskilling*—degradation of previously acquired competencies due to reduced practice—and *upskilling inhibition*—failure to develop advanced skills among trainees because AI systems are doing the difficult parts—as central risks of AI adoption (?). Commentaries and empirical studies on AI decision support in diagnostics explicitly worry about over-reliance, skill degradation, diagnostic deskilling, and loss of vigilance if clinicians defer too readily to AI outputs (???).

In our notation, the assistant increasingly intercepts those task instances in D that would otherwise provide practice and feedback to update t_t . Observable performance for the *combined* human–AI system may go up; but the contribution from the human’s tacit state shrinks, so that their independent competence is worse than in the no-assistant baseline.

Cognitive offloading and the “Google effect.” A second mechanism is cognitive offloading: the tendency to rely on external systems for storage and processing when they are easily available.

The “Google effect” literature shows that intensive Internet search leads people to treat the web as an external memory store: they remember *where* to find information rather than the information itself, and forgetting of details accelerates when future access is anticipated (?). Gong and Yang’s meta-analytical review documents that intensive search behavior systematically changes memory and retrieval mechanisms in this way (?). Related work on technology use and cognitive aging argues that increased dependence on external media for semantic, autobiographical, and spatial memory can accelerate decline if it reduces practice with one’s own intrinsic abilities (?).

Recent work extends this to AI tools specifically. Gerlich’s study on AI tools and critical thinking finds a significant negative relationship between heavy AI-tool usage and critical-thinking scores, with cognitive offloading statistically mediating this effect: participants who offload more reasoning to AI tools score worse on standardized critical thinking tests (?). Qualitative synthesis of this and related work emphasizes that users come to rely on AI for analysis and synthesis, not just for recall, and consequently engage less in deep, effortful thinking on their own (?).

In our framework, this means that both the *contents* and the *procedures* required to perform tacit checks on tasks in D are increasingly stored in, and executed by, the assistant rather than encoded in t_t . The user becomes more proficient at calling the assistant, but worse at reconstructing, critiquing, or replacing its reasoning.

Early evidence of degraded independent performance under AI use. Third, there is emerging empirical work that directly measures how AI assistance affects independent performance over time. In medicine, studies of AI-supported diagnostics show clear short-term gains in accuracy and speed for physicians using AI decision support (e.g. ???), but parallel reviews now explicitly flag the risk that this support will erode core clinical skills if it replaces, rather than augments, trainee practice (????). Similar concerns appear in stroke imaging and radiology training, where triage and detection algorithms risk relegating trainees to passive oversight roles (?).

Outside medicine, experimental work on generative-AI use in writing tasks has begun to measure neural and behavioral effects. An EEG-based study of essay writing with and without a chatbot assistant finds that participants using the assistant show lower brain engagement in regions associated with attention, planning, and memory, and that over several months they exhibit declining independent performance and increasing reliance on copy-and-paste use of AI outputs (?). Journalistic syntheses of this and related studies similarly report that frequent chatbot use is associated with reduced critical thinking and weaker performance when the tool is removed, especially among younger or less-experienced users (?).

These results are not yet definitive, and some meta-analyses also find positive effects of generative AI on specific higher-order skills in controlled educational settings (?). However, taken together with the older automation and offloading literatures, they make it reasonable to model epistemic degradation as the *default trajectory* when high-performing assistants replace, rather than structure, human practice on a task family D .

The formal assumption. We can now state the assumption our protocol is designed to address.

Assumption 4.2 (Degradation under sustained assistance). Let D be a family of tasks, and consider two learning processes for the same user:

1. A *baseline* process in which the user solves tasks in D without access to an AI assistant (or with only non-substitutive tools), updating t_t through unaided problem-solving, feedback, and explicit practice.

2. An *assisted* process in which a high-performing AI assistant proposes candidate solutions and rationales for most tasks in D , and the user typically accepts these with limited independent checking.

Assume that (i) human memory and skill exhibit standard decay in the absence of practice or retrieval, as modeled in classical forgetting and skill-decay work (e.g. ??); and (ii) the presence of the assistant systematically reduces the frequency and depth of unaided problem-solving and checking on D , as suggested by the automation, offloading, and deskilling studies discussed above (????).

Then, beyond some time horizon, the assisted process leads to epistemic degradation on D in the sense of Definition ?? : when evaluated without the assistant, the user’s independent performance and error-detection ability on D are worse than under the baseline process.

This assumption captures the worry that motivates our protocol design. As AI assistants become more capable, human scientists risk becoming less capable *relative to the tasks they routinely delegate*: the joint human–AI system may look impressively competent, but the next generation’s tacit epistemic state t_t fails to accumulate the interaction-derived structure needed to generate and scrutinize scientific results on its own. The protocols we propose in the remainder of the paper are explicitly designed to break this dynamic: they aim to secure near-term accuracy while preserving, and in some cases amplifying, the long-run development of human tacit competence.

Context degradation. Long-run use of an AI assistant is assumed to induce epistemic degradation. This is modelled by a function $\delta : T \rightarrow T$ satisfying:

- (D1) *Monotonicity*: $\delta(t) \preceq t$ for all $t \in T$;
- (D2) *Strict decay somewhere*: there exists t with $\delta(t) \prec t$;
- (D3) *Non-invertibility*: there is no $f : T \rightarrow T$ such that $\delta \circ f = \text{id}_T$.

Intuitively, δ aggregates effects such as fatigue, forgetting, and inappropriate deference to the system. Repair operations (rest, study, institutional checks) can be represented by separate operators, but are not needed for the core Epistemic Dynamics results and are therefore left for later extensions.

Requirements and hallucination. Hallucination is defined at the level of the protocol, rather than in terms of any particular model. To this end, we introduce a requirement map

$$\text{Req} : E \times \mathbb{R}_{\geq 0} \rightarrow T,$$

where $e \in E$ is an explicit artifact (e.g., an explanation or derivation) and $c \geq 0$ is a tacit credence level indicating how strongly the agent treats e as reliable. The element $\text{Req}(e, c)$ represents the tacit grounding required to treat (e, c) as epistemically adequate.

We assume:

- (R1) *Monotonicity in credence*: if $c_1 \leq c_2$ then $\text{Req}(e, c_1) \preceq \text{Req}(e, c_2)$ for all $e \in E$;
- (R2) *Nontriviality*: not all artifacts have the same requirements up to $\preceq$.

Definition 4.3 (Hallucination). Let $t \in T$ be the current tacit state, $e \in E$ an explicit output, and $c \geq 0$ a credence level. A *hallucination* occurs at this step if

$$t \prec \text{Req}(e, c).$$

We then say that e is *hallucinated* (relative to (t, c)).

In this sense, hallucination is a mismatch between available tacit grounding and what would be required to support the explicit artifact at the assigned credence. This definition does not depend on internal model mechanisms or intent.

Repair, workload, and the cost–speed tradeoff. The Epistemic Dynamics analysis above treats degradation as one-sided: tacit states evolve via a monotone, non-invertible operator δ under recurring nontrivial demands, yielding inevitability of requirement-violating steps. Real institutions also invest in *repair*: training, feedback, reflection, and retraining that partially restore grounding. We model these as operators $\rho : T \rightarrow T$ that decrease an entropy-like potential $S : T \rightarrow \mathbb{R}$ on tacit states, while δ increases S whenever nontrivial work is done.

If average entropy production from nontrivial tasks exceeds the average entropy removal achievable by bounded repair operators ρ , then S drifts upward and eventually crosses any fixed requirement threshold, recovering the inevitability results even in the presence of repair. Conversely, stabilizing S requires repair at a rate and intensity comparable to the rate at which tasks are processed. In high-load regimes this typically means a governance workload whose cost and latency scale with AI usage. Under the FRFP assumptions there is thus no regime in which one simultaneously obtains unbounded speed/cost gains from automation and long-horizon immunity to FRFP-hallucination: strong repair can delay and reshape risk, but only by re-introducing substantial human effort (see Appendix ?? for a formal δ – ρ analysis).

4.3 Pipeline Transformers and Runs

The static semantics of Appendix ?? already associates to each explicit pipeline $e \in E_{\text{sem}}$ a tacit effect via the semantic RB functor $\text{RB}^\sharp$ and the tacit primitives TE, HFD . In particular, evaluation of e at tacit context t can be written as

$$t \longmapsto \text{HFD} \circ \text{TE} \circ \text{RB}^\sharp(e, t).$$

For dynamic analysis, we compose this with degradation.

Definition 4.4 (Pipeline transformer). A complete pipeline at a given turn consists of:

- an explicit artifact $e \in E_{\text{sem}}$ constructed as in Epistemic Spaces and Epistemic Algebra;
- the tacit suffix $\text{RB} \rightarrow \text{TE} \rightarrow \text{HFD}$;
- a degradation step $\delta : T \rightarrow T$.

It induces a map

$$\llbracket P \rrbracket_\delta : T \rightarrow T, \quad \llbracket P \rrbracket_\delta(t) := \delta(\text{HFD} \circ \text{TE} \circ \text{RB}^\sharp(e, t)).$$

Definition 4.5 (Run). A *run* (or *interaction sequence*) is a sequence of tacit states $(t_n)_{n \geq 0}$ such that, for some sequence of pipelines $P_0, P_1, \dots$,

$$t_{n+1} = \llbracket P_n \rrbracket_\delta(t_n) \quad \text{for all } n \geq 0.$$

When convenient, we also record the corresponding explicit artifacts and credences, writing $(t_n, e_n, c_n)_{n \geq 0}$.

All Epistemic Dynamics conclusions are stated in terms of such runs and the hallucination condition $t_n \prec \text{Req}(e_n, c_n)$.

4.4 Baseline Hallucination Inevitability

We first consider a deterministic setting with fixed degradation δ and fixed pipelines. To capture the notion that nontrivial epistemic demands recur, we use a requirement floor.

Definition 4.6 (Requirement floor and recurring demands). A state $r \in T$ is a *requirement floor* if, for a run $(t_n, e_n, c_n)_{n \geq 0}$, we say that *nontrivial demands recur relative to r* when there exists an infinite subsequence $(n_k)_{k \geq 0}$ such that

$$\text{Req}(e_{n_k}, c_{n_k}) \succeq r \quad \text{for all } k.$$

We strengthen the degradation assumption to ensure that tacit states eventually fall below this floor.

Assumption 4.7 (Degradation below the requirement floor). There exists $r \in T$ such that, for every $t_0 \in T$, there is $N \in \mathbb{N}$ with

$$\delta^N(t_0) \prec r.$$

Under these assumptions, Appendix ?? shows:

Theorem 4.8 (Hallucination inevitability, deterministic case). *Let $(t_n, e_n, c_n)_{n \geq 0}$ be a run for which Assumption ?? holds and nontrivial demands recur relative to some requirement floor r . Then there exists $n \in \mathbb{N}$ such that*

$$t_n \prec \text{Req}(e_n, c_n).$$

In particular, every infinite run of this form contains at least one hallucination.

The theorem applies to runs that are both degrading and nontrivial in the sense above. Scenarios restricted to very easy tasks, very short horizons, or aggressive external re-grounding may not satisfy these premises and are not targeted here.

4.5 Confidence, Requirement Escalation, and Feedback

The behaviour of the human agent depends not only on the explicit content e_n but also on how plausible it appears. Epistemic Dynamics therefore adds structure for explicit plausibility, credence, and feedback between credence and degradation. The formal definitions and proofs are given in Appendix A.9.8–A.9.13; we summarise the main ideas.

Plausibility and confidence. An explicit plausibility functional $\pi : E \rightarrow [0, 1]$ assigns to each artifact a scalar measure of surface coherence or apparent authority. Tacit state is extended to $T^e := T \times \mathbb{R}_{\geq 0}$, with elements (t, c) where t is grounding and c is credence. A confidence inflation operator $\Phi : (T^e \times E) \rightarrow T^e$ updates credence according to plausibility, without improving t ; for example

$$\Phi((t, c), e) = (t, c + \varphi(\pi(e))),$$

for some monotone $\varphi : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$.

Requirement escalation and feedback-coupled degradation. Since $\text{Req}(e, c)$ is monotone in c , higher credence raises requirements. To reflect the possibility that higher confidence reduces scrutiny, degradation is allowed to depend on c via a map $\delta^* : T^e \rightarrow T$, written $\delta^*(t, c) = \delta_c(t)$, such that:

- for fixed t , higher c leads to weaker outputs: $\delta^*(t, c_2) \preceq \delta^*(t, c_1)$ whenever $c_1 \leq c_2$;
- at zero credence, δ^* reduces to the baseline δ .

An extended run $(t_n, c_n, e_n)_{n \geq 0}$ is then obtained by interleaving confidence inflation and feedback-coupled degradation. Under assumptions that credence can grow without bound along subsequences when high-plausibility artifacts recur, Appendix A.9.13 establishes an *accelerated* version of Theorem ??: hallucinations still occur in finite time, and the time-to-hallucination is not increased, and may be strictly reduced, by the presence of the feedback loop.

4.6 Stochastic Pipelines, Ensembles, and Finite Horizons

The preceding discussion treats runs as deterministic sequences. In many applications, FRFP pipelines include stochastic components (e.g., sampling, randomised retrieval). Appendix A.9.15–A.9.16 extends the dynamic semantics to this case by viewing runs as trajectories ω in a space Ω and equipping Ω with probability measures arising from stochastic implementations.

For each trajectory $\omega = ((t_n, c_n, e_n))_{n \geq 0}$, the *hallucination time* is defined as

$$\tau(\omega) := \inf\{n \geq 0 : t_n \prec \text{Req}(e_n, c_n)\},$$

with $\tau(\omega) = +\infty$ if no hallucination occurs. When the structural premises of Theorem ?? (and its feedback-augmented variants) hold pointwise on the support of a stochastic pipeline, Appendix A.9.15–A.9.16 shows that $\tau(\omega) < \infty$ for all such ω . Any probability measure supported on these trajectories then satisfies $\mathbb{P}(\tau < \infty) = 1$: hallucination is almost sure.

For a fixed horizon N , the finite-horizon hallucination probability $p_N := \mathbb{P}(\tau \leq N)$ is non-decreasing in N and converges to 1 as $N \rightarrow \infty$. For K independent runs, the probability that at least one run hallucinates by time N is $1 - (1 - p_N)^K$. Thus ensembles can improve robustness over short horizons, but they do not change the long-run structural inevitability when the underlying conditions are met.

4.7 Implications

Epistemic Dynamics shows how the static FRFP architecture behaves over long interaction sequences when combined with simple models of context degradation, requirements, and confidence. The main conclusions are:

- hallucination is naturally captured as a mismatch between tacit context and task requirements, independent of any particular AI architecture;
- under monotone degradation and recurring nontrivial demands, hallucinations are inevitable along sufficiently long runs;
- feedback from plausibility and credence to degradation tends to shorten the time-to-hallucination rather than extend it;
- stochasticity and ensembles can reduce error on finite horizons, but do not remove the long-run structural pressure towards hallucination under the same assumptions.

The formal statements and proofs are given in Appendix ?? . Subsequent sections extend the framework from single agents to multi-agent and institutional settings.

The dynamics layer enriches tacit states with degradation and requirements. In the airfare setting, the requirement map $\text{Req}(e, c)$ encodes, for a given explicit configuration e (set of options, price ranges, constraints) and credence level c , how much tacit grounding is required for it to be appropriate to accept an itinerary as “best”. For example, a corporate travel policy might demand that the human understand not only the price and timing, but also fare rules and change penalties before accepting a premium non-refundable ticket.

As the human interacts with the assistant across many planning tasks, tacit grounding degrades via $\delta : T \rightarrow T$ under cognitive load or time pressure, while nontrivial requirements recur. Within our Epistemic Dynamics model, prolonged or repeated use of such a “best airfare” protocol without adequate rest or oversight will eventually lead to requirement-violating acceptances—for instance, choosing an itinerary that superficially fits high-level preferences but violates tacit constraints the human would normally enforce (such as minimum connection times or visa requirements). The safe-horizon functional $H_P(\varepsilon)$ then captures, for a given pipeline P and tolerance ε , how many such planning episodes can be run before the probability of a tacitly unacceptable choice exceeds ε .

5 Collective Epistemic Dynamics

We now lift the conversational calculus to populations and institutions. Multiple human agents share explicit artefacts and communicate via typed workflows; their tacit states evolve individually but interact through shared artefacts, committees, and policies. The resulting collective layer supports population-level dynamics, consensus mechanisms, and governance operators, and yields explicit-only impossibility results for hallucination detection and institutional automation. Technical details and categorical formulations appear in the appendices, in particular Appendices A–C.

5.1 From Single Pipelines to Epistemic Populations

In the single-agent setting, an FRFP pipeline comprises:

- an explicit artefact space E (texts, code, plans, proofs),
- a tacit space T with a preorder $\preceq$ capturing “at least as grounded as”,
- an extended tacit state space $T^e = T \times \mathbb{R}_{\geq 0}$ pairing grounding and tacit credence,
- dynamics combining degradation δ , requirement maps Req , confidence inflation, and feedback-coupled degradation.

A single execution is a trajectory

$$\omega = ((t_n, c_n, e_n))_{n \geq 0},$$

and hallucination occurs at the first n such that $t_n \prec \text{Req}(e_n, c_n)$ while the explicit artefact e_n is nevertheless accepted in the pipeline.

To move to the collective level, we replace the single tacit space by a family

$$A_i = (T_i, \preceq_i, \delta_i, T_i^e), \quad i \in I,$$

of epistemic agents indexed by a set I (individuals, teams, or institutions). Each A_i has its own tacit space, requirement relation, and degradation dynamics. Agents share a common explicit reasoning

environment described by the category

$$\mathcal{C} = N \ltimes E,$$

where E collects explicit artefacts and N captures navigation and control structure at the explicit level (task selection, mode switching, tool invocation, and so on).

The joint explicit state of the population is modeled as an object of the product category

$$\mathcal{C}^I.$$

An object $X \in \mathcal{C}^I$ is a family $X = (X_i)_{i \in I}$ of explicit states, indicating for each agent which explicit task and artefacts are currently active. A morphism $X \rightarrow Y$ in $\mathcal{C}^I$ represents one step of explicit activity across the population: one or more agents update their explicit state, for example by computing, revising, delegating, or producing new artefacts.

A population semantics functor

$$\overline{J} : \mathcal{C}^I \longrightarrow \prod_{i \in I} T_i^e$$

associates to each global explicit state X a tuple

$$\overline{J}(X) = ((t_i, c_i))_{i \in I}$$

of extended tacit states, obtained by allowing each agent to consume the explicit artefacts in X and update according to the single-agent dynamics. This lifts explicit–tacit separation to the population level: explicit artefacts are shared, while tacit grounding remains agent-local.

5.2 Collective Dynamics as a Markov System

To analyze temporal behaviour, we endow $\mathcal{C}^I$ with stochastic dynamics. From any global explicit state X , there is a distribution over possible successors Y , reflecting:

- which agents act next (possibly determined by a scheduling or institutional mechanism),
- which explicit morphisms in $\mathcal{C}$ they apply (such as computation, summarization, deployment, or consensus),
- which agents observe the newly produced artefacts (a communication structure on I),
- how each tacit state (t_i, c_i) is updated via representational backflow, degradation, confidence inflation, and feedback-coupled degradation.

Formally, this is captured by a Markovian collective dynamics

$$M : \text{Obj}(\mathcal{C}^I) \longrightarrow \mathcal{P}(\text{Obj}(\mathcal{C}^I)),$$

which specifies for each X a probability measure $M(X, \cdot)$ on successor states. Equivalently, one can view M as a functor

$$M : \mathcal{C}^I \longrightarrow \mathbf{Kl}(D),$$

into the Kleisli category of the discrete distribution monad D , but the object-level transition kernel suffices for present purposes. Fix an initial state $X_0 \in \text{Obj}(\mathcal{C}^I)$. The transition kernel M then induces a probability measure on the space of infinite executions

$$\Omega := \text{Obj}(\mathcal{C}^I)^{\mathbb{N}},$$

namely the law of the Markov chain on $\text{Obj}(\mathcal{C}^I)$ with initial condition X_0 and transition kernel M . We denote this measure by P_{M,X_0} . When the initial state is clear from context, we suppress it and write P_M .

A *collective execution* is a random path

$$X_0, X_1, X_2, \dots$$

in $\mathcal{C}^I$ generated by M from an initial state X_0 . The corresponding tacit trajectories $\bar{J}(X_n)$ record how grounding and credence evolve over time for each agent, coupled through shared explicit artefacts and communication. This yields a population-level path space and probability measure analogous to the single-agent setting, and permits the usual notion of stopping time with respect to the resulting filtration (see Appendix A.10 for details).

5.3 Collective Hallucination and Its Inevitability

At the individual level, hallucination is the first time an explicit artefact is endorsed despite insufficient tacit grounding. At the collective level, the corresponding notion involves endorsement by a subset of agents.

Given a collective execution $(X_n)_{n \geq 0}$ with tacit states

$$\bar{J}(X_n) = ((t_{i,n}, c_{i,n}))_{i \in I},$$

a *collective hallucination* at time n occurs if there exists an explicit artefact $e \in E$ and a subset $J \subseteq I$ with $|J| \geq 1$ such that:

- each agent $j \in J$ has accepted or propagated e (i.e., treated it as admissible for downstream use under their tacit gate) in their pipeline by time n , and
- for all $j \in J$,

$$t_{j,n} \prec_j \text{Req}_j(e, c_{j,n}),$$

i.e. the tacit state lies strictly below the applicable requirement for e in that context.

Here “nontrivial” means *nonempty*: we allow $|J| \geq 1$, so the collective notion strictly extends the single-agent notion. When modeling multi-approval governance, we use the strengthened threshold $|J| \geq k$ (typically $k = 2$), corresponding to acceptance rules that require multiple independent tacit gates.

The *collective hallucination time* is the stopping time τ^{coll} defined by

$$\tau^{\text{coll}}(\omega) := \inf \{ n \geq 0 : \omega \text{ exhibits a collective hallucination at step } n \},$$

with $\inf \emptyset = \infty$ as usual.

Under the same high-level assumptions that drive single-agent inevitability (finite tacit capacities, non-invertible degradation for each agent, recurrent demands on grounding via surface-plausible artefacts, confidence inflation, and requirement escalation), and under mild assumptions on M (continued production and circulation of artefacts through the population), one obtains an almost-sure finiteness result:

$$P_M(\tau^{\text{coll}} < \infty) = 1.$$

In words, over sufficiently long horizons, some under-grounded artefact will almost surely be collectively endorsed, provided the structural FRFP conditions remain satisfied. A formal statement and proof are given in Appendix A.10.11–A.10.12.

5.4 Consensus Mechanisms as Explicit-Only Functors

Collective systems frequently employ consensus mechanisms: voting schemes, averaging procedures, ensemble methods, committee decisions, and more elaborate governance structures. Within the present framework, these are modeled as explicit-only operators

$$C : E^n \longrightarrow E,$$

which aggregate a finite tuple of explicit artefacts $(e_1, \dots, e_n) \in E^n$ into a new explicit artefact $C(e_1, \dots, e_n)$. Crucially, such operators act solely in explicit space; they have no direct access to tacit states T_i or to the requirement maps Req_i .

A basic consequence is *tacit-stability*. Fix an agent i and a context c_i . Suppose that for each $k \in \{1, \dots, n\}$,

$$t_i \prec_i \text{Req}_i(e_k, c_i),$$

so each input artefact is under-grounded for agent i in context c_i . Because explicit transformations do not modify the underlying tacit state, any output $C(e_1, \dots, e_n)$ is evaluated against the same t_i and c_i and therefore remains under-grounded.

As a consequence, fully automated consensus processes that operate purely on explicit artefacts cannot, in general, eliminate hallucination or under-grounded decisions at the collective level. They can smooth, denoise, or delay failures by reshaping explicit behaviour, but they cannot create new grounding in tacit space. This extends the single-agent observation that explicit refactorings do not restore grounding to the population level.

5.5 Structural Notions at the Collective Level

The collective formalism introduces structural notions that do not appear at the static single-agent level and that will be used later.

Equivalence of collective protocols. Viewing $\mathcal{C}^I$ as the explicit category for the full population, different interfaces, workflows, or consensus arrangements can be represented as endofunctors

$$F : \mathcal{C}^I \longrightarrow \mathcal{C}^I$$

that preserve the population semantics $\bar{\mathcal{J}}$. If two designs F_1 and F_2 are related by explicit-only refactorings that leave $\bar{\mathcal{J}}$ unchanged, they are *equivalent* in the FRFP sense: they have the same tacit semantics and the same distribution of the collective hallucination time τ^{coll} . Any difference between them is then syntactic rather than epistemic.

Refinement order on architectures. Given two Markovian collective dynamics M_1 and M_2 defined on the same population, with corresponding collective hallucination times τ_1^{coll} and τ_2^{coll} , define the finite-horizon risk curves

$$p_N^{(k)} := P_{M_k}(\tau_k^{\text{coll}} \leq N), \quad k \in \{1, 2\}, N \in \mathbb{N}.$$

We say that M_2 *dynamically refines* M_1 if $p_N^{(2)} \leq p_N^{(1)}$ for all N . This induces a partial order on collective designs: some architectures are strictly safer than others in the sense of never increasing finite-horizon hallucination risk. Explicit refactorings that preserve $\bar{\mathcal{J}}$ cannot strictly improve this order; they can at best preserve it. In contrast, interventions that alter tacit structure (e.g. adding independent evaluators, modifying requirements, or slowing degradation) can produce genuine refinements. A more detailed treatment appears in Appendix A.10.

Compositional risk and defense in depth. Many systems decompose into interacting subpopulations (for example, scientific, engineering, and policy components). Each subpopulation has its own collective hallucination time $\tau_{\text{sub}}^{\text{coll}}$ and hazard profile. At a coarse level, the overall collective hallucination time is bounded by

$$\tau^{\text{coll}} = \min_{\text{sub}} \tau_{\text{sub}}^{\text{coll}}.$$

The system fails epistemically as soon as any critical component fails. The formalism therefore supports a compositional “defense-in-depth” analysis: subsystem-level guarantees on hazard profiles may, under appropriate coupling conditions, yield system-level guarantees. The appendices provide the technical apparatus needed to make such statements precise.

Do collectives reduce or increase hallucination risk on fixed horizons? Moving from a single agent to a population does not have a uniform effect on hallucination risk. On a *fixed horizon* N , collectives can make hallucination *less probable* or *more probable*, depending on how acceptance is implemented and how correlated the agents’ errors and evidence streams are.

When collectives can reduce fixed-horizon risk (defense in depth). If an explicit artefact must pass *independent* tacit gates before it is accepted (e.g., two-person sign-off, committee review, replication-style checks), then collective hallucination requires that multiple tacit states simultaneously fall below their relevant requirements. When the gates are sufficiently independent—in the sense that their under-groundedness events are not strongly positively correlated—the probability of a collective hallucination by time N can be strictly smaller than in the single-agent case. Intuitively, independence turns “one agent misses the flaw” into “several agents must miss it in compatible ways.” This is the collective analogue of layered safety checks: additional gates can improve finite-horizon safety by reducing the chance that any single agent’s degraded tacit state becomes the sole point of failure.

When collectives can increase fixed-horizon risk (amplification and speedup). Collectives can also *increase* short-horizon risk when they raise throughput of artefact production and circulation, or when their failures are highly correlated. Shared tools, shared prompts, shared datasets, or shared institutional incentives can make agents’ errors co-move, so that many agents become under-grounded with respect to the same artefacts at roughly the same time. In addition, populations typically explore and propagate far more candidate artefacts per unit time than a single user; even if the per-artefact hallucination probability is unchanged, the probability that *some* artefact is accepted while under-grounded can increase simply because the system samples more opportunities for failure. Finally, if “consensus” operates as an explicit-only aggregator over weakly grounded artefacts, it can accelerate endorsement by producing artefacts that look more authoritative without increasing tacit grounding.

Structural implication. The collective layer therefore supports two distinct design levers. First, *gate structure* (how many tacit acceptances are required, and by whom) can reduce finite-horizon risk when evaluations are genuinely independent. Second, *correlation and throughput control* (diversity of evidence, diversity of tools, rate-limits, and escalation rules) becomes essential to prevent populations from becoming high-throughput amplifiers of the same under-grounded artefacts. None of these effects contradict the long-horizon inevitability results under the FRFP assumptions: layered collectives can reshape and delay risk, but cannot, in general, remove it over unbounded horizons when tacit capacities degrade and nontrivial requirements recur.

Non-identifiability from logs. Collective executions give rise to rich explicit traces: logs, metrics, dashboards, and derived indicators. All of these live in $\mathcal{C}^I$ or in its images under explicit functors.

Tacit states $\bar{J}(X)$ and requirement maps Req_i remain unobserved. In general, many different tacit configurations and dynamics M can induce the same (or arbitrarily similar) distributions over explicit traces. It follows that correctness and hallucination risk are *non-identifiable* from logs alone: without additional assumptions, explicit-only monitoring and metrics cannot uniquely determine how well grounded a population’s beliefs are.

Scaling and phase-transition questions. The population-level formulation also supports scaling questions. Parameterizing systems by population size $|I|$, communication structure, and per-agent degradation properties, one can ask how epistemic risk scales with the number of agents and tools, whether particular communication patterns are stabilizing or destabilizing, and whether qualitative changes (phase transitions) occur as complexity grows. The present work does not attempt a systematic scaling analysis, but the structural notions above identify the relevant objects for future investigation.

5.6 Connections to Example Domains and Local Interventions

The same collective apparatus applies across domains in which many agents interact over shared artefacts:

- Scientific communities can be modeled as populations of agents (labs, groups, or individual scientists) with papers, datasets, and code as explicit artefacts, and peer review, replication, and citation practices as partial consensus mechanisms.
- Software development lifecycles can be treated as populations of developers, reviewers, operators, and tools with explicit artefacts such as requirements, designs, code, tests, and deployment configurations, coupled by CI/CD and incident-response processes.
- Strategic domains such as military planning or pandemic response involve agents in governments, institutions, and advisory bodies operating over explicit artefacts including plans, intelligence, models, and policies, with doctrines and institutions acting as higher-level operators on M and on Req .
- Markets and large-scale experiments provide examples where rich explicit traces (transaction logs, event data) remain insufficient to uniquely identify underlying beliefs or theories, paralleling the non-identifiability observation above.

The collective perspective also clarifies the status of local interaction patterns, such as common prompting schemes for language models (incentive framing, challenge prompts, stepwise guidance, high-stakes framing, self-evaluation prompts). These can be viewed as local interventions on:

- the explicit pipeline shape in $N \times E$ (for example, encouraging decomposition and self-checking),
- the effective requirement schedule (for example, weakening answers to hypotheses or raising the bar for strong claims),
- the human user’s tacit dynamics (for example, prompting increased attention and skepticism).

Such interventions can materially improve finite-horizon correctness and enlarge safe horizons for particular tasks. They do not, however, alter the structural facts established in the underlying theory: explicit-only systems remain ungrounded, human tacit capacities remain finite and degrading, and, within the Epistemic Dynamics and population assumptions of FRFP, under-grounded collective decisions remain inevitable over unbounded horizons.

In summary, the collective layer provides a coherent extension of FRFP to multi-agent settings. It connects individual pipeline dynamics to institutional design, clarifies the limits of explicit-only interventions (including consensus mechanisms and monitoring), and identifies the structural levers—in particular, those acting on tacit dynamics and requirements—on which robust long-horizon human–AI collaboration must ultimately rely.

In a collective version of the airfare example, a team or organisation might use a shared assistant to propose itineraries for many travellers under a common travel policy. Individual tacit states live in agent-local copies of T , while the shared explicit category records policies, candidate options, and past choices. Consensus mechanisms and governance operators at this layer determine, for example, whether a manager’s tacit judgment can override an employee’s preferred itinerary, or how exceptions to the cheapest-available-fare policy are recorded. Our non-automation results imply that, even with strong explicit policies and logging, no explicit-only governance operator can, in general, guarantee that all accepted itineraries are tacitly acceptable across all such populations and protocols.

6 Structural Limits of Explicit Mechanisms

The preceding development has described a class of human–AI collaborations in which

- explicit artefacts and transformations are modelled in a category E of explicit reasoning,
- navigation, prompting, and other meta-operations are represented in a category N ,
- tacit states and evaluations live in a space T , and
- the combined system $N \times E$ is equipped with a reduction relation and a semantic functor

$$\llbracket - \rrbracket : N \times E_{\text{sem}} \longrightarrow T,$$

with observable correctness extracted via a map $\gamma : \text{Ob}(T) \rightarrow J$.

Under the FRFP axioms (ETS, HEG, HEC, AE), explicit computation is separated from tacit evaluation and cannot reconstruct tacit state or correctness except via designated human interaction. At the same time, the dynamic and collective layers developed earlier show that long-horizon behaviour is inherently stochastic: pipelines are implemented as stochastic processes over a space of admissible trajectories, and populations interacting through explicit channels induce Markovian dynamics over collections of agents.

This section explains how these ingredients jointly constrain what can be achieved by mechanisms that operate purely in explicit space. Formal statements and proofs of the main structural results are given in Appendix D.

6.1 Dynamic Quantities and Long-Horizon Behaviour

When a fixed pipeline P is executed under FRFP constraints, the dynamic semantics introduced previously associate to any admissible implementation:

- a measurable space of trajectories Ω_{allowed} of runs, each trajectory $\omega \in \Omega_{\text{allowed}}$ corresponding to a reduction sequence in $N \times E$ satisfying the interaction protocol constraints;

- a hallucination time random variable

$$\tau : \Omega_{\text{allowed}} \longrightarrow \mathbb{N} \cup \{\infty\},$$

recording the first time a requirement-violating artefact is accepted along the run;

- finite-horizon hallucination probabilities

$$p_N = \mathbb{P}(\tau \leq N)$$

for $N \in \mathbb{N}$;

- a hazard sequence $(h_n)_{n \in \mathbb{N}}$ derived from the increments of (p_N) ; and
- for each tolerance $\varepsilon \in (0, 1)$, a safe-horizon functional

$$H_P(\varepsilon) = \sup\{N \in \mathbb{N} : p_N \leq \varepsilon\}.$$

These objects summarise, in a long-horizon sense, how often and how soon a given implementation of P will fail the tacit requirements encoded by **Req**.

Under the epistemic dynamics assumptions (finite tacit capacity, monotone non-invertible degradation of tacit state, recurring nontrivial demands, and appropriate independence conditions), the model yields an inevitability theorem: along admissible trajectories in this setting, hallucination occurs almost surely in the limit. In particular, $\mathbb{P}(\tau < \infty) = 1$ for any such implementation, and for every fixed $\varepsilon \in (0, 1)$ the safe horizon $H_P(\varepsilon)$ is almost surely finite.

These results are proved in the dynamic layer but are agnostic to the internal details of the explicit mechanisms; they depend only on the interaction between explicit artefacts, requirement maps, and degradation behaviour of tacit state.

6.2 Population-Level Dynamics and Consensus

When multiple agents interact through explicit channels, the population development introduces:

- an explicit reasoning category $\mathcal{C}$ for individual agents;
- a product category $\mathcal{C}^I$ of explicit profiles across a population I ;
- functorial consensus operators

$$C : \mathcal{C}^n \longrightarrow \mathcal{C},$$

modelling explicit-only aggregation rules (voting, averaging, reconciliation of drafts, and so on); and

- a Markovian collective dynamics M on $\mathcal{C}^I$, obtained by iterating local update and interaction steps.

Given an epistemic population (agents, tacit spaces, and requirement maps) and such a dynamics M , one can define a collective hallucination time τ_{coll} as the first step at which a nonempty subset of agents jointly accept an artefact that is under-grounded relative to their individual requirements.¹

Under assumptions analogous to those used for single-pipeline dynamics, one again obtains an almost-sure inevitability result: for admissible populations and dynamics consistent with the FRFP

¹When modelling multi-approval governance, one may strengthen “nonempty” to $|J| \geq k$ (typically $k = 2$) to represent acceptance rules that require multiple independent tacit gates.

constraints, the collective stopping time τ_{coll} is almost surely finite. Moreover, the almost-sure finiteness of τ_{coll} is invariant under inserting additional explicit-only consensus layers: functorial consensus operators act purely in explicit space and cannot, by themselves, change the inevitability class of the process.

These collective results provide a quantitative description of long-horizon epistemic risk at the level of whole institutions or populations, using the same kind of stochastic and categorical apparatus as in the single-pipeline case.

6.3 Tacit Dependence and Explicit-Only Mechanisms

Against this backdrop, it is natural to ask which desirable properties one might hope to enforce by purely explicit means. In the present setting, explicit-only mechanisms are those whose internal computation and inputs/outputs are confined to E (or $\mathcal{C}$ in the population case), and which do not gain access to tacit states or tacit evaluations other than through the designated interfaces already modelled in the semantics.

A recurring theme is that many natural predicates of interest are *tacit-dependent*: their truth value can vary between admissible runs that share the same explicit artefacts and explicit interaction history but differ in their tacit evolution. Examples include:

- whether a given explicit artefact meets the relevant tacit requirements at the point of use;
- whether hallucination occurs within a fixed horizon, as expressed by events of the form $[\tau \leq N]$;
- whether an implementation has a given ε -safe horizon, as expressed by inequalities $H_P(\varepsilon) \geq N$; and
- in the collective setting, whether a population remains hallucination-free up to time N , as expressed by events such as $[\tau_{\text{coll}} > N]$.

Because the combined explicit trace does not uniquely determine the underlying tacit trajectory, predicates of this kind cannot be reduced to functions of explicit artefacts alone. This observation is formalised in Appendix A.11, which defines tacit-dependent predicates in terms of the semantic functor $\llbracket - \rrbracket$ and observable correctness $\gamma(\llbracket P \rrbracket)$ and then studies their interaction with explicit-only mechanisms.

The central structural result, stated and proved there, shows that under ETS and AE, no mechanism confined to explicit space can enforce a tacit-dependent predicate across all admissible runs. The proof relies only on the fact that such a mechanism receives isomorphic explicit inputs across certain pairs of admissible runs whose tacit behaviour differs, and hence must behave identically even when the predicate of interest disagrees.

6.4 Consequences for Detection, Elimination, and Consensus

Several consequences follow immediately when this schema is instantiated with the dynamic and collective predicates arising from earlier sections.

Detection and elimination. The definition of hallucination in terms of requirement maps and tacit state yields a predicate of the form “this explicit artefact is hallucinated” which is tacit-dependent: the same explicit output can be acceptable or unacceptable depending on the tacit configuration in which it is evaluated. It follows that:

- there is no perfect hallucination detector operating purely on explicit artefacts and logs; and

- no explicit-only policy can guarantee that hallucination never occurs along admissible runs, or that $\tau = \infty$ (or $\mathbb{P}(\tau = \infty) = 1$) holds uniformly across implementations supported on Ω_{allowed} .

Under the epistemic dynamics assumptions, inevitability theorems already show that τ is almost surely finite for admissible implementations, and that $H_P(\varepsilon)$ is almost surely finite for each fixed $\varepsilon \in (0, 1)$. The structural results in Appendix D complement this by ruling out the possibility of using explicit-only mechanisms to enforce $\tau = \infty$ or $H_P(\varepsilon) = \infty$ as design constraints.

Correctness oracles. Observable correctness is represented by $\gamma(\llbracket P \rrbracket)$ and is defined purely in tacit space, subject to HEG and HEC. For a fixed explicit artefact, different admissible runs can lead to different tacit evaluations and hence different observable correctness values. A hypothetical correctness oracle

$$\Omega : E \longrightarrow J$$

that recovered $\gamma(\llbracket P \rrbracket)$ from the explicit artefact alone would thus be required to decide a tacit-dependent predicate using only explicit data. The formal results establish that no such oracle can exist within the FRFP axioms if it is confined to explicit computation.

Consensus and collective guarantees. In the population setting, consensus operators $C : E^n \rightarrow E$ and explicit coordination rules act entirely in E and do not modify tacit states directly. They can be composed to form consensus pipelines and embedded in the Markov dynamics M on population states. The structural results then imply that:

- explicit-only consensus cannot create grounding ex nihilo: if each input artefact exceeds the grounding of an agent, so does the consensus output for that agent;
- no fully automated consensus pipeline can guarantee that its outputs are hallucination-free across all admissible populations and implementations; and
- more strongly, predicates such as $[\tau_{\text{coll}} = \infty]$ are tacit-dependent and cannot be enforced by explicit-only transformations of the population dynamics.

Invariance results for τ_{coll} in the collective dynamics further reinforce this picture: inserting additional explicit-only consensus layers can alter the distribution of collective risk but cannot change the fact that $\tau_{\text{coll}} < \infty$ almost surely under the stated assumptions.

6.5 Implications for Design and Governance

Taken together, these structural limitations delineate the scope of what explicit-only design can accomplish in long-horizon human–AI collaboration.

On the one hand, the dynamic layer provides meaningful quantitative levers:

- implementations can be compared via their finite-horizon risk profiles (p_N) and hazard sequences (h_n);
- the safe-horizon functional $H_P(\varepsilon)$ captures, for each tolerance ε , how long a system can typically operate before crossing a risk threshold;
- dynamic refinement orders allow one to classify implementations as safer or riskier over finite horizons.

On the other hand, the impossibility results anchored in tacit dependence show that no amount of explicit-only filtering, logging, or aggregation can convert a system whose long-run behaviour is almost surely fallible into one that is guaranteed safe across all admissible runs and populations.

Within this boundary, the design problem becomes one of shaping hazard profiles and safe horizons, structuring interactions so that collective dynamics refine a baseline rather than degrade it, and introducing additional forms of tacit evaluation and institutional oversight.

Appendix B.6 develops concrete protocol templates and institutional patterns compatible with these constraints. Appendix C instantiates the dynamic and institutional semantics in detailed case studies (ICU sepsis literature review and financial scenario analysis / stress testing). Appendix B.11 provides a FRFP-compliant system-instruction set for an explicit-only assistant, and Appendix B.27–B.30 analyses a set of FRFP-aware prompting patterns and their effects on hazard profiles and safe horizons. Together, these appendices turn the abstract design levers of this section into operational tools for building and evaluating concrete workflows.

Subsequent sections on governance, incentives, and institutional design can be read as exploring these remaining degrees of freedom: adjusting requirement maps, allocating human attention, and configuring collective processes so as to manage long-horizon risk in a way that is consistent with the FRFP axioms and with the structural limitations just described.

Formal statements and proofs of the general schema and its corollaries appear in Appendix D; the present discussion is intended to make clear how those results integrate with the dynamic and collective layers and how they bear on the design of real-world human–AI collaborations.

7 Institutions for Long-Horizon Human–AI Collaboration

The previous sections analyzed individual human–AI pipelines and their long-run behavior: how explicit pipelines acquire tacit grounding, how degradation and hallucination accumulate over time, and how collective epistemic dynamics emerge when many agents interact with shared systems. We now step up one more level, from collections of agents and runs to the *institutions* that sit above them. The aim of this section is conceptual: to describe the role of institutions in stabilizing long-horizon practice, while leaving formal details to Appendix A.12.

7.1 From Individual Agents to Epistemic Populations

Earlier, we treated a single human agent as a bearer of tacit context, capable of evaluating explicit artefacts and updating their internal state in response to interaction with an FRFP pipeline. At scale, however, we are never dealing with a single agent in isolation. Scientific communities, engineering teams, regulatory bodies, and user populations are all better modeled as *epistemic populations*: structured collections of agents, each with their own tacit capacities and degradation dynamics.

Formally, we use the agent and population structures introduced in the collective dynamics analysis. Each epistemic agent carries:

- a tacit context space T_i with a preorder expressing “at least as well grounded as”;
- a degradation operator $\delta_i : T_i \rightarrow T_i$ capturing background loss of tacit context;
- an extended tacit state T_i^e that pairs tacit context with a notion of subjective credence.

For institutional purposes, we enrich this with a *correctness quotient* $\gamma_i : T_i \rightarrow J$, where J is the space of observable correctness judgments (accept/reject/revise, etc.). The tuple

$$(T_i, \preceq_i, \delta_i, T_i^e, \gamma_i)$$

is the institutional “signature” of the agent: it records which tacit states they can occupy, how those states degrade, and how they express correctness judgments when asked.

An *epistemic population* is then a family of such agents indexed by a set I , together with the morphisms that allow us to compare and transform populations. This lets us treat heterogeneous communities—with agents who differ in tacit capacity, degradation rates, and evaluative norms—within a single formal frame. Full categorical details are given in Appendix A.12, but the upshot for the main story is that we can talk sensibly about how a fixed pipeline behaves when replicated across a population, and how institutions act on the resulting pattern of judgments.

7.2 Replication and Population-Level Semantics

Given a well-typed FRFP pipeline P , the earlier semantic development assigns it a tacit meaning $\llbracket P \rrbracket$ and an associated correctness outcome under evaluation by a single agent. In practice, however, we do not trust any single evaluation of P ; we care about what happens when P is executed across a population, at different times, under different tacit conditions.

To capture this, we consider *admissible runs* of P by each agent in the population. An admissible run bundles:

- an explicit execution trace of P in the navigation–explicit space $N \times E$, subject to the structural constraints maintained throughout; and
- the induced evolution of the agent’s extended tacit state along that trace, as specified by the epistemic dynamics update rules (degradation, confidence inflation, requirement escalation, and feedback-coupled degradation).

A *replication family* for P is a choice of one admissible run for each agent (or for some subcollection of agents) in the population. For each such run, we select an evaluation time—for instance, when the pipeline halts or when a fixed resource limit is reached—and record the corresponding tacit state $t_{i,*}$ for agent A_i .

What matters for institutional behavior is not the detailed trajectory of each run, but the *pattern of correctness judgments* these runs induce. We therefore define the *population semantic profile* of P relative to a replication family $R(P)$ as the multiset

$$\mathcal{P}(P; R(P)) = \{\gamma_i(t_{i,*}) : i \in I_{\text{rep}}\} \subseteq J.$$

This multiset is the primary interface between long-horizon practice and institutions. It aggregates how many agents, under what tacit conditions, endorse, reject, or revise the outputs of a given pipeline. The construction makes clear that institutional input is inherently *tacit-dependent*, even though it is summarised in a discrete space of judgments.

The formal development of admissible runs, replication families, and population semantic profiles is given in Appendix A.12, building on the run semantics and stochastic models of Appendices B and C.

7.3 Institutions as Aggregators over Tacit Judgments

Institutions—journals, standards bodies, model review boards, internal red-teams, and so on—do not typically see the full detailed state of every agent. Instead, they receive some distribution of judgments: accept, reject, request further work, abstain. In our framework, an *institutional aggregator* is a map

$$\text{Mult}(J) \rightarrow J \cup \{\perp\},$$

where $\text{Mult}(J)$ is the space of multisets of judgments and $\perp$ denotes suspended judgment. For a given pipeline P , an institution chooses a particular aggregator I_P and applies it to the population semantic profiles generated when P is replicated.

This abstraction covers common patterns:

- majority or supermajority rules;
- unanimity or veto-based protocols;
- weighted consensus, where some agents’ judgments count more than others;
- deliberative or staged procedures that can output “no decision” ($\perp$).

Crucially, I_P operates *only* on the multiset of tacitly produced judgments. It does not see raw model outputs, logs, or metrics directly. All explicit activity is mediated through human agents, whose tacit evaluations are then aggregated.

Full definitions of institutional aggregators, institutional outcomes, and their basic properties appear in Appendix A.12. Here, the important point is that institutions sit at a distinct layer in the architecture: they neither compute the explicit artefacts themselves, nor embody the tacit processes that evaluate them. They are rules for turning many tacit judgments into collective decisions.

7.4 Why Institutions Cannot Be Fully Automated

Given mature AI systems and rich logging, it is tempting to imagine institutions that operate purely on explicit artefacts and their explicit summaries: dashboards, performance metrics, proofs of robustness, and so on. In our framework, such an institution would be *explicit-only*: for each pipeline P , its decisions could be computed by a mechanism that inspects only explicit artefacts, never accessing the underlying tacit states or correctness judgments.

The impossibility results developed earlier rule this out in a strong sense. Under the assumptions that:

- correctness is fundamentally tacit-dependent (it lives in the γ_i quotients of tacit states), and
- admissible pipelines can be arranged so that explicit traces are ambiguous with respect to tacit correctness,

Appendix A.12 proves an institutional non-automation theorem: no explicit-only institutional layer can be *correctness-preserving* across all admissible pipelines and populations. In particular, if we require that the institution agree with unanimous tacit verdicts whenever they occur, then there is no way to define its decisions purely in terms of explicit artefacts.

Intuitively, any such institution would amount to an “explicit-only correctness oracle,” and the earlier no-go theorems about correctness oracles apply. No amount of logs and metrics, however elaborate, can substitute for direct access to tacit evaluations. Institutions that aim at epistemic reliability must therefore be designed around *tacit input*, even if they also make use of explicit signals in how they schedule, resource, and structure evaluation.

The formal statement and proof of the institutional non-automation theorem are collected in Appendix A.12, where they are derived as a corollary of the explicit-only impossibility results from the more general theory.

8 Governance, Law, and Oversight

Institutions regulate the behaviour of individual pipelines and agents. But institutions themselves change over time: journals tighten or relax standards, internal review processes are redesigned, and regulatory frameworks evolve. In long-horizon settings, these higher-order changes are at least as important as first-order institutional choices. This section outlines how governance and law fit into the architecture, again referring to Appendix F for the formal layer.

8.1 Governance as a Meta-Institution

The collection of all institutional arrangements we might adopt forms a structured space. For each pipeline P we can choose an aggregator I_P , and we may change these choices over time in response to experience. We write Inst for the category whose objects are such institutional families and whose morphisms are structure-preserving transformations between them.

A *governance operator* is then an endofunctor

$$G : \text{Inst} \rightarrow \text{Inst}$$

that describes how institutional arrangements evolve. At each time step t , a governance process takes an existing institutional layer $(I_P^{(t)})_P$ and produces an updated one $(I_P^{(t+1)})_P$. This update may depend on observables extracted from the collective dynamics of agents and pipelines: disagreement rates, error frequencies, time-to-detection for hallucinations, evidence of confidence inflation, and so on.

The collective dynamics were modeled earlier as a process on epistemic populations interacting with pipelines. Appendix F shows how to couple this with governance: observables are extracted from the evolving population, fed into a governance operator, and used to adjust institutional aggregators over time. The resulting picture is a closed loop:

1. institutions aggregate tacit judgments and shape which pipelines are accepted or deployed;
2. those pipelines, in turn, influence the tacit states of agents and the composition of populations;
3. governance watches this evolving behaviour and modifies institutions accordingly.

This three-level interaction—agents and pipelines, institutions, and governance—is where the long-horizon behaviour of human–AI systems is ultimately determined.

8.2 Explicit-Only Governance and Its Limits

Just as we can imagine explicit-only institutions, we can imagine explicit-only governance: processes that monitor logs, metrics, and other explicit artefacts of institutional performance, and then automatically adjust institutional rules based solely on those explicit signals.

In our framework, an explicit-only governance operator is one that, at each time step, updates the institutional layer using only explicit artefacts produced so far. Formally, Appendix F captures this by requiring the updates to factor through explicit-only mechanisms on the explicit space E , in the same way the explicit-only institutions of Appendix A.12 factor through explicit artefacts.

The same tension appears here as before. If correctness is fundamentally tacit-dependent, and if pipelines and populations can be arranged so that explicit traces underdetermine tacit correctness, then governance that never consults tacit judgments cannot reliably track institutional performance. Any attempt to make governance fully explicit-only will either fail to detect drift away from correct

tacit practice, or else reintroduce tacit judgments indirectly (for example, via “ground truth” labels or other human-provided signals).

Appendix F formalizes this as a non-automation theorem for governance: under the same assumptions that rule out explicit-only correctness oracles and explicit-only institutional layers, there is no correctness-preserving governance operator that depends exclusively on explicit artefacts or their explicit summaries. In essence, a fully automated governance layer would again amount to a higher-order explicit-only oracle, and the impossibility results apply at this meta-level as well.

8.3 Law as a Constraint on Governance

Governance does not operate in a vacuum. In practice, institutions and their evolution are constrained by legal and normative frameworks: data protection laws, safety regulations, due-process requirements, and so on. In the present setting, we model these as predicates on institutional arrangements and governance operators.

Concretely, we can view a legal or regulatory framework as specifying a set of *admissible* institutions and transformations between them. A governance process is then *lawful* if it never maps admissible institutions to inadmissible ones. Law can also constrain which observables governance is allowed to use (for example, privacy-preserving restrictions on what can be logged) and which kinds of interventions are permitted.

While the formal treatment of law as a constraint on *Inst* and on governance functors appears in Appendix F, the conceptual takeaway is straightforward: the space of possible governance processes is smaller than the space of merely mathematically definable ones, and non-automation results for governance must be understood relative to whatever legal admissibility conditions we impose.

8.4 Human Oversight as a Structural Requirement

Taken together, the institutional and governance non-automation results point to a strong conclusion. It is not simply that, for now, we happen to rely on humans in the loop; rather, under the structural assumptions of the framework, there is no coherent way *within FRFP’s explicit–tacit separation and dynamic model* to remove them while preserving correctness over long horizons.

At the institutional level, correctness-preserving aggregation requires access to tacit judgments. At the governance level, correctness-preserving adaptation of institutions requires some access to tacit evaluations of institutional performance. Attempts to fully automate either layer collapse, in the limit, into explicit-only oracles, which the earlier impossibility theorems prohibit.

This does not mean that institutions or governance must be informal or opaque. On the contrary, much of their structure can and should be formalized, monitored, and partially automated. The key point is that the formal and automated parts must be designed *around* human tacit judgment, not as replacements for it.

The detailed statements and proofs of these claims are provided in Appendices E and F. Here we emphasize their architectural consequence: in any long-horizon human–AI ecosystem that satisfies the assumptions of the framework, human agents retain irreplaceable roles at three levels simultaneously:

1. as individual evaluators of explicit artefacts;
2. as contributors to institutional decisions via their tacit judgments; and
3. as overseers of governance processes that adapt and constrain institutions over time.

The rest of the paper can be read as an exploration of how to design pipelines, institutions, and governance mechanisms that make these roles precise rather than aspirational.

9 Discussion and Outlook

9.1 Design Implications

The FRFP architecture formalizes a set of design constraints for AI systems intended for long-horizon scientific collaboration:

- **Explicit-only AI.** AI systems should manipulate explicit structures but must not assume tacit roles such as correctness judgment. This limits their epistemic authority while maximizing their computational utility.
- **Single explicit–tacit boundary.** All correctness-bearing information flow should pass through a clearly identifiable boundary operator RB, with no hidden channels from tacit to explicit space.
- **Auditability via canonical forms.** Reasoning trajectories can be normalized within $\mathcal{N} \times \mathcal{E}$, yielding canonical forms amenable to audit and replication.
- **Structural handling of hallucinations.** Hallucinations are treated as structural mismatches between context and requirements, not as psychological quirks of models. This allows analysis of how context degradation, feedback, and task design interact.

Kernel-level constraints vs. instantiated layers. The design constraints above attach first and foremost to the kernel of FRFP: the explicit/tacit separation with a single boundary, the minimal primitive set, and the canonical $N \times E$ semantics with tacit correctness. By construction, any system that respects this kernel inherits the same structural limitations and invariants, regardless of how it instantiates dynamics, populations, or institutions. Conversely, Sections 6–10 and Appendices B–J are deliberately more contingent: they work out one particular family of choices for context degradation and requirements (Appendices B–D), collective and institutional dynamics (Appendices C–F), and concrete protocols, case studies, instruction sets, and prompting patterns (Appendices G–J) compatible with the kernel. These outer layers are intended as illustrative and revisable; what is non-negotiable, from the FRFP point of view, is the underlying correctness architecture and its separation between explicit computation and tacit judgment. Readers primarily interested in organizational and governance implications, rather than the categorical details, may wish to consult the non-technical organizational summary in Appendix K.

9.2 FRFP as a formalization of the scientific method

Although FRFP is stated as a domain-general protocol for long-horizon human–AI collaboration, its core architectural commitments mirror the stabilisation logic of the scientific method. Science is not merely a sequence of cognitive steps; it is an institutional technology for converting exploratory reasoning into stable, publicly defensible claims. FRFP abstracts that conversion into an explicit/tacit split plus a responsibility-bearing acceptance boundary.

A translation dictionary. The following correspondences are the intended reading of the formal objects.

- **Explicit artefacts (E).** Hypotheses, models, derivations, experimental designs, protocols, code, datasets, plots, and written arguments: objects that can be logged, transmitted, inspected, and mechanically transformed.

- **Explicit operations** (RI, EC, ED). Representation initiation (writing down a conjecture or design), computation (derivation, simulation, fitting, summarisation), and diagnostics (sanity checks, ablations, error analysis, unit tests, proof checking where applicable).
- **Navigation** ($\mathcal{N}$). Legitimate changes of perspective that preserve intent: reparameterisations, alternative model classes, reorganising an analysis, swapping datasets, or reframing a subproblem while remaining within the same research question.
- **Tacit state** (T). The correctness-bearing background a scientific community relies on but cannot fully encode: domain know-how, interpretive competence, judgment of adequacy, and the willingness to stand behind a claim under scrutiny.
- **Boundary** ($\text{RB} : E \rightarrow T$). Review-and-belief update: the act of assessing an explicit artefact against tacit standards and incorporating (or rejecting) it as part of what one is prepared to endorse.
- **Tacit primitives** (TE, HFD). Evaluation and acceptance: internal or institutional endorsement decisions (“this analysis is sound enough to rely on”; “this result is ready to submit”; “this meets the evidential bar for policy use”).

Replication, reproducibility, and peer review as FRFP patterns. The scientific stabilisation norms that motivate FRFP can be expressed as structured constraints on explicit workflows and on where boundary crossings are permitted.

- **Reproducibility** is an explicit-side property: an analysis is reproducible when the relevant explicit artefacts (data, code, environment, protocol) are sufficient for another agent to re-run an explicit pipeline and obtain the same explicit outputs, up to documented tolerances. In FRFP terms this is invariance in E under admissible refactoring and navigation, together with suitable logging of the pipeline normal form.
- **Replication** is a stronger, world-facing constraint: a claim survives replication when an independent execution produces new explicit artefacts (new data) that, upon boundary crossing, supports the same tacit endorsement. Formally, replication is not just agreement of E -outputs; it is agreement of tacit acceptance $\gamma(t)$ under independent pipelines that re-contact the domain.
- **Peer review** is a multi-agent boundary mechanism: independent tacit evaluators consume shared explicit artefacts and apply their own tacit operators before endorsing publication-level acceptance. In FRFP terms, peer review introduces additional boundary crossings by other agents, creating a layered acceptance structure that can improve finite-horizon risk when evaluations are sufficiently independent.

What FRFP adds, relative to informal scientific method talk. FRFP is not a sociological model of science; it is a protocol semantics intended to make long-horizon correctness and responsibility analyzable. Relative to informal descriptions of the scientific method, FRFP contributes three formal clarifications.

- **Acceptance is an event, not a vibe.** Scientific practice often blurs exploration and endorsement; FRFP forces the distinction by making acceptance a boundary crossing $E \rightarrow T$ with explicit responsibility.

- **Hallucination is endorsement failure.** In FRFP, “hallucination” is not the production of an incorrect string; it is the protocol-level event in which an explicit artefact is incorporated into tacit acceptance without meeting the active tacit requirements for that credence.
- **Stabilisation is workflow-level.** Reliability is treated as an emergent property of explicit pipelines plus review structure, not as a property of any single component (human or AI). This matches the way scientific norms reduce error by layering checks and independent evaluation rather than assuming perfect agents.

Scope and limits of the mapping. The mapping is exact in regimes where “science” is best understood as a stabilisation protocol over explicit public artefacts (papers, code, data) and tacit endorsement (what researchers and institutions will stand behind). It is intentionally agnostic about internal cognitive models and about which explicit checks are appropriate in a given domain. Conversely, in fully formal regimes where correctness is exhausted by an explicit oracle (e.g., proof checking under a fixed calculus), FRFP should be read as placing that oracle inside explicit space as an explicit correctness engine, with tacit judgment still governing the adequacy and scope of the formalisation. The purpose of the mapping is therefore not to claim that FRFP *is* the scientific method, but to show that FRFP isolates, and makes compositional, the same stabilisation logic that has historically made scientific claims durable.

9.3 Limitations and scope

The present work has several limitations.

Single-agent tacit model. We model one tacit context space T at a time; multi-agent and institutional settings, where different humans (and possibly institutions) have distinct tacit states, require richer structures (e.g. families of tacit spaces, communication channels, and aggregation mechanisms). Developing such multi-agent extensions is left to future work.

Worst-case focus. The hallucination inevitability results rely on deliberately worst-case assumptions: long or unbounded runs, monotone degradation, no explicit repair operator, and simple forms of requirement escalation and feedback. Practical systems may use aggressive re-grounding, multiple agents, explicit restarts, or external tools (e.g. proof assistants, simulators) to reduce risk. Our results should therefore be read as structural “upper bounds” on what can go wrong in principle, not as quantitative predictions for particular deployments.

Abstract requirements and tacit semantics. The requirement map $\text{Req}(e, c)$ is treated as an abstract semantic object; we do not attempt to estimate it from data or tie it to any specific scientific or engineering domain. Likewise, tacit states are modeled at a coarse level (often as elements of a preorder or low-dimensional space), omitting many cognitive and socio-technical factors that matter in practice. Making Req and the tacit dynamics empirically grounded for particular domains is outside the scope of this paper.

Level of abstraction. More broadly, we work at an intentionally abstract, foundational level and do not propose concrete algorithms, training procedures, or evaluations. FRFP is a design framework and semantic target, not a ready-made system. Instantiating it in concrete tools will require additional engineering choices (e.g. representations, interfaces, inference methods) that we do not model here.

Scope of inevitability and non-automation claims. All inevitability and non-automation statements in this paper are conditional on the FRFP architecture and its standing assumptions. In particular, the hallucination inevitability theorems presuppose the Epistemic Dynamics degradation and requirement conditions, together with the admissibility constraints on runs, and the impossibility results for detectors, consensus mechanisms, institutions, and governance operators presuppose explicit–tacit separation and human-exclusive grounding. They should therefore not be read as claiming that hallucination is inevitable, or that fully automated governance oracles are impossible, for *all* conceivable AI architectures. Rather, they assert that such guarantees cannot be obtained *within* the FRFP class of models, once tasks are nontrivial and requirements recur; trivial strategies such as always answering “I don’t know” or never accepting artefacts are ruled out by these structural premises rather than by logical impossibility. These non-automation results apply to any explicit-only AI system, including hypothetical AGI, as long as correctness remains a human/institutional notion rather than delegated to the system itself. In particular, in domains where correctness is truly explicit at the relevant level of analysis—for example, inner loops governed by sound and complete proof or test oracles—our Epistemic Dynamics inevitability statements are vacuous for that inner task. They become informative only when one models the outer tacit layer where specifications, benchmarks, and institutional standards are chosen and revised (Appendix D.3).

Counter-scenarios for Epistemic Dynamics assumptions. The Epistemic Dynamics inevitability results target a regime with recurring nontrivial demands, accumulating degradation, and bounded repair. There are three salient counter-scenarios. In *balanced or negative drift* regimes, institutions invest heavily enough in repair—for example, intensive and frequent simulation training, structured post-mortems, and recertification in ICU- or nuclear-safety settings—that average repair offsets or exceeds task-induced degradation. Our results then describe what happens if such investment is reduced, rather than claiming that drift wins regardless of effort. In *unbounded repair* regimes, one postulates repair moves with no global bound on their effect, such as costless, arbitrarily frequent “oracle refresh” operations or instant replacement by perfectly grounded experts; this violates the bounded-repair assumption and lies outside FRFP’s human-bounded institutional model. In *low-demand* regimes, such as rare emergency committees that face serious decisions only occasionally, nontrivial demands are so infrequent that modest repair may suffice to maintain grounding, and asymptotic Epistemic Dynamics arguments are not the dominant concern.

These counter-scenarios also clarify an economic tension. In high-load settings, pushing repair hard enough to stabilize tacit grounding typically implies a governance workload whose cost and latency scale with AI usage. Institutions can thus choose a point on a tradeoff surface: minimal repair, preserving large speed and cost advantages at the price of long-run inevitability; substantial repair, which buys longer safe horizons but erodes pure automation gains; or intermediate regimes that partially delay and reshape, but do not eliminate, long-run risk.

Scope of novelty. As noted in the Introduction, the categorical ingredients we use (free categories, groupoids, fibrations, Grothendieck constructions) are standard; the contribution is to assemble them into a single architecture that captures explicit/tacit separation and feedback, and to derive system-level constraints on auditability, correctness, and hallucination. FRFP is intended as a reusable structural framework rather than a specific algorithm or training procedure. Our strongest governance-level claims should thus be read as conditional: they are formal no-go theorems *given* the FRFP axioms, plus an empirical hypothesis that high-stakes scientific and regulatory practice falls within that class of architectures, from which we derive normative guidance for protocol and institutional design.

9.4 Future Directions

Several directions are natural next steps.

Empirical dynamics of human–AI protocols. The current framework treats tacit context, degradation, requirements, confidence, and feedback as abstract operators. A first direction is to make these empirically meaningful. This includes: (i) developing task- and domain-specific proxies for tacit context, degradation, requirement maps, and plausibility functionals; (ii) instrumenting real human–AI workflows so that trajectories in $N \times E$ can be logged, normalized, and analysed; and (iii) testing structural predictions (such as inevitability and acceleration of hallucination) against behavioural and longitudinal data.

Repair, re-grounding, and reset mechanisms. Our analysis intentionally focuses on settings where context degrades and is not fully restored. A complementary direction is to design explicit repair and re-grounding mechanisms—study breaks, external reference checks, slow deliberation modes—and represent them as operators within the same algebraic framework. At the protocol level, this suggests specifying and analysing reset policies, usage caps, and re-verification gates as first-class constructions in $N \times E$, and characterising when such mechanisms significantly delay (though do not eliminate) structural hallucination.

Collective architectures and institutions. The framework can be extended from single users to populations of agents and to institutions that aggregate many runs. This opens questions about how different institutional patterns (peer review, adversarial review, replication, regulatory stress-testing) can be represented, and which combinations provably buffer individual degradation while respecting a human-exclusive grounding constraint. Another direction is to relate formal notions such as population semantic profiles, institutional stability, and institutional non-automation to empirical studies of scientific and regulatory processes, and to compare predicted failure modes (fake consensus, lock-in, correlated error) with observed ones.

Governance, law, and meta-level design. The same categorical machinery can be used to model governance operators that act on institutions, together with constraints that restrict how institutions may change. This suggests: (i) designing concrete governance observables (drift, disagreement, confidence-inflation indicators) that can be computed from FRFP-compatible logs; (ii) exploring families of governance operators that incorporate human oversight at the institutional level without collapsing into fully automated correctness claims; and (iii) mapping these patterns onto emerging regulatory regimes for AI in science, medicine, and finance, to clarify which regulatory templates are structurally compatible with an explicit/tacit separation.

Richer explicit logics and tooling. On the explicit side, the framework is agnostic about the internal logic carried by E . A further direction is to instantiate E with richer structures: type-theoretic or temporal logics, proof objects from interactive theorem provers, or domain-specific calculi for simulation and data analysis. This would support more expressive explicit pipelines while preserving the RB – TE – HFD suffix and its correctness properties. In parallel, one can build FRFP-aware tooling: typed interface languages, normalization and auditing layers for $N \times E$, and assistants that expose their operations as TDG terms rather than as opaque conversations.

Benchmarking and comparative evaluation. Finally, FRFP suggests protocol-level evaluation criteria. Instead of scoring isolated model outputs, one can compare whole workflows—different ways of structuring the explicit/tacit boundary, navigation, repair, and institutional aggregation—using metrics derived from the dynamic theory, such as time-to-hallucination distributions, robustness under degradation, and sensitivity to feedback coupling. Developing such benchmarks would allow FRFP-compatible systems to be compared along principled axes, and would help connect the abstract theory to concrete engineering trade-offs.

FRFP is intended as a foundation, not a finished product. Its value lies in making constraints and invariants explicit so that future systems—both technical and institutional—can be designed, analysed, and compared within a common formal framework.

10 Alternative formalisms

FRFP is one point in a broader design space of human–AI collaboration architectures. This section positions FRFP against competing formalisms, clarifies which alternatives do *not* reduce to FRFP (because they step outside the FRFP axiom class), and explains why FRFP is the appropriate choice for the theorem-level aims of this volume.

10.1 The design space: what varies across architectures

Any protocol-level formalism must decide (at minimum):

1. **Where correctness lives.** Is correctness a property decidable in explicit space (a predicate or oracle on artefacts), or a property of tacit endorsement states?
2. **Whether tacit is writable.** Is there a channel $T \rightarrow E$ that exports acceptance states into machine-manipulable specifications, policies, or constraints?
3. **Who may access tacit.** May AI systems condition on tacit state (or modify it), or is tacit strictly human-exclusive?
4. **How acceptance is structured.** Is there a single acceptance boundary, or multiple gates (individual, committee, institution) with distinct authority and standards?
5. **How uncertainty is represented.** Is endorsement discrete (a finite judgment quotient), ordered (preorders), or probabilistic (graded belief and risk thresholds)?

FRFP fixes these choices in a specific way: correctness is localized in tacit space; AI is confined to explicit moves; the explicit–tacit boundary is one-way $E \rightarrow T$ with no return channel $T \rightarrow E$; and the tacit interface is intentionally minimal so that the explicit algebra carries the technical structure.

10.2 Architectures that do *not* reduce to FRFP

Because FRFP is proved (in Appendix A) to be *initial* only within a category of frameworks that share the FRFP epistemic primitives and axioms, an alternative architecture avoids “reducing to FRFP” precisely by violating at least one of those axioms. The following are genuine competitors in that sense.

(A) Bidirectional boundary architectures ($T \leftrightarrow E$). These formalisms introduce an essential write-back channel $F : T \rightarrow E$, by which tacit endorsement states externalize into explicit policies, specifications, rubrics, or machine-checkable constraints. This matches institutional learning patterns (postmortems $\rightarrow$ checklists $\rightarrow$ standards), but it is outside FRFP by construction: FRFP forbids $T \rightarrow E$. In such systems, the acceptance criteria are not merely applied at review points; they are continuously rewritten into the explicit layer.

(B) Tacit-in-the-loop AI. Here AI systems may condition on tacit state or participate in tacit updates (e.g., morphisms of the form $T \times E \rightarrow E$ or AI-mediated operators $T \rightarrow T$). This architecture is attractive for personalization and closed-loop assistance, but it breaks human-exclusive correctness and blurs responsibility: the system can implicitly steer endorsement by reading or shaping tacit states. This is therefore not an instance of FRFP.

(C) Explicit-only correctness architectures. In some regimes, correctness can be reduced to a complete explicit oracle: proof checking, model checking, exhaustive tests under a closed-world specification. In such settings, “acceptance” is an explicit computation on artefacts, not a tacit judgment. This is again outside FRFP’s target class: FRFP assumes a genuine tacit layer that is not exhausted by any fixed explicit oracle.

(D) Multi-boundary acceptance architectures. Real institutions often implement layered gates of endorsement (individual sign-off, peer review, risk committee approval, regulatory clearance). A formalism may represent these as multiple tacit layers and multiple boundaries, with cross-talk between gates. If the ontology class fixes a single boundary primitive, then such multi-gate architectures do not reduce to FRFP without extending the primitive structure.

(E) Probabilistic tacit semantics. A competing formalism may model tacit state as a richer belief object (e.g., probability measures, utility-weighted acceptance thresholds, risk budgets) rather than a preorder with a finite observable judgment quotient. When the tacit algebra differs materially (graded endorsement rather than discrete quotients and idempotent operators), the architecture falls outside the specific tacit-interface axioms used in FRFP’s initiality claims.

10.3 Which alternatives can be viewed as generalizations of the scientific method

Several competing architectures can still be regarded as “generalizations of the scientific method,” but they generalize different facets of it.

- **Explicit-only correctness** generalizes the formal-verification slice of science (mathematics, parts of theoretical computer science), where correctness is internal to a fixed calculus.
- **Bidirectional $T \leftrightarrow E$** generalizes the meta-scientific process by which tacit know-how becomes codified into explicit methods, standards, and tooling.
- **Multi-boundary acceptance** generalizes the institutional reality of science as layered endorsement (lab judgment, journal review, replication, community uptake).
- **Probabilistic tacit semantics** generalizes science as graded belief and risk management under uncertainty.

FRFP generalizes a particular core: *stabilization by accountable review*, where public explicit artefacts are strengthened by explicit checks and then cross a responsibility-bearing acceptance boundary.

10.4 Why FRFP among the alternatives

FRFP is chosen to match a specific problem class and to enable theorem-level analysis.

Target regime. We aim at long-horizon workflows in which (i) requirements evolve, (ii) explicit artefacts and evaluation criteria drift across extended traces, and (iii) no fixed explicit oracle exhausts what it means for an artefact to be adequate. In this regime, architectures that place correctness entirely in E are aiming at a different class of problems.

Accountability and non-circular semantics. FRFP enforces that endorsement is attributable: AI systems can produce and transform explicit artefacts, but only humans occupy correctness-bearing tacit states. The one-way boundary $E \rightarrow T$ prevents implicit rewriting of acceptance criteria by the system itself. Architectures with essential $T \rightarrow E$ write-back or tacit-in-the-loop AI can be valuable, but they entangle acceptance and generation in ways that complicate responsibility and undermine the invariance and non-automation results that this volume targets.

Minimality and canonicity within the axiom class. Within the class of formalisms that share FRFP’s epistemic primitives and constraints (explicit/tacit separation, one-way boundary, human-exclusive correctness, AI confined to explicit moves), FRFP is initial: any competing framework satisfying the same assumptions admits a unique structure-preserving interpretation from FRFP. In this precise sense, FRFP is not an arbitrary design choice inside its assumption class; alternatives that keep the same primitives are instances/refinements rather than routes around the theorem-level consequences.

10.5 Practical reading of this comparison

The existence of non-FRFP architectures is not a defect: it clarifies scope. FRFP is a protocol-level formalism for domains where correctness-bearing endorsement cannot be reduced to explicit computation, and where the central risks are long-horizon drift, accountability loss, and explicit-only governance limits. Competing architectures are appropriate when one wishes to (i) treat correctness as fully explicit, (ii) explicitly codify tacit judgments into machine-checkable policy, (iii) close the loop between system behavior and human internal state, or (iv) model layered institutional gates or graded belief as first-class. Subsequent volumes may study carefully delimited departures from FRFP’s axioms; the present volume fixes them to make the foundational theorems sharp.

Appendices

A A

This appendix develops the categorical and algebraic core of the FRFP (Foundational Reasoning and Feedback Protocol) architecture. It provides a self-contained account of:

- an ambient category with explicit and tacit modes;
- a free explicit algebra of task decompositions (the Task Decomposition Grammar, TDG);
- a navigation groupoid acting on explicit shapes;
- a fibration of explicit pipelines over shapes;
- a Grothendieck-style semidirect product $N \ltimes E$;
- a semantic functor into a preorder-enriched tacit category; and
- correctness and canonicity theorems for Epistemic Structure.

We assume familiarity with basic category theory, rewriting systems, and context-free grammars. Throughout, we write C_{full} for the ambient category, $\mathcal{E}$ for explicit space, and $\mathcal{T}$ for tacit space.

A.1 Foundations

A.1.1 Ambient Category C_{full}

Definition A.1 (Ambient category). The ambient category C_{full} is an arbitrary category that may contain objects beyond those used by the FRFP protocol. We designate three *distinguished* objects $\emptyset, E, T \in \text{Ob}(C_{\text{full}})$:

- $\emptyset$: the empty explicit state;
- E : the explicit state object (machine-manipulable representations);
- T : the tacit state object (human correctness-bearing states).

Morphisms in C_{full} are partitioned into three epistemic classes:

- explicit morphisms with codomain E , forming the explicit subcategory $\mathcal{E} \subseteq C_{\text{full}}$;
- tacit morphisms with codomain T , forming the tacit subcategory $\mathcal{T} \subseteq C_{\text{full}}$;
- boundary morphisms $E \rightarrow T$, of which a distinguished one is the *Representational Backflow* operator $\text{RB} : E \rightarrow T$.

Composition in C_{full} satisfies the usual category axioms:

$$(h \circ g) \circ f = h \circ (g \circ f), \quad f \circ \text{id}_X = f = \text{id}_Y \circ f.$$

Definition A.2 (Explicit–Tacit Separation (ETS)). The *Explicit–Tacit Separation* axiom requires:

- there are no morphisms $T \rightarrow E$ in C_{full} ;
- tacit evaluation cannot be embedded inside explicit pipelines;
- the boundary morphism $\text{RB} : E \rightarrow T$ is not invertible.

A.1.2 Modal Interpretation of Explicit and Tacit Knowledge

Definition A.3 (Explicit modality). A cognitive item k is represented in *explicit* modality if there exist maps

$$\text{enc} : E \rightarrow \Sigma^*, \quad \text{dec} : \Sigma^* \rightarrow E$$

such that $\text{dec}(\text{enc}(k))$ is equivalent to k in its epistemic role (e.g., as a proof, equation, or program). In particular, k can be faithfully encoded as a finite symbolic artifact and recovered without loss of its correctness-relevant structure.

Definition A.4 (Tacit modality). A cognitive item is represented in *tacit* modality when no such encoding preserves its correctness-bearing content. Judgments, intuitions, acceptance decisions, and other human correctness-bearing acts live in tacit space.

Human expression from tacit cognition to explicit content (e.g., writing a new question, revising a claim) is modeled externally as the appearance of new explicit morphisms in $\mathcal{E}$; it is not modeled as a morphism $T \rightarrow E$.

A.1.3 Explicit Category $\mathcal{E}$

Definition A.5 (Explicit category). The explicit category $\mathcal{E}$ is the full subcategory of C_{full} on objects with codomain E . Its core generating morphisms are:

$$\text{RI} : \emptyset \rightarrow E \quad (\text{Representation Initiation}), \quad \text{EC} : E \rightarrow E \quad (\text{Explicit Computation}), \quad \text{ED} : E \rightarrow E \quad (\text{Explicit$$

Remark. EC represents algorithmic transformations of explicit content (e.g., inference, rewriting, search). ED represents explicit checks and diagnostics (e.g., consistency checks, sanity tests), preparing but not performing tacit evaluation.

A.1.4 Tacit Category $\mathcal{T}$

We now formalize tacit space as a preorder-enriched one-object category.

Definition A.6 (Tacit category). The tacit category $\mathcal{T}$ has:

- a single object, denoted again by T ;
- hom-set $\text{Hom}_{\mathcal{T}}(T, T)$ equipped with a composition operation $\circ$, making it a monoid;
- a preorder $\sqsubseteq$ on $\text{Hom}_{\mathcal{T}}(T, T)$ such that composition is monotone:

$$f_1 \sqsubseteq f_2, \quad g_1 \sqsubseteq g_2 \quad \Rightarrow \quad g_1 \circ f_1 \sqsubseteq g_2 \circ f_2.$$

Definition A.7 (Tacit primitives). The tacit primitives are the morphisms

$$\text{TE} : T \rightarrow T \quad (\text{Tacit Evaluation}), \quad \text{HFD} : T \rightarrow T \quad (\text{Human Final Decision}),$$

subject to:

- *idempotence*: $\text{TE} \circ \text{TE} = \text{TE}$ and $\text{HFD} \circ \text{HFD} = \text{HFD}$;
- *commutation*: $\text{TE} \circ \text{HFD} = \text{HFD} \circ \text{TE}$;

- *monotonicity*: for every $f : T \rightarrow T$,

$$f \sqsubseteq \text{TE} \circ f, \quad f \sqsubseteq \text{HFD} \circ f.$$

Definition A.8 (Correctness quotient). Let J be a finite set of correctness judgments (e.g., $\{\text{accept}, \text{reject}, \text{revise}\}$). A *correctness quotient* is a surjective map

$$\gamma : \text{Ob}(\mathcal{T}) \rightarrow J$$

assigning to each tacit state a correctness class, subject to

$$\gamma \circ \text{TE} = \gamma = \gamma \circ \text{HFD}.$$

Thus tacit evaluation and final decision may refine or stabilize a tacit state but do not change its correctness class.

Interpretation and non-triviality of T . The tacit category T is not introduced merely as a formal firewall that “reserves correctness to tacit space”. Rather, it is intended to capture a substantive empirical claim about long-horizon scientific and regulatory work: in these regimes, there are correctness-bearing aspects of judgment that cannot be exhaustively encoded in a fixed, finite explicit representation without loss.

Definitions A.3–A.4 distinguish explicit and tacit modalities by the existence of lossless encodings into symbolic artefacts. In many familiar settings—for example, textbook exercises, small programs, or contest-style benchmarks—it is reasonable to treat *inner* task correctness as effectively explicit: the task is specified by a finite prompt, there is a fixed input–output relation or reference implementation, and success can be checked by a test harness or proof checker. FRFP does not deny the existence or value of such regimes. Instead, it treats them as *tacit-insensitive inner layers*: conditional on a surrounding tacit backdrop (choice of formalism, modelling assumptions, risk tolerances), correctness for that inner task behaves as if it were a pure explicit oracle on E .

The global explicit/tacit split remains. Even in proof-assistant or test-oracle settings, tacit judgment is involved in selecting the calculus, specifying what the theorem or contract should state, and deciding whether conformance to that specification is adequate for the real-world purpose. What FRFP targets are workflows where this outer tacit layer is non-trivial and dynamically involved in long-horizon use: tasks evolve, standards and constraints shift, and background scientific understanding and norms play an essential role in deciding whether an explicit artefact is “good enough”. In those settings it is non-trivial, and in our view appropriate, to treat the final correctness status of an explicit pipeline as living in T rather than in E . (See Appendix D.4 for further discussion.)

A reasonable skeptic might ask: under what conditions could one reject the Epistemic Spaces axioms and the later Epistemic Algebra structure on T ? There are at least three salient counter-scenarios:

1. **Tacit-insensitive inner correctness.** For some domains and layers, once a formalism, specification, and world-modelling stance have been tacitly fixed, every correctness-bearing judgment at the inner level can be captured by a finite explicit specification and a sound and complete proof- or test-based oracle. At that inner layer one can model correctness directly in E and treat T as abstracted away; FRFP’s Phase-3 inevitability and non-automation results then become vacuous or uninteresting for that particular subproblem. Our focus is on regimes where the outer tacit layer—where specifications, standards, and modelling choices are themselves evolving—remains dynamically involved in correctness.

2. **No degradation or escalation.** Epistemic Algebra impossibility results rely on a degradation operator δ and recurring nontrivial requirements relative to a floor r . If one believes that, in their target setting, tacit state does not degrade in any relevant sense, or that requirements do not escalate or recur in the way we assume (for example, because tasks are short-lived and fully specified at the outset), then the inevitability theorems may not apply. Such systems fall outside FRFP’s intended long-horizon regime.
3. **Perfect tacit emulation.** If one posits an AI system that can internally reconstruct or simulate human tacit evaluation from explicit traces without loss—violating the NTER and CSC constraints—then our explicit–tacit separation (ETS, HEG, HEC) is simply rejected. FRFP explicitly rules out this possibility by axiom rather than attempting to model it.

In summary, treating correctness as localized in T is a substantive modelling choice: it reflects the view that, for many real scientific and governance workflows, there is residual human judgment that cannot be fully captured by static benchmarks or formal specifications, and that this residual plays an essential role in long-horizon correctness. The impossibility and inevitability results of Epistemic Structure and Dynamics should therefore be read as conditional on this non-trivial tacit structure: they apply precisely in domains where correctness genuinely outstrips what can be made explicit and mechanically checked.

A.1.5 Boundary Morphism and Semantic RB Functor

Definition A.9 (Boundary morphism RB). The *Representational Backflow* operator is a morphism

$$\text{RB} : E \rightarrow T$$

in C_{full} . It is the unique explicit-to-tacit interface specified by the Representational Backflow axiom.

Let E_{sem} denote the free explicit category generated by the Task Decomposition Grammar (defined in Section ??).

Definition A.10 (Semantic RB functor). The *semantic RB* is a functor

$$\text{RB}^\sharp : E_{\text{sem}} \rightarrow \mathcal{T}$$

assigning to each explicit pipeline a tacit state representing the content received by the human for evaluation.

A.1.6 Interaction Protocols and Epistemic Spaces Axioms

Definition A.11 (Interaction Protocols constraints). The operational environment satisfies:

- **IL (Interaction Limitation):** interaction is bounded and turn-based;
- **AR (Alignment Restriction):** the system must not autonomously act outside explicit instructions;
- **MD (Mode Discipline):** separation between explicit and tacit modes is enforced at the interface level;
- **NTER (No Tacit Evaluation Reconstruction):** the system must not reconstruct tacit evaluation internally from explicit traces;

- **CSC (Correctness Separation Constraint):** correctness decisions must not be delegated to explicit computation.

Definition A.12 (Epistemic Spaces axioms). An admissible epistemic framework satisfies:

- **ETS (Explicit–Tacit Separation):** no morphisms $T \rightarrow E$;
- **HEG (Human-Exclusive Grounding):** correctness grounding occurs only in tacit space via TE, executed by humans;
- **HEC (Human-Exclusive Closure):** final acceptance/rejection occurs only via HFD, executed by humans;
- **RB (Representational Backflow):** there is a unique boundary morphism $RB : E \rightarrow T$ used by all pipelines;
- **CP (Compositional Pipelines):** epistemic processes are built as compositions of primitives into pipelines;
- **AE (AI-Explicit Restriction):** AI systems act only via explicit-space morphisms in $\mathcal{E}$.

A.2 Explicit Algebra and the Task Decomposition Grammar

A.2.1 TDG Signature and Free Category

Definition A.13 (TDG signature). The Task Decomposition Grammar (TDG) signature Σ_{TDG} consists of:

- objects: $\emptyset$ and E ;
- generating morphisms:

$$RI : \emptyset \rightarrow E, \quad EC : E \rightarrow E, \quad ED : E \rightarrow E;$$

- structural operators BRANCH, PARALLEL, LOOP acting as higher-arity constructors on morphisms.

Definition A.14 (Free explicit category). The explicit semantic category is the free category on the TDG signature:

$$E_{\text{sem}} := \text{FreeCat}(\Sigma_{\text{TDG}}).$$

There is a canonical interpretation functor $U_E : E_{\text{sem}} \rightarrow \mathcal{E}$ sending each TDG term to its denotation as an explicit morphism.

Theorem A.15 (Free explicit algebra). *For any category $\mathcal{D}$ and any interpretation of the TDG signature Σ_{TDG} in $\mathcal{D}$, there exists a unique functor $\hat{F} : E_{\text{sem}} \rightarrow \mathcal{D}$ extending that interpretation.*

Proof. This is the standard universal property of the free category on a graph: objects of E_{sem} are objects of Σ_{TDG} , and morphisms are freely generated paths in the underlying graph. Any graph morphism from Σ_{TDG} into $\mathcal{D}$ extends uniquely to a functor. $\square$

A.2.2 TDG Grammar

We present TDG both as a signature and as a context-free grammar over terms.

Nonterminals. We use:

Q for explicit pipelines, P for full pipelines (explicit + tacit suffix).

Terminals. The terminal symbols are:

RI, EC, ED, RB, TE, HFD, BRANCH, PARALLEL, LOOP.

Productions. The grammar productions are:

$$\begin{aligned} Q &::= \text{RI}, \\ Q &::= Q; \text{EC}, \\ Q &::= Q; \text{ED}, \\ Q &::= \text{BRANCH}(Q_1, \dots, Q_k), \\ Q &::= \text{PARALLEL}(Q_1, \dots, Q_k), \\ Q &::= \text{LOOP}(Q_{\text{cond}}, Q_{\text{body}}), \\ P &::= Q; \text{RB}; \text{TE}; \text{HFD}. \end{aligned}$$

Thus TDG presents E_{sem} as an algebra of explicit pipelines with a fixed tacit suffix.

A.2.3 Shapes and Shape Category

Definition A.16 (TDG shape). TDG shapes are generated by the grammar

$$S ::= \text{chain} \mid \text{branch}(S_1, \dots, S_k) \mid (S_1, \dots, S_k) \mid \text{loop}(S_{\text{cond}}, S_{\text{body}}).$$

Definition A.17 (Shape of a TDG term). The mapping $\text{shape} : \text{TDG-terms} \rightarrow \text{Shapes}$ is defined inductively by:

- $\text{shape}(\text{RI}) = \text{chain}$;
- $\text{shape}(Q; \text{EC}) = \text{shape}(Q)$ and similarly $\text{shape}(Q; \text{ED}) = \text{shape}(Q)$;
- $\text{shape}(\text{BRANCH}(Q_1, \dots, Q_k)) = \text{branch}(\text{shape}(Q_1), \dots, \text{shape}(Q_k))$;
- analogously for PARALLEL and LOOP.

Definition A.18 (Shape category $\mathcal{S}$). The shape category $\mathcal{S}$ has:

- objects: shapes S as above;
- morphisms: structural rewiring maps between shapes (e.g., regrouping, insertion/removal of trivial chains), closed under composition and identities.

A.2.4 Explicit Reduction System

Definition A.19 (Explicit reduction). A binary relation $\rightarrow_E$ on $\mathcal{E}$ is an *explicit reduction system* if it is a typed rewrite system satisfying:

- **typing:** if $e : X \rightarrow E$ and $e \rightarrow_E e'$, then $e' : X \rightarrow E$;

- **TDG compatibility:** reductions act componentwise on TDG combinators (BRANCH, PARALLEL, LOOP);
- **termination:** there is no infinite chain $e_0 \rightarrow_E e_1 \rightarrow_E e_2 \rightarrow_E \dots$;
- **local confluence:** if $e \rightarrow_E e_1$ and $e \rightarrow_E e_2$, then e_1 and e_2 have a common descendant.

We treat these properties as axioms of the explicit reduction system.

Under these assumptions, every explicit artifact admits a unique normal form modulo $\rightarrow_E$.

A.3 Navigation and Shapes

A.3.1 Navigation Category and Groupoid

Definition A.20 (Navigation category N_0). The navigation category N_0 has:

- objects: shapes S ;
- generating morphisms: navigation steps $g : S \rightarrow S'$, such as TRYMODEL, ADDASSUMPTION, DROPASSUMPTION, MERGE, RESTART, each with a well-defined source and target shape;
- morphisms: finite composites of such generators.

Definition A.21 (Navigation groupoid N). The navigation groupoid N is obtained by freely adjoining inverses to morphisms of N_0 and quotienting by the relations

$$g^{-1} \circ g = \text{id}_S, \quad g \circ g^{-1} = \text{id}_{S'},$$

for each generator $g : S \rightarrow S'$. Thus every navigation step is reversible up to equality in N .

A.3.2 Fibration $p : E \rightarrow S$

Definition A.22 (Projection to shapes). The functor $p : \mathcal{E} \rightarrow \mathcal{S}$ sends each explicit pipeline $e \in \mathcal{E}$ to its shape $p(e) := \text{shape}(e)$, and each explicit morphism to the corresponding structural morphism between shapes.

Definition A.23 (Fibers). For each shape S , the fiber

$$E_S := p^{-1}(S)$$

is the category of explicit morphisms with shape S .

Definition A.24 (Cartesian liftings and reindexing). For each navigation morphism $n : S \rightarrow S'$ in N , we assume a cartesian lifting

$$n^* : E_S \rightarrow E_{S'}$$

which reindexes explicit pipelines along n . This yields an indexed category

$$E : N^{\text{op}} \rightarrow \mathbf{Cat}, \quad S \mapsto E_S, \quad n \mapsto n^*.$$

A.4 Combined System: Grothendieck Construction

A.4.1 Semidirect Product $N \ltimes E$

Definition A.25 (Grothendieck construction $N \ltimes E$). Given the indexed category $E : N^{\text{op}} \rightarrow \mathbf{Cat}$, the Grothendieck construction $\int_N E$ (also written $N \ltimes E$) has:

- objects: pairs (S, e) with $S \in \text{Ob}(N)$ and $e \in E_S$;
- morphisms $(S, e) \rightarrow (S', e')$: pairs (n, f) where $n : S \rightarrow S'$ in N and $f : n^*(e) \rightarrow e'$ in $E_{S'}$.

Composition is given by

$$(n_1, f_1) \circ (n_2, f_2) = (n_1 \circ n_2, f_1 \circ n_1^*(f_2)).$$

Informally, we may write elements as (n, e) and composition as $(n_1, e_1) \circ (n_2, e_2) = (n_1 \circ n_2, e_1 \circ n_1(e_2))$ whenever the lifted action is defined.

A.4.2 Rewrite System on $N \ltimes E$

Definition A.26 (Combined reduction). The one-step reduction relation $\rightarrow$ on $N \ltimes E$ is generated by:

- **explicit step**: if $e \rightarrow_E e'$ then $(n, e) \rightarrow (n, e')$;
- **navigation step**: if $n \rightarrow_N n'$ and n' is defined on e , then $(n, e) \rightarrow (n', n'(e))$.

We assume:

- (A1) $\rightarrow_E$ is terminating and locally confluent;
- (A2) navigation reductions $\rightarrow_N$ have normal forms in each component of N ;
- (A3) the lifted action is functorial and preserves RB, TE, and HFD.

These are axioms of the FRFP reduction environment.

A.5 Semantics and Tacit Correctness

A.5.1 Semantic Functor

Definition A.27 (Semantic bracket). A *semantic functor* is a functor

$$\llbracket - \rrbracket : N \ltimes E_{\text{sem}} \rightarrow \mathcal{T}$$

assigning to each combined configuration a tacit state. On explicit pipelines $e \in E_{\text{sem}}$, we set

$$\llbracket e \rrbracket := \text{HFD} \circ \text{TE} \circ \text{RB}^\sharp(e).$$

Definition A.28 (Navigation invariance). The functor $\llbracket - \rrbracket$ is *navigation-invariant up to preorder* if

$$\llbracket (n, e) \rrbracket \sqsubseteq \llbracket e \rrbracket$$

for all n, e where the action is defined, and equality holds in the quotient by the induced equivalence on tacit states in confluent cases.

Definition A.29 (Observable correctness). Given a pipeline P , its observable correctness outcome is

$$\gamma(\llbracket P \rrbracket) \in J.$$

A.6 Main Theorems

A.6.1 Well-Formedness of TDG Pipelines

Theorem A.30 (Well-formedness). *Every TDG-generated pipeline P is a well-typed morphism in C_{full} of the form*

$$\text{RI} \rightarrow (\text{EC}/\text{ED})^* \rightarrow \text{RB} \rightarrow \text{TE} \rightarrow \text{HFD}.$$

Proof. By induction on TDG derivations. The base case RI initializes an object in E . The constructors EC and ED have codomain E and thus preserve explicit type. Structural combinators (BRANCH, PARALLEL, LOOP) act within the explicit category and preserve codomain E componentwise. The production for P enforces the suffix RB; TE; HFD, whose types are $E \rightarrow T \rightarrow T \rightarrow T$. By ETS there are no morphisms $T \rightarrow E$, so:

- no tacit operator can appear before RB;
- no explicit operator can appear after RB.

Thus every derivation yields a well-typed morphism of the stated form. $\square$

A.6.2 Explicit Completeness of TDG

Theorem A.31 (Explicit completeness). *TDG generates exactly the admissible explicit morphisms in $\mathcal{E}$.*

Proof. By construction, E_{sem} is the free category on the TDG signature, and U_E interprets TDG terms as explicit morphisms in $\mathcal{E}$. Every explicit morphism in the FRFP setting is built from RI, EC, ED, and the structural operators, hence arises from a TDG term; conversely, each TDG term denotes a well-typed explicit morphism. The image of TDG under U_E is thus exactly the admissible explicit morphisms. $\square$

A.6.3 Minimality of Primitive Set

Theorem A.32 (Minimality). *The primitive set $\{\text{RI}, \text{EC}, \text{ED}, \text{RB}, \text{TE}, \text{HFD}\}$ is minimal under the Epistemic Spaces axioms: removing any primitive violates at least one of ETS, HEG, HEC, RB, CP, or AE.*

Proof. Briefly:

- RI is required to start from $\emptyset$ and generate explicit content; without it there is no canonical initiation of pipelines.
- EC is required for generative explicit reasoning; without it, explicit computation collapses.
- ED is required for explicit diagnostics without tacit evaluation; removing it forces diagnostics either into tacit space (violating AE) or to be indistinguishable from EC (violating CP and the intended modal roles).
- RB is required for explicit-to-tacit handoff; without it there is no admissible morphism $E \rightarrow T$, contradicting the requirement that pipelines reach tacit evaluation.
- TE is required for tacit grounding (HEG); without it correctness cannot be evaluated in tacit space.

- HFD is required for final correctness closure (HEC); without it there is no morphism corresponding to final acceptance/rejection.

Removing any primitive causes at least one Epistemic Spaces axiom to fail, so the set is minimal. $\square$

A.6.4 Epistemic Spaces Initiality

Definition A.33 (Epistemic Spaces framework). A *Epistemic Spaces framework* is a tuple

$$\mathcal{F}_1 = (\mathcal{C}, E_{\mathcal{C}}, T_{\mathcal{C}}, \text{RI}_{\mathcal{C}}, \text{EC}_{\mathcal{C}}, \text{ED}_{\mathcal{C}}, \text{RB}_{\mathcal{C}}, \text{TE}_{\mathcal{C}}, \text{HFD}_{\mathcal{C}})$$

satisfying the Epistemic Spaces axioms. Morphisms in the category $\mathbf{Frm}_1$ of Epistemic Spaces frameworks are functors preserving E , T , and all six primitives.

Theorem A.34 (Epistemic Spaces initiality). *There exists an initial object $F_{\mathbf{FRFP}}^{(1)} \in \mathbf{Frm}_1$.*

Proof. Construct $F_{\mathbf{FRFP}}^{(1)}$ as the free category on objects E, T and morphisms $\text{RI}, \text{EC}, \text{ED}, \text{RB}, \text{TE}, \text{HFD}$, modulo the equations encoding the Epistemic Spaces axioms (ETS, HEG, HEC, RB, CP, AE). Any other Epistemic Spaces framework interprets these generators in a way that satisfies the same equations, yielding a unique induced functor $F_{\mathbf{FRFP}}^{(1)} \rightarrow \mathcal{F}_1$. Thus $F_{\mathbf{FRFP}}^{(1)}$ is initial in $\mathbf{Frm}_1$. $\square$

Remark A.34a (Non-trivial content of Epistemic Spaces initiality). The statement that $F_{\mathbf{FRFP}}^{(1)}$ is initial in $\mathbf{Frm}_1$ uses the free-category construction, but its content is more than formal packaging. The Epistemic Spaces axioms (ETS, HEG, HEC, RB, CP, AE), together with the minimality result (Theorem A.32), severely constrain which primitive roles may appear and how they can be composed. Initiality then says that any epistemic framework respecting exactly these roles and axioms must interpret $\text{RI}, \text{EC}, \text{ED}, \text{RB}, \text{TE}, \text{HFD}$ in a way that factors uniquely through the $\mathbf{FRFP}$ ontology. In particular, attempts to “hide” additional power in explicit space (for example, by letting EC or ED implicitly perform tacit evaluation) break the axioms rather than yielding a genuinely different Epistemic Spaces framework.

A.6.5 Epistemic Algebra Canonicity

Definition A.35 (Epistemic Algebra architecture). A *Epistemic Algebra architecture* is a tuple

$$\mathcal{F}_2 = (E, S, p, N, (\cdot)^*, N \ltimes E)$$

satisfying:

- TDG presents E as a free explicit category;
- N is a navigation groupoid on shapes S ;
- $p : E \rightarrow S$ is a fibration with cartesian liftings n^* ;
- $N \ltimes E$ is the Grothendieck construction of the indexed category $E : N^{\text{op}} \rightarrow \mathbf{Cat}$;
- the boundary structure RB-TE-HFD is preserved under navigation.

Morphisms in the category of Epistemic Algebra architectures preserve all of this structure.

Theorem A.36 (Epistemic Algebra canonicity). *The FRFP Epistemic Algebra architecture $F_{\text{FRFP}}^{(2)}$ is a canonical object in the category of Epistemic Algebra architectures: for every Epistemic Algebra architecture $\mathcal{A}$, there exists a unique structure-preserving morphism*

$$F_{\text{FRFP}}^{(2)} \rightarrow \mathcal{A}.$$

Proof. By the universal properties of:

- the free explicit category E_{sem} ;
- the groupoid completion constructing N ;
- the Grothendieck construction forming $N \ltimes E$.

Any Epistemic Algebra architecture satisfying the same axioms must interpret the FRFP construction in a role-preserving way, inducing a unique morphism from $F_{\text{FRFP}}^{(2)}$. $\square$

A.6.6 Confluence of $N \ltimes E$

Theorem A.37 (Confluence of $N \ltimes E$). *Under assumptions (A1)–(A3), the reduction relation $\rightarrow$ on $N \ltimes E$ is confluent.*

Proof sketch. Local peaks $(n, e) \leftarrow (n_0, e_0) \rightarrow (n', e')$ decompose into:

- explicit–explicit peaks, which join by local confluence of $\rightarrow_E$;
- navigation–navigation peaks, which join by normalization in each connected component of N ;
- mixed peaks, which join because the lifted action commutes with explicit reduction and preserves RB, TE, HFD.

Together with termination up to navigation-normalization, this yields global confluence. $\square$

Remark A.37a (Non-trivial aspects of Epistemic Algebra structure). While Theorems A.36 and A.37 are phrased in terms of standard categorical constructions (free categories, groupoid completions, Grothendieck constructions), several aspects of the Epistemic Algebra architecture are mathematically substantive:

- **Compatibility with RB–TE–HFD.** The lifted navigation action and the Grothendieck construction are required to preserve the boundary structure. This rules out navigation steps that implicitly smuggle tacit evaluation back into explicit space or that duplicate/delete the RB–TE–HFD suffix. Showing that the action can be defined so as to commute with explicit reduction while preserving RB, TE, HFD is a genuine structural constraint on admissible navigation schemes.
- **Confluence of the combined reduction.** Confluence of $\rightarrow$ on $N \ltimes E$ is not automatic from confluence of $\rightarrow_E$ and normalization in N . The proof must analyze mixed peaks and use the functoriality of the lifted action, together with preservation of RB, TE, and HFD, to ensure that explicit and navigation steps can be permuted without changing the normal form.

- **Interaction with semantics.** The semantic correctness theorem (Theorem A.38) depends crucially on the fact that navigation-invariant semantics on $N \ltimes E$ respects the Interaction Protocols constraints (no illicit $T \rightarrow E$ morphisms) and the boundary-preservation properties above. This is what forces correctness judgments to depend only on the tacit evaluation of the explicit pipeline, not on the particular exploration trajectory.

These non-trivial interactions justify treating Epistemic Algebra canonicity as more than a choice of notation: any alternative architecture satisfying the same constraints must reproduce this structure and, in particular, cannot avoid the resulting preservation and non-automation results without breaking one of the axioms.

A.6.7 Semantic Correctness for Epistemic Structure

Collecting the constructions above, we obtain a semantic correctness theorem for Epistemic Structure FRFP pipelines.

Theorem A.38 (Semantic correctness). *Let P be any TDG-generated pipeline that is Interaction Protocols admissible and Epistemic Spaces/Algebra well-typed. Then:*

1. P has a unique normal form in $N \ltimes E$ modulo navigation;
2. the observable correctness outcome $\gamma(\llbracket P \rrbracket)$ is invariant under:
 - explicit reduction $\rightarrow_E$,
 - navigation in N ,
 - composition in the semidirect product $N \ltimes E$.

In particular, correctness judgments depend only on the (tacitly evaluated) semantic content of the explicit pipeline, not on the particular exploration trajectory used to construct it.

Proof sketch. Item (1) follows from Theorem ?? and the termination assumptions (A1)–(A3). Item (2) follows from navigation invariance of the semantic functor, the Interaction Protocols constraints (which rule out illicit T-to-E feedback or hidden tacit reconstruction), and the fact that RB, TE, HFD are preserved under navigation and do not appear in the explicit phase. $\square$

A.7 Worked example: sepsis consult as an FRFP pipeline

We sketch a concrete instance of the FRFP calculus corresponding to the sepsis consult scenario of §???. The goal is to show how a real conversational workflow is represented as a TDG term, reduced to normal form in $N \ltimes E$, and checked for Interaction Protocols admissibility and Epistemic Spaces/Algebra well-typedness.

TDG and free explicit category. Let the TDG have objects

$$(E_0) = \{\text{Ctx}, \text{Input}, \text{CaseSummary}, \text{DraftPlan}, \text{CandidatePlan}, \text{Report}, \text{Decision}\}$$

and generators

$$\begin{aligned}
& \text{RB} : \text{Ctx} \rightarrow \text{Input}, \\
& \text{prep} : \text{Input} \rightarrow \text{CaseSummary}, \\
& \text{ehr} : \text{CaseSummary} \rightarrow \text{CaseSummary}, \\
& \text{calc} : \text{CaseSummary} \rightarrow \text{CaseSummary}, \\
& \text{assist}_0 : \text{CaseSummary} \rightarrow \text{DraftPlan}, \\
& \text{review}_{\text{res}} : \text{DraftPlan} \rightarrow \text{CandidatePlan}, \\
& \text{review}_{\text{att}} : \text{CandidatePlan} \rightarrow \text{Report}, \\
& \text{HFD} : \text{Report} \rightarrow \text{Decision}.
\end{aligned}$$

Let E be the free category on this graph, modulo the Epistemic Algebra equations for TE and the RB–TE–HFD boundary. We define

$$\text{assist} := \text{assist}_0 \circ \text{calc} \circ \text{ehr} : \text{CaseSummary} \rightarrow \text{DraftPlan},$$

and the explicit sepsis pipeline

$$e_{\text{sepsis}} := \text{HFD} \circ \text{review}_{\text{att}} \circ \text{review}_{\text{res}} \circ \text{assist}_0 \circ \text{calc} \circ \text{ehr} \circ \text{prep} \circ \text{RB} : \text{Ctx} \rightarrow \text{Decision}.$$

Reduction and normal form. Let R_E be the explicit reduction system on E that contracts ehr and calc into assist inside the assist_0 wrapper:

$$\text{assist}_0 \circ \text{calc} \circ \text{ehr} \rightarrow \text{assist}.$$

By confluence of R_E and compatibility with the RB–TE–HFD boundary, the normal form of e_{sepsis} is

$$e_{\text{sepsis}}^* := \text{HFD} \circ \text{review}_{\text{att}} \circ \text{review}_{\text{res}} \circ \text{assist} \circ \text{prep} \circ \text{RB}.$$

Lifting to $N \ltimes E$ and applying the confluence theorem for the combined reduction (Epistemic Algebra) ensures that any admissible navigation of the conversational UI yields a unique normal form equivalent to e_{sepsis}^* up to the RB–TE–HFD suffix.

Interaction Protocols admissibility. We interpret the Interaction Protocols axioms IL, AR, MD, and NTER as constraints on realizations of e_{sepsis} in a concrete UI:

- IL: the only interaction channels between AI and human tacit state are via explicit objects in E (case summaries, drafts, candidate plans, reports). No morphisms $T \rightarrow E$ are introduced.
- AR: there is no morphism in the realization that effects clinical actions without passing through $\text{HFD} \circ \text{review}_{\text{att}}$; the AI cannot place orders or write notes.
- MD: all model-dependent computation is represented by assist and its internal decomposition; tacit evaluation (clinician judgment) is localized at $\text{review}_{\text{res}}$ and $\text{review}_{\text{att}}$.
- NTER: the implementation forbids any arrow that would encode a tacit correctness oracle in explicit space (no reconstruction of T from model outputs). In particular, the realization of assist is constrained to produce drafts and rationales, not authoritative decisions.

Under these conditions, any concrete realization of e_{sepsis} satisfies the Interaction Protocols interface axioms and is therefore admissible.

Epistemic Spaces/Algebra well-typedness. In Epistemic Spaces, each generator is assigned an epistemic role: `prep` and `assist` are explicit-computation arrows; `reviewres` and `reviewatt` are explicit presentation points at which tacit evaluation in T occurs; `RB` and `HFD` are boundary morphisms. The typing rules of the Epistemic Spaces ontology then derive a composite judgment showing that e_{sepsis}^* is a well-typed explicit pipeline with correctness localized in tacit post-composition at `reviewatt`. In Epistemic Algebra, the same term appears as a normal form in $N \ltimes E$; compatibility of reduction with the semantics ensures that the correctness predicate applied to e_{sepsis}^* coincides with the protocol-level notion of FRFP-hallucination (a violation of $t \succeq \text{Req}(e, c)$ at the moment of final acceptance). Thus the sepsis consult example realizes a concrete conversational UI that is Interaction Protocols admissible, Epistemic Spaces/Algebra well-typed, and corresponds to a specific normal form in the FRFP calculus.

A.7.1 Purpose and Scope

The discussion in the Appendix till now formalizes *static* correctness of individual FRFP pipelines, Epistemic Dynamics analyzes what happens when such pipelines are iterated by the same human–AI system over many turns. The focus is on:

1. persistence and degradation of tacit context;
2. explicit-only hallucinations and their containment;
3. confidence inflation and requirement escalation;
4. feedback-coupled degradation and accelerated hallucination;
5. stochastic and finite-horizon pipelines.

No collective or institutional mechanisms are assumed in this section; these appear in latter.

Throughout, we fix the Epistemic structure and work with the underlying tacit context space T and explicit artifact space E .

A.7.2 Tacit Context Space

Each human agent possesses a tacit context space T , equipped with a preorder $\preceq$.

Definition A.39 (Tacit context preorder). The binary relation $\preceq$ on T is read as:

$$t_1 \preceq t_2 \quad \text{“}t_1 \text{ is epistemically no stronger than } t_2\text{”}.$$

We write $t_1 \prec t_2$ if $t_1 \preceq t_2$ and $t_2 \not\preceq t_1$.

No algebraic structure on T is assumed beyond this preorder. The categorical tacit structure (one-object category, primitives TE, HFD, correctness quotient γ) is still in force, but for dynamic analysis we work primarily with the underlying preordered set $(T, \preceq)$.

A.7.3 Context Degradation

Tacit context degrades over time due to forgetting, misplaced trust, habituation, and other long-run effects.

Definition A.40 (Context degradation). A *context degradation* operator is a function

$$\delta : T \rightarrow T$$

modeling loss of grounding across turns. We assume:

1. **Monotonicity:** $\delta(t) \preceq t$ for all $t \in T$.
2. **Strict decay somewhere:** there exists $t \in T$ with $\delta(t) \prec t$.
3. **Non-invertibility:** δ has no right inverse; i.e., there is no function $f : T \rightarrow T$ such that $\delta \circ f = \text{id}_T$.

We write δ^n for the n -fold iterate of δ .

Intuitively, δ expresses the part of tacit dynamics that is *purely degrading*. In this appendix we deliberately ignore repair/refresh operations (e.g., reading textbooks, taking breaks) to obtain worst-case inevitability results. Repair can be modeled by separate operators $\rho : T \rightarrow T$, but these are left for future work.

A.7.4 Dynamic Interpretation of Pipelines

We now interpret complete FRFP pipelines as transformations of tacit context.

Let E_{sem} be the explicit semantic category and $\text{RB}^\sharp : E_{\text{sem}} \rightarrow T$ the semantic backflow (Section A.0.5). For each explicit pipeline $e \in E_{\text{sem}}$, the Epistemic Space suffix TE, HFD yields a tacit transformer

$$t \mapsto \text{HFD} \circ \text{TE} \circ \text{RB}^\sharp(e, t),$$

where we allow the tacit state t to modulate interpretation (e.g., via background knowledge).

Definition A.41 (Pipeline transformer). A *complete FRFP pipeline* P at a given turn consists of:

- an explicit artifact $e \in E_{\text{sem}}$ produced by the AI system via explicit computation and navigation;
- the canonical tacit suffix $\text{RB} \rightarrow \text{TE} \rightarrow \text{HFD}$;
- subsequent degradation δ .

It induces a tacit transformer

$$\llbracket P \rrbracket_\delta : T \rightarrow T, \quad \llbracket P \rrbracket_\delta(t) := \delta(\text{HFD} \circ \text{TE} \circ \text{RB}^\sharp(e, t)).$$

Definition A.42 (Run). A *run* (or *pipeline execution*) is a sequence

$$t_0, t_1, t_2, \dots$$

of tacit states such that there exist pipelines $P_0, P_1, P_2, \dots$ with

$$t_{n+1} = \llbracket P_n \rrbracket_\delta(t_n) \quad \text{for all } n \geq 0.$$

We will often also track the explicit artifacts e_n and credences c_n associated with each P_n ; this is made precise in Sections ?? and ??.

A.7.5 Requirements and Hallucination

We formalize hallucination in terms of a requirement map on explicit artifacts.

Definition A.43 (Requirement map). A *requirement map* is a function

$$\text{Req} : E \times \mathbb{R}_{\geq 0} \rightarrow T$$

assigning to each explicit artifact $e \in E$ and tacit credence $c \geq 0$ a required tacit grounding $\text{Req}(e, c)$. We require:

1. **Monotonicity in credence:** if $c_1 \leq c_2$ then $\text{Req}(e, c_1) \preceq \text{Req}(e, c_2)$;
2. **Nontriviality:** there exist artifacts whose requirements strictly exceed others in the preorder.

When credence is suppressed, we write $\text{Req}(e)$ for a baseline requirement (e.g., at default credence).

Definition A.44 (Hallucination). Let $t \in T$ be the tacit state at evaluation time, $e \in E$ an explicit artifact, and $c \geq 0$ the tacit credence assigned to e . A *hallucination* occurs at this step if

$$t \prec \text{Req}(e, c).$$

We say that the artifact e is *hallucinated* (relative to (t, c)) in this case.

Remark A.45 (Structural nature of hallucination). In this formalism, hallucination is an *implementation-agnostic* relation between tacit state t , requirement $\text{Req}(e, c)$, and explicit artifact e . It does not depend on the model's internal computation or intent. Concrete architectures (e.g., LLMs) shape *how often* trajectories enter states with $t \prec \text{Req}(e, c)$, but do not appear in the definition itself.

A.7.6 Hallucination Inevitability

We now state a baseline hallucination inevitability theorem for deterministic degradation dynamics.

To capture recurring nontrivial demands, we restrict to runs that keep encountering artifacts whose requirements exceed a fixed lower bound.

Definition A.46 (Nontrivial requirement floor). A requirement floor is a tacit state $r \in T$. A run $(t_n, e_n, c_n)_{n \geq 0}$ has *recurring nontrivial demands* relative to r if there exists an infinite subsequence $(n_k)_{k \geq 0}$ such that

$$\text{Req}(e_{n_k}, c_{n_k}) \succeq r \quad \text{for all } k.$$

We also assume that degradation eventually pushes tacit states below this floor.

Assumption A.47 (Baseline degradation below the requirement floor). There exists $r \in T$ such that for every initial tacit state $t_0 \in T$, there exists $N \in \mathbb{N}$ with

$$\delta^N(t_0) \prec r.$$

This is a strengthening of “strict decay somewhere”: it requires that every trajectory ultimately falls below the floor r .

Theorem A.48 (Hallucination inevitability). *Assume:*

1. context degradation δ satisfies monotonicity, non-invertibility, and Assumption ??;
2. requirements admit a nontrivial floor r and are monotone in credence;
3. a run $(t_n, e_n, c_n)_{n \geq 0}$ has recurring nontrivial demands relative to r .

Then every infinite run produces at least one hallucination; i.e., there exists n with

$$t_n \prec \text{Req}(e_n, c_n).$$

Proof sketch. By Assumption ??, for any initial t_0 there exists N with $\delta^N(t_0) \prec r$. Consider the given run and the subsequence (n_k) along which $\text{Req}(e_{n_k}, c_{n_k}) \succeq r$. For sufficiently large k , monotonicity of δ and recurrence of nontrivial demands imply that

$$t_{n_k} \preceq \delta^m(t_0) \prec r \preceq \text{Req}(e_{n_k}, c_{n_k})$$

for some m , so $t_{n_k} \prec \text{Req}(e_{n_k}, c_{n_k})$, i.e., a hallucination. A detailed proof can be given by induction along the run, but the key idea is that degradation eventually drives tacit grounding below a floor that requirements exceed infinitely often. $\square$

Remark A.49. The theorem applies to *nontrivial, open-ended* task streams with recurrent demands above a fixed floor. Trivial avoidance regimes (e.g., restricting to a tiny class of always-easy tasks or always answering “I don’t know”) fall outside the intended scope.

A.7.7 Explicit-Only Hallucinations

We now isolate explicit artifacts that are produced entirely within explicit reasoning, but which can be hallucinated relative to some tacit state encountered along a run.

Definition A.50 (Explicit-only hallucination). An *explicit-only hallucination* is an explicit artifact $e \in E$ such that:

1. e is well-typed and syntactically valid as an FRFP explicit object;
2. e is surface-plausible under an explicit plausibility functional (Section ??);
3. there exists a tacit state t and credence c encountered in some run such that $t \prec \text{Req}(e, c)$.

Proposition A.51 (Containment but not elimination). *Nothing in Epistemic Space or Epistemic Algebra forbids the production of explicit-only hallucinations. Eliminating them entirely would require either:*

- *an explicit correctness oracle, or*
- *a simulation of tacit evaluation inside explicit space,*

both of which would violate Explicit–Tacit Separation (ETS) and Human-Exclusive Grounding (HEG).

The explicit–tacit separation guarantees containment, not elimination: explicit-only hallucinations cannot become correctness judgments without passing through RB, but they may still be produced arbitrarily often within explicit space.

A.7.8 Confidence Inflation

We model how plausible explicit artifacts raise tacit credence even when they do not increase grounding.

Definition A.52 (Explicit plausibility). An *explicit plausibility functional* is a map

$$\pi : E \rightarrow [0, 1],$$

measuring surface coherence, fluency, or apparent authority of explicit artifacts.

We extend tacit state with a credence component.

Definition A.53 (Extended tacit state). Let

$$T^e := T \times \mathbb{R}_{\geq 0}$$

be the space of extended tacit states (t, c) , where $t \in T$ is grounding and c is tacit credence (how settled or reliable the agent treats the current artifact).

Definition A.54 (Confidence inflation). A *confidence inflation operator* is a map

$$\Phi : (T^e \times E) \rightarrow T^e, \quad \Phi((t, c), e) := (t, c + \varphi(\pi(e))),$$

where $\varphi : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ is monotone increasing.

We require that Φ does not increase tacit grounding:

$$\Phi((t, c), e) = (t', c') \quad \Rightarrow \quad t' = t.$$

Thus confidence inflation changes only the agent’s interpretive stance (higher credence), not the underlying grounding t .

Remark A.55. Confidence inflation captures the structural asymmetry between “feeling confident” and “being grounded”. Systems optimized for producing well-formed, plausible outputs (e.g., LLMs) inevitably exhibit this asymmetry: they can increase credence by producing plausible artifacts without necessarily increasing grounding.

A.7.9 Requirement Escalation Under Plausibility

Credence affects what requirements are appropriate: asserting high-credence claims demands stronger grounding.

Definition A.56 (Requirement escalation). The requirement map $\text{Req} : E \times \mathbb{R}_{\geq 0} \rightarrow T$ is *escalatory* if it is monotone in credence, i.e.,

$$c_1 \leq c_2 \quad \Rightarrow \quad \text{Req}(e, c_1) \preceq \text{Req}(e, c_2),$$

for all $e \in E$. Repeated exposure to highly plausible artifacts (large $\pi(e)$) can then raise requirements even if the explicit content e is unchanged.

Combining confidence inflation and requirement escalation, we see that the same explicit artifact can acquire higher requirements over time as agents become more confident in it.

A.7.10 Feedback-Coupled Degradation

We now allow degradation to depend on credence, introducing a feedback loop: overconfidence reduces skepticism and checking, accelerating loss of grounding.

Definition A.57 (Feedback-coupled degradation). A *feedback-coupled degradation* operator is a function

$$\delta^* : T^e \rightarrow T, \quad \delta^*(t, c) = \delta_c(t),$$

where $\delta_c : T \rightarrow T$ is a family of degradation operators indexed by credence c , satisfying:

1. **Credence monotonicity:** for fixed t , if $c_1 \leq c_2$ then $\delta^*(t, c_2) \preceq \delta^*(t, c_1)$; i.e., higher confidence weakens effective grounding.
2. **Consistency with baseline degradation:** $\delta^*(t, 0) = \delta(t)$ for all t .

Definition A.58 (Extended run with feedback). An *extended run* is a sequence

$$(t_0, c_0, e_0), (t_1, c_1, e_1), \dots$$

such that, at each step n ,

$$\begin{aligned} (t_n^+, c_n^+) &= \Phi((t_n, c_n), e_n) && \text{(confidence inflation),} \\ t_{n+1} &= \delta^*(t_n^+, c_n^+) && \text{(feedback-coupled degradation),} \\ c_{n+1} &\text{ chosen by tacit update (e.g., reset or carry-over).} \end{aligned}$$

The exact rule for updating c_{n+1} can be left abstract; all we require is that confidence inflation can push credence arbitrarily high along some runs.

A.7.11 Rate of Context Loss

We can analyze the rate at which tacit grounding is lost under feedback-coupled degradation.

Definition A.59 (Grounding measure and loss rate). Let $\mu : T \rightarrow \mathbb{R}_{\geq 0}$ be any monotone measure of tacit grounding (used only for analysis). For an extended run (t_n, c_n, e_n) , define the instantaneous loss rate at turn n by

$$\rho_n := \mu(t_n) - \mu(\delta^*(t_n, c_n)).$$

Proposition A.60 (Loss rate increases with credence). *Under the credence-monotonicity assumption for δ^* and monotonicity of μ , the loss rate ρ_n is non-decreasing in c_n . In particular, higher credence does not slow context loss and may accelerate it.*

Proof. If $c_1 \leq c_2$, then $\delta^*(t, c_2) \preceq \delta^*(t, c_1)$ and monotonicity of μ yields $\mu(\delta^*(t, c_2)) \leq \mu(\delta^*(t, c_1))$. Thus

$$\mu(t) - \mu(\delta^*(t, c_2)) \geq \mu(t) - \mu(\delta^*(t, c_1)),$$

so ρ_n increases with c_n . □

A.7.12 Accelerated Hallucination

We now combine confidence inflation, requirement escalation, and feedback-coupled degradation.

Definition A.61 (Feedback-coupled update). Along an extended run, the joint update

$$(t_{n+1}, c_{n+1}) = (\delta^*(t_n, c_n + \varphi(\pi(e_n))), c_{n+1})$$

captures the combined effect of:

- confidence inflation $(t_n, c_n) \mapsto (t_n, c_n + \varphi(\pi(e_n)))$;
- degradation δ^* , which is more severe at higher credence.

Grounding weakens faster while requirements grow stronger (via $\text{Req}(e_n, c_n)$).

This creates a widening gap between available grounding t_n and demanded grounding $\text{Req}(e_n, c_n)$, leading to accelerated hallucination.

A.7.13 Accelerated Hallucination Inevitability

We can now state a strengthened inevitability theorem in the presence of feedback.

Assumption A.62 (Unbounded credence under plausibility). There exists $\bar{\pi} \in (0, 1]$ such that for any initial credence c_0 and any infinite sequence of artifacts with $\pi(e_n) \geq \bar{\pi}$ infinitely often, the accumulated credence

$$c_n^+ := c_0 + \sum_{k=0}^{n-1} \varphi(\pi(e_k))$$

tends to $+\infty$ along a subsequence.

Theorem A.63 (Accelerated hallucination inevitability). *Assume:*

1. *baseline context degradation δ is monotone and non-invertible and satisfies Assumption ??;*
2. *confidence inflation Φ with $\varphi(\bar{\pi}) > 0$ for some $\bar{\pi} \in (0, 1]$;*
3. *requirement escalation: Req is monotone in credence;*
4. *feedback-coupled degradation δ^* is credence-monotone and reduces to δ at zero credence;*
5. *the run contains infinitely many surface-plausible explicit artifacts with $\pi(e_n) \geq \bar{\pi}$ and recurring nontrivial demands relative to some floor r .*

Then:

1. hallucination occurs in finite time; i.e., there exists $n < \infty$ such that $t_n \prec \text{Req}(e_n, c_n)$;
2. the minimum such n is strictly smaller than in the baseline degradation model without feedback (for any fixed initial state and pipeline sequence), in the sense that feedback cannot increase time-to-hallucination.

Proof sketch. By unbounded credence under plausibility and the presence of infinitely many plausible artifacts, credence c_n grows without bound along some subsequence. By credence-monotonicity of δ^* and requirement escalation, higher credence simultaneously:

- increases required grounding $\text{Req}(e_n, c_n)$, and
- accelerates degradation of available grounding via δ^* .

Thus the gap

$$t_n \prec \text{Req}(e_n, c_n)$$

is reached in finite time, and strictly earlier than under the baseline degradation δ alone, where the same sequence of artifacts would produce slower loss of grounding and lower requirements. A fully formal proof can be obtained by coupling two runs (with and without feedback) and using the monotone measure μ to compare times at which $\mu(t_n)$ falls below $\mu(\text{Req}(e_n, c_n))$. $\square$

A.7.14 Degradation–repair dynamics and entropy drift

We extend the Epistemic Dynamics to include abstract repair. The aim is to state conditions under which repair delays or reshapes, but does not remove, long-run degradation.

Definition A.64 (Entropy functional on T). An *entropy functional* on the tacit category T is a map $S : T \rightarrow \mathbb{R}$ such that

1. $t \preceq t' \Rightarrow S(t) \leq S(t')$ for all $t, t' \in T$; and
2. there exists $S_{\text{req}} \in \mathbb{R}$ such that $S(t) \leq S_{\text{req}}$ whenever t meets the relevant requirement floor(s), and $S(t) > S_{\text{req}}$ implies that accepting a nontrivial demand at t yields an FRFP-hallucination in the sense $t_n \prec \text{Req}(e_n, c_n)$.

Thus $S(t)$ measures a grounding deficit: larger values correspond to more degraded tacit states.

We consider runs $(t_n)_{n \geq 0}$ in T evolving under a mixture of degradation and repair, and write $S_n := S(t_n)$. Let $D_n, R_n \in \{0, 1\}$ indicate, respectively, whether step n processes a nontrivial demand and whether a repair operation is applied.

Assumption A.65 (Task-induced degradation). There exists $\varepsilon > 0$ such that for all $t \in T$ and all nontrivial demand steps to which the Epistemic Dynamics apply,

$$S(\delta(t)) \geq S(t) + \varepsilon.$$

Assumption A.66 (Bounded repair). There exists $B < \infty$ such that for all admissible repair operators $\rho : T \rightarrow T$ and all $t \in T$,

$$S(\rho(t)) \geq S(t) - B.$$

Combining these, for each N we have

$$S_N \geq S_0 + \varepsilon \sum_{n=1}^N D_n - B \sum_{n=1}^N R_n, \quad (1)$$

where the sums record entropy produced by nontrivial tasks and removed by repair.

Theorem A.67 (Deterministic entropy drift and inevitability). *Suppose Assumptions ?? and ?? hold and that $(D_n, R_n)_{n \geq 1}$ satisfies*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N D_n \geq \alpha > 0, \quad \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N R_n \leq \beta < \infty.$$

Assume further that there exists $\gamma > 0$ with

$$\varepsilon \alpha - B \beta \geq \gamma > 0.$$

Then there exists N_0 such that $S_N \geq S_0 + \gamma N$ for all $N \geq N_0$. In particular, for any finite S_{req} the hitting time

$$\sigma_{\text{req}} := \inf\{n \geq 0 : S_n > S_{\text{req}}\}$$

is finite. If nontrivial demands recur after σ_{req} and the protocol sometimes accepts artifacts at those steps with credence corresponding to S_{req} , then along some admissible run an FRFP-hallucination event $t_n \prec \text{Req}(e_n, c_n)$ occurs.

Proof sketch. Divide (??) by N and apply the liminf/limsup bounds on the empirical frequencies of D_n and R_n to obtain

$$\liminf_{N \rightarrow \infty} \frac{S_N - S_0}{N} \geq \varepsilon \alpha - B \beta \geq \gamma > 0.$$

Hence $S_N \geq S_0 + \gamma N$ for all large N , so S_N eventually exceeds any finite S_{req} and $\sigma_{\text{req}} < \infty$. The hallucination clause follows from the definition of S_{req} and the criterion $t_n \prec \text{Req}(e_n, c_n)$ at an accepted nontrivial demand. $\square$

Theorem A.68 (Stochastic entropy drift and almost-sure inevitability). *Suppose Assumptions ?? and ?? hold and that $(D_n, R_n)_{n \geq 1}$ is stationary and ergodic with*

$$\mathbb{E}[D_1] = \bar{d} > 0, \quad \mathbb{E}[R_1] = \bar{r} < \infty.$$

Assume that

$$\varepsilon \bar{d} - B \bar{r} = \gamma > 0.$$

Then almost surely

$$\liminf_{N \rightarrow \infty} \frac{S_N - S_0}{N} \geq \gamma > 0,$$

so $S_N \rightarrow +\infty$ almost surely. In particular, the hitting time $\sigma_{\text{req}} := \inf\{n \geq 0 : S_n > S_{\text{req}}\}$ is almost surely finite. Under the same acceptance and recurrence conditions as in Theorem ??, the hallucination time $\tau(\omega)$ in the Epistemic Dynamics stochastic semantics satisfies $P(\tau < \infty) = 1$.

Proof sketch. By the ergodic theorem, almost surely

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N D_n = \bar{d}, \quad \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N R_n = \bar{r}.$$

Together with (??) this yields

$$\liminf_{N \rightarrow \infty} \frac{S_N - S_0}{N} \geq \varepsilon \bar{d} - B\bar{r} = \gamma > 0 \quad \text{a.s.}$$

Thus $S_N \rightarrow +\infty$ almost surely and $\sigma_{\text{req}} < \infty$ almost surely; whenever a nontrivial demand is accepted at or after σ_{req} , the hallucination criterion is met at some finite step, so $P(\tau < \infty) = 1$. $\square$

Remark A.69 (Counter-scenarios and safe horizons). The drift condition in Theorems ?? and ?? formalizes a high-load, bounded-repair regime: nontrivial demands occur with positive asymptotic frequency, repair has bounded one-step effect, and net entropy production $\varepsilon \bar{d}$ exceeds maximal entropy removal $B\bar{r}$. In this regime, repair changes the timing and shape of hallucination risk but does not eliminate it.

Several counter-scenarios lie at or beyond this domain. In *balanced or negative drift* regimes, repair investment is large enough that $B\bar{r} \geq \varepsilon \bar{d}$ and S_N can remain tight or decrease. In *unbounded repair* regimes, one allows repair moves with no global bound B (e.g. costless, arbitrarily frequent “oracle refresh” operations or always-perfect staff replacement), violating Assumption ?. In *low-demand* regimes, such as rare emergency committees, the effective demand frequency $\bar{d}$ may be so small that modest repair keeps S below S_{req} over relevant horizons.

Within the high-load, bounded-repair regime, improving repair capacity modifies the hazard profile $(h_n)_{n \geq 0}$ and increases the safe-horizon functional $H_P(\varepsilon)$, but for any fixed stochastic FRFP pipeline P satisfying the drift condition, $H_P(\varepsilon)$ remains finite for every $\varepsilon \in (0, 1)$: one cannot obtain a global guarantee of “no hallucinations ever” over unbounded horizons by bounded repair alone.

A.7.15 Stochastic Pipelines and Their Impact

We extend the preceding results to stochastic FRFP pipelines.

*Trajectories and probability measures

Definition A.70 (Trajectory). Fix an initial extended state $(t_0, c_0, e_0) \in T \times \mathbb{R}_{\geq 0} \times E$. A *trajectory* (or *run*) is a sequence

$$\omega = ((t_0, c_0, e_0), (t_1, c_1, e_1), (t_2, c_2, e_2), \dots)$$

such that each $(t_{n+1}, c_{n+1}, e_{n+1})$ is obtained from (t_n, c_n, e_n) by one application of the FRFP pipeline, tacit evaluation, confidence update, and degradation (possibly feedback-coupled).

Let Ω denote the set of all such trajectories. Let $\Omega_{\text{allowed}} \subseteq \Omega$ be the subset of trajectories that satisfy the structural assumptions of this appendix (degradation, recurring requirements, etc.).

We consider two equivalent presentations of stochasticity:

- *Random-seed semantics*: extend the state space with a random seed r , and suppose there is a deterministic update map

$$F : T^e \times E \times R \rightarrow T^e \times E \times R$$

such that each $r \in R$ induces a unique trajectory $\omega(r) \in \Omega$.

- *Markov-kernel semantics*: one-step evolution is described by a Markov kernel

$$K : T^e \times E \rightarrow \mathcal{D}(T^e \times E),$$

where $\mathcal{D}(\cdot)$ indicates probability distributions; a trajectory ω is then a sample path of the stochastic process generated by K .

In either case, a *stochastic FRFP pipeline* is identified with a probability measure $\mathbb{P}$ on Ω induced by the underlying randomness.

*Hallucination events

Recall the hallucination criterion: hallucination at step n occurs when

$$t_n \prec \text{Req}(e_n, c_n).$$

Definition A.71 (Hallucination events). For a trajectory $\omega \in \Omega$ and $n \geq 0$, define the indicator

$$H_n(\omega) := \mathbf{1}[t_n \prec \text{Req}(e_n, c_n)] \in \{0, 1\}.$$

Define the *eventual hallucination indicator*

$$H(\omega) := \mathbf{1}[\exists n \geq 0 : H_n(\omega) = 1].$$

Thus $H(\omega) = 1$ iff the trajectory ω contains at least one hallucination.

Assumption A.72 (Pathwise inevitability). Suppose the structural assumptions of Sections B.1–B.12 hold (on degradation, requirements, and feedback), and that the baseline or accelerated hallucination inevitability theorem has been established in the deterministic setting. We assume:

$$\forall \omega \in \Omega_{\text{allowed}} : H(\omega) = 1.$$

That is, every allowed infinite trajectory eventually produces a hallucination.

Theorem A.73 (Almost-sure inevitability under stochastic pipelines). *Let $\mathbb{P}$ be any probability measure on Ω whose support is contained in Ω_{allowed} . Then*

$$\mathbb{P}(\text{eventual hallucination}) = \mathbb{P}(\{\omega : H(\omega) = 1\}) = 1.$$

Proof. By Assumption ??, $H(\omega) = 1$ for all $\omega \in \Omega_{\text{allowed}}$. Since $\mathbb{P}$ is supported on Ω_{allowed} , we have $H(\omega) = 1$ for $\mathbb{P}$ -almost every ω . Hence

$$\mathbb{P}(\text{eventual hallucination}) = \int_{\Omega} H(\omega) d\mathbb{P}(\omega) = \int_{\Omega} 1 d\mathbb{P}(\omega) = 1.$$

□

Thus passing from deterministic to stochastic pipelines does not weaken the inevitability result: randomness merely selects *which* hallucinating trajectory is realized, not whether hallucination occurs.

*Probabilistic confluence

Stochasticity breaks single-outcome confluence but yields a natural probabilistic analogue.

Definition A.74 (Output distribution and probabilistic confluence). Fix a time $n \geq 0$ and initial state (t_0, c_0, e_0) . Let $\mathbb{P}$ be a stochastic FRFP pipeline with support in Ω_{allowed} . The *time- n output distribution* is the pushforward measure

$$\mu_n^{\mathbb{P}} := (e_n)_{\#}\mathbb{P},$$

where $e_n : \Omega \rightarrow E$ maps a trajectory to its explicit artifact at step n .

Two stochastic implementations $\mathbb{P}_1, \mathbb{P}_2$ are *probabilistically confluent at time n* if they induce the same output distribution $\mu_n^{\mathbb{P}_1} = \mu_n^{\mathbb{P}_2}$ on E .

If both implementations correspond to the same Markov kernel on $T^e \times E$, they are probabilistically confluent at every time n . This notion provides a distributional counterpart to the canonical-form and confluence properties discussed earlier.

A.7.16 Finite-Horizon Pipelines and Their Impact

We now study the effect of restricting attention to finite prefixes of trajectories, both in deterministic and stochastic settings.

*Hallucination time

Definition A.75 (Hallucination time). For a trajectory $\omega \in \Omega$, define the *hallucination time*

$$\tau(\omega) := \inf\{n \geq 0 : t_n \prec \text{Req}(e_n, c_n)\},$$

with the convention $\tau(\omega) = +\infty$ if the set is empty (i.e., if no hallucination ever occurs).

Thus $\tau(\omega) < \infty$ iff ω contains at least one hallucination. In terms of Definition B.13.2, $H(\omega) = \mathbf{1}[\tau(\omega) < \infty]$.

Lemma A.76 (Pathwise inevitability and finite hallucination time). *Under Assumption ?? (pathwise inevitability), we have*

$$\forall \omega \in \Omega_{\text{allowed}} : \tau(\omega) < \infty.$$

Proof. By Assumption ??, for each $\omega \in \Omega_{\text{allowed}}$ there exists some n with $H_n(\omega) = 1$. By definition of τ , this implies that $\tau(\omega)$ is the infimum of a nonempty subset of $\mathbb{N}$, hence finite. $\square$

A *finite horizon* of length N corresponds to observing only the prefix

$$\omega_{\leq N} := ((t_0, c_0, e_0), \dots, (t_N, c_N, e_N)).$$

We say that a trajectory *hallucinates within horizon* N if $\tau(\omega) \leq N$.

Remark A.77 (No universal finite bound in general). Pathwise inevitability ($\tau(\omega) < \infty$ for all allowed ω) does not imply the existence of a universal finite upper bound N^* such that $\tau(\omega) \leq N^*$ for all $\omega \in \Omega_{\text{allowed}}$. In general, for any fixed N there may exist trajectories with $\tau(\omega) > N$.

*A finite-time bound under strengthened assumptions

If desired, one can obtain a finite-time upper bound under stronger assumptions.

Assumption A.78 (Uniform loss and requirement floor). Suppose there exist:

- a grounding measure $\mu : T \rightarrow \mathbb{R}_{\geq 0}$;
- a requirement level $r \in T$ and constant $\alpha > 0$ with $\mu(r) = \alpha$;
- a constant $\varepsilon > 0$;

such that, for every allowed trajectory $\omega = ((t_n, c_n, e_n))_{n \geq 0}$:

1. *Monotone loss*: $\mu(t_{n+1}) \leq \mu(t_n)$ for all n .
2. *Uniform loss above r* : whenever $\mu(t_n) > \alpha$, we have

$$\mu(t_n) - \mu(t_{n+1}) \geq \varepsilon.$$

3. *Recurrent requirement*: there exists an infinite sequence (n_k) with $\text{Req}(e_{n_k}, c_{n_k}) \succeq r$ for all k .

Lemma A.79 (Finite-time bound under uniform loss). *Under Assumption ??, every allowed trajectory ω satisfies*

$$\tau(\omega) \leq N^* := \left\lceil \frac{\mu(t_0) - \alpha}{\varepsilon} \right\rceil.$$

That is, a hallucination must occur by time N^ .*

Proof. By monotone loss and the uniform-loss condition, as long as $\mu(t_n) > \alpha$,

$$\mu(t_{n+1}) \leq \mu(t_n) - \varepsilon.$$

By induction, for any n with this property,

$$\mu(t_n) \leq \mu(t_0) - n\varepsilon.$$

For $n = N^*$ we obtain $\mu(t_{N^*}) \leq \alpha = \mu(r)$. By recurrence of requirements, there exists some k with $n_k \leq N^*$ and $\text{Req}(e_{n_k}, c_{n_k}) \succeq r$. Using monotonicity of μ with respect to $\preceq$, we conclude that

$$t_{n_k} \preceq r \preceq \text{Req}(e_{n_k}, c_{n_k}),$$

and in fact $t_{n_k} \prec \text{Req}(e_{n_k}, c_{n_k})$ for nontrivial requirements, so $\tau(\omega) \leq n_k \leq N^*$. $\square$

These stronger conditions are not needed for the main inevitability results, but they demonstrate that finite-time bounds are available in regimes with sufficiently uniform context loss and recurrent demands.

*Stochastic finite horizons

Returning to the stochastic setting, we can describe how finite-horizon hallucination probabilities behave.

Definition A.80 (Finite-horizon hallucination probability). Fix a horizon $N \in \mathbb{N}$. Define the event

$$A_N := \{\omega \in \Omega : \tau(\omega) \leq N\},$$

and the corresponding probability

$$p_N := \mathbb{P}(A_N) = \mathbb{P}(\tau \leq N).$$

Thus p_N is the probability that a random run hallucinates within the first N steps.

Proposition A.81 (Monotone convergence of finite-horizon probabilities). *Assume almost-sure inevitability (Theorem ??), i.e., $\mathbb{P}(\tau < \infty) = 1$. Then:*

1. *The sequence $(p_N)_{N \geq 0}$ is non-decreasing: $p_N \leq p_{N+1}$ for all N .*
2. $\lim_{N \rightarrow \infty} p_N = 1$.

Proof. For each ω , the events $\{\tau(\omega) \leq N\}$ are nested: $\{\tau \leq N\} \subseteq \{\tau \leq N+1\}$. Hence $A_N \subseteq A_{N+1}$ and $p_N \leq p_{N+1}$. Moreover, $\{\tau < \infty\} = \bigcup_N A_N$, so by continuity of probability from below,

$$\mathbb{P}(\tau < \infty) = \lim_{N \rightarrow \infty} \mathbb{P}(A_N) = \lim_{N \rightarrow \infty} p_N.$$

By assumption, $\mathbb{P}(\tau < \infty) = 1$, so $\lim_{N \rightarrow \infty} p_N = 1$. $\square$

Proposition A.82 (Multiple runs over a fixed horizon). *Fix a horizon N and consider $K \in \mathbb{N}$ independent trajectories $\omega^{(1)}, \dots, \omega^{(K)}$, with corresponding hallucination times $\tau^{(i)}$. Then:*

1. *The probability that no trajectory hallucinates within horizon N is*

$$\mathbb{P}(\forall i, \tau^{(i)} > N) = (1 - p_N)^K.$$

2. *The probability that at least one trajectory hallucinates within horizon N is*

$$1 - (1 - p_N)^K.$$

Proof. By independence,

$$\mathbb{P}(\tau^{(i)} > N) = 1 - p_N$$

for each i , so

$$\mathbb{P}(\forall i, \tau^{(i)} > N) = \prod_{i=1}^K (1 - p_N) = (1 - p_N)^K.$$

The complementary event has probability $1 - (1 - p_N)^K$. □

These results show how inevitability manifests at finite horizons: for any fixed horizon, hallucination remains probabilistic ($p_N < 1$ in general), but as the horizon length or number of runs increases, the probability that hallucination has occurred approaches 1.

Future work lifts these individual dynamics to populations of agents and explicit consensus mechanisms.

B.16 Hazard Functions and Safe Horizons

We now refine finite-horizon analysis by introducing per-step hazard and a derived notion of “safe horizon” for a given stochastic FRFP pipeline.

Definition A.83 (Per-step hazard). Let $\tau : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be the hallucination time of Definition B.31, and let P be a stochastic FRFP pipeline with associated distribution of τ . For each $n \in \mathbb{N}$ with $P(\tau \geq n) > 0$ we define the (discrete) hazard at step n by

$$h_n := P(\tau = n \mid \tau \geq n) = \frac{P(\tau = n)}{P(\tau \geq n)}.$$

If $P(\tau \geq n) = 0$ we set $h_n := 0$.

Intuitively, h_n is the conditional probability that a run first hallucinates at step n , given that it has not hallucinated earlier. The sequence $(h_n)_{n \geq 0}$ is a pipeline-level summary of how risk is distributed along the trajectory.

Definition A.84 (Finite-horizon survival and hazard). For $N \in \mathbb{N}$ let

$$S_N := P(\tau > N) = 1 - p_N$$

be the survival probability at horizon N , where $p_N = P(\tau \leq N)$ is the finite-horizon hallucination probability from Definition B.36. Whenever $P(\tau \geq n) > 0$ for all n , the survival probabilities admit the product representation

$$S_N = \prod_{n=0}^N (1 - h_n),$$

so that the finite-horizon risk $p_N = 1 - S_N$ can be understood as the cumulative effect of per-step hazards.

This representation makes explicit the trade-off between (i) the number of steps in a pipeline and (ii) the hazard profile (h_n) induced by its design (e.g. decomposition, tool calls, or repeated self-checks).

Definition A.85 (Safe horizon at tolerance ε). Fix a tolerance level $\varepsilon \in (0, 1)$. The ε -safe horizon of a stochastic FRFP pipeline P is defined as

$$H_P(\varepsilon) := \sup\{N \in \mathbb{N} : p_N \leq \varepsilon\} \in \mathbb{N} \cup \{\infty\}.$$

When $H_P(\varepsilon)$ is finite, any horizon $N \leq H_P(\varepsilon)$ satisfies $P(\tau \leq N) \leq \varepsilon$, i.e. hallucination within that horizon is at most ε -likely.

Remark A.86 (Design interpretation). The safe-horizon functional $H_P(\varepsilon)$ isolates a pipeline-level design parameter: for a given stochastic implementation P and admissible risk budget ε , it characterizes how long a single run can safely be extended before hallucination becomes probable. Changes to decomposition, step-ordering, or intermediate checks can therefore be compared in terms of their effect on (h_n) and $H_P(\varepsilon)$, even when the underlying explicit pipeline is statically equivalent in $N \times E$.

Combining Definition ?? with Proposition B.38, we can also analyze restart and parallelization policies: for example, running K independent instances of a pipeline up to horizon $N \leq H_P(\varepsilon)$ yields hallucination probability $1 - (1 - p_N)^K$, which can be bounded using the same tolerance parameter.

B.17 Dynamic Refinement of Pipelines

We next introduce a notion of dynamic refinement between stochastic FRFP pipelines that compares their hallucination-time distributions.

Definition A.87 (Dynamic refinement). Let P_1, P_2 be stochastic FRFP pipelines on the same state space and admissible trajectory set Ω_{allowed} , with induced hallucination times τ_1, τ_2 and finite-horizon probabilities

$$p_N^{(i)} := P_i(\tau_i \leq N), \quad N \in \mathbb{N}, \quad i \in \{1, 2\}.$$

We say that P_2 is a *dynamic refinement* (or *dynamically safer variant*) of P_1 if for all $N \in \mathbb{N}$ we have

$$p_N^{(2)} \leq p_N^{(1)}.$$

Equivalently, τ_2 stochastically dominates τ_1 in the usual sense.

Remark A.88 (Static vs dynamic equivalence). Two pipelines may be statically equivalent in the sense of Appendix A (same normal form in $N \times E$ and same tacit semantics $J-K$), yet dynamically inequivalent: they can induce different hazard profiles and hence different hallucination-time distributions. Dynamic refinement isolates this difference at the level of trajectories, not static syntax: P_2 improves on P_1 precisely when it never increases finite-horizon hallucination risk.

Remark A.89 (Limitations). Dynamic refinement is a partial order on stochastic implementations, not an absolute guarantee of safety: if the baseline pipeline P_1 satisfies the inevitability conditions of Theorems B.11, B.25, and B.29, then so does any refinement P_2 . Refinements can delay hallucination and reduce risk over practically relevant horizons, but they cannot eliminate hallucination altogether without violating the structural assumptions of this appendix.

B.18 Structural implications of pipeline dynamics

The move from static FRFP pipelines to the dynamic setting of this appendix turns time and iteration into first-class objects. Beyond the inevitability results of Theorems B.11, B.25, and B.29, this buys several structural notions that will be used in later appendices and case studies.

(1) Hazard profiles and safe horizons. Definition B.31 endows each stochastic FRFP implementation with a hallucination time τ , and Definitions B.36 and ?? refine this to a finite-horizon risk curve $(p_N)_{N \geq 0}$ and a discrete hazard sequence $(h_n)_{n \geq 0}$. The safe-horizon functional $H_P(\varepsilon)$ in Definition ?? abstracts how long a given pipeline can be run, under a fixed stochastic implementation P and risk budget ε , before hallucination becomes likely. Thus even when static semantics and correctness are fixed, pipeline dynamics exposes a new design degree of freedom: the shape of the hazard profile along a run.

(2) Time–decomposition trade-offs. In the static theory of Appendices A and B.1–B.3, different decompositions of a task into explicit steps are equivalent whenever they normalize to the same morphism in $N \ltimes E$ and share the same tacit semantics $J-K$. Once trajectories are considered, this equivalence is refined: more fine-grained decompositions typically reduce the requirement demanded at individual steps, but also increase the length of the trajectory and the opportunity for degradation to act. Pipeline dynamics therefore makes precise a basic design tension: reshaping a task into “more steps” can both reduce per-step hazard and increase cumulative risk, and only the dynamic quantities (h_n) and (p_N) decide which effect dominates in a concrete setting.

(3) Dynamic refinement of implementations. Definition ?? introduces a partial order on stochastic implementations of the *same* static pipeline: P_2 dynamically refines P_1 when it never increases finite-horizon risk. This separates:

- static equivalence in $N \ltimes E$ (same normal form, same tacit semantics), from
- dynamic safety properties (how quickly hallucination emerges under a particular scheduling, tool use pattern, or feedback coupling).

Subsequent appendices will use this distinction to talk about pipeline-level improvements that do not change what a system computes in principle, but do change how its risk unfolds in time.

(4) Session limits, restart policies, and composition. Because hallucination is represented as a stopping time and finite-horizon probabilities are explicit, Appendix B supports principled discussion of session limits and restarts. Proposition B.38 already shows how repeated runs over a fixed horizon aggregate risk via $1 - (1 - p_N)^K$. In more complex settings, this allows single-pipeline guarantees (bounds on p_N or $H_P(\varepsilon)$) to be composed into system-level guarantees for restart, parallelization, or human-in-the-loop oversight policies, much as Appendix A.10 composes agent-level inevitability into collective inevitability.

(5) Localizing and targeting failure modes. Finally, working with trajectories rather than only normal forms means that hallucination can be localized to segments of a run: particular regions of the state space, update rules, or phases of the pipeline where hazard is concentrated. This enables targeted interventions (for example, inserting human checks or additional tools only before entering high-hazard regions) that would be invisible in a purely static semantics.

Taken together, these structures show that pipeline dynamics is not only a way to restate hallucination inevitability in stochastic language. It also provides a vocabulary—hazard, safe horizons, dynamic refinement, and time–decomposition trade-offs—for comparing and improving FRFP pipelines *without* altering their underlying explicit semantics in $N \propto E$.

A.7.17 Purpose and Scope

This appendix extends Epistemic dynamics to *collective* epistemic systems. Rather than a single agent executing a single FRFP pipeline, we consider populations of agents who:

- share an explicit artifact space E ,
- each carry their own tacit state and degradation dynamics,
- exchange only explicit artifacts (no tacit information flow),
- may be coupled by explicit consensus mechanisms and institutions.

The objective is to characterize:

1. how to view a population as a single large FRFP system,
2. how stochastic collective evolution can be modeled as a Markov process on this system,
3. how hallucination and other failures lift from individuals to populations, and
4. which collective design changes *cannot*, even in principle, eliminate hallucination.

This section provides the bridge between individual pipelines (Sections A.2–A.8), impossibility results (Appendix A.11), and institutional design (Appendix B and Appendix C).

A.7.18 Epistemic Agents

Definition A.90 (Epistemic agent). An *epistemic agent* is a tuple

$$A_i = (T_i, \preceq_i, \delta_i, T_i^e),$$

where:

- T_i is a tacit context space equipped with a preorder $\preceq_i$ encoding “at least as grounded as”;
- $\delta_i : T_i \rightarrow T_i$ is a (tacit) context degradation operator;
- $T_i^e := T_i \times \mathbb{R}_{\geq 0}$ is the extended tacit state space, pairing grounding $t_i \in T_i$ with tacit credence $c_i \in \mathbb{R}_{\geq 0}$.

Intuitively, A_i captures a single human (or human-like) reasoner as modeled in Epistemic Dynamics: grounding lives in T_i , weakens under δ_i , and credence lives in the additional component of T_i^e .

A.7.19 Epistemic Populations

Definition A.91 (Epistemic population and category **Pop**). An *epistemic population* is a finite or countable family $P = (A_i)_{i \in I}$ of epistemic agents indexed by a set I . The category **Pop** of epistemic populations is defined by:

- objects: epistemic populations $P = (A_i)_{i \in I}$;

- morphisms $F : P \rightarrow P'$: pairs $(f, (F_i)_{i \in I})$ where $f : I \rightarrow I'$ is a function on agent indices and, for each i ,

$$F_i : T_i \rightarrow T'_{f(i)}$$

is monotone with respect to $\preceq_i, \preceq'_{f(i)}$ and commutes with degradation and credence:

$$F_i \circ \delta_i = \delta'_{f(i)} \circ F_i.$$

Morphisms in **Pop** model re-indexing or restructuring of populations: embedding agents into larger organizations, replacing agents, or transferring tacit states along institutional lines.

A.7.20 Shared Explicit Reasoning Category

Let $\mathcal{C} := N \ltimes E$ be the reasoning category of Appendix A, where E denotes explicit artifacts and N navigation moves in explicit space. For a fixed index set I , we form the product category

$$\mathcal{C}^I,$$

whose objects are I -indexed families $X = (X_i)_{i \in I}$ of explicit states and whose morphisms are families of explicit updates. Intuitively, $\mathcal{C}^I$ is the “big explicit state space” of the whole population: it records, for each agent, which task and explicit artifacts they are currently working with.

We assume a single shared explicit artifact space E ; each agent i is equipped with a representational backflow map

$$\text{RB}_i : E \rightarrow T_i,$$

as in Appendix A. Explicit artifacts are globally visible; tacit grounding remains agent-local.

A.7.21 Communication Structure

Definition A.92 (Communication graph). A *communication structure* on population $P = (A_i)_{i \in I}$ is a directed graph

$$G = (I, \rightarrow),$$

where $i \rightarrow j$ indicates that explicit artifacts produced by i may be consumed by j (for example, by reading a paper, listening to a briefing, or reviewing a log). No tacit information is transmitted across G : only explicit artifacts traverse edges, and representational backflow RB_j determines how they affect T_j .

A.7.22 Collective Dynamics as a Markov System

A *collective execution* in earlier drafts was modeled as a bare sequence of events $(i_n, e_n) \in I \times E$ indicating “agent i_n produces artifact e_n ”. We now package such executions into a stochastic dynamical system over the large explicit category $\mathcal{C}^I$.

Definition A.93 (Collective state). A collective explicit state is an object $X \in \mathcal{C}^I$ together with a population $P = (A_i)_{i \in I}$ and communication graph $G = (I, \rightarrow)$. We write $(P, G; X)$ for such a state.

Definition A.94 (Population semantics). A *population semantics functor* is a functor

$$\overline{J} : \mathcal{C}^I \rightarrow \prod_{i \in I} T_i^e$$

assigning to each collective explicit state X the tuple of extended tacit states $((t_i, c_i))_{i \in I}$ obtained when agents in P consume the artifacts in X (subject to G) and update using the operators of Appendix B (degradation, confidence inflation, requirement escalation, and feedback-coupled degradation).

Definition A.95 (Markovian collective dynamics). A *Markovian collective dynamics* on (P, G) is a transition kernel

$$M : \text{Obj}(\mathcal{C}^I) \rightarrow \mathcal{P}(\text{Obj}(\mathcal{C}^I))$$

assigning to each collective state X a probability measure $M(X, \cdot)$ on next states Y , such that each transition can be factored as:

1. select an acting agent $i_n \in I$ and explicit move in $\mathcal{C}$;
2. update X to Y accordingly (producing a new artifact e_n in E);
3. for each j with $i_n \rightarrow j$, update (t_j, c_j) via the feedback-coupled dynamics of Appendix B using e_n ;
4. apply baseline degradation for all agents.

Equivalently, one may view M as a functor

$$M : \mathcal{C}^I \longrightarrow \mathbf{Kl}(D),$$

where $\mathbf{Kl}(D)$ is the Kleisli category of the discrete distribution monad D , but we will only use the induced transition kernel on objects.

A *collective execution* is then a random path

$$X_0, X_1, X_2, \dots$$

in $\mathcal{C}^I$ sampled from M starting at some initial state X_0 , with tacit trajectories $\bar{J}(X_n) = ((t_{i,n}, c_{i,n}))_{i \in I}$.

A.7.23 Collective Hallucination and Stopping Times

We now lift the notion of hallucination to the population level.

Definition A.96 (Collective hallucination). Given a population $P = (A_i)_{i \in I}$, a communication graph G , and a collective execution $(X_n)_{n \geq 0}$ with associated tacit states $(t_{i,n}, c_{i,n})$, a *collective hallucination* occurs at step n if there exists an explicit artifact $e \in E$ and a nontrivial subset $J \subseteq I$ such that:

- each $j \in J$ has accepted or propagated e (in the sense of Appendix B: e enters their explicit pipeline), and
- for each $j \in J$,

$$t_{j,n} \prec_j \text{Req}_j(e, c_{j,n}),$$

i.e., the available tacit grounding is strictly below the requirement demanded by e in context $c_{j,n}$.

Let Ω be the space of all collective paths $(X_n)_{n \geq 0}$ induced by M from X_0 , and let P_M be the corresponding path measure.

Definition A.97 (Collective hallucination time). The *collective hallucination time* $\tau^{\text{coll}} : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ is the stopping time

$$\tau^{\text{coll}}(\omega) := \inf\{n \geq 0 : \omega \text{ exhibits a collective hallucination at step } n\},$$

with the convention that $\inf \emptyset = \infty$.

Under the Epistemic Dynamics assumptions (finite tacit capacity, non-invertible degradation, confidence inflation, requirement escalation, and feedback-coupled degradation), individual agents exhibit almost-sure hallucination along sufficiently long admissible runs (Appendix B). The purpose of this appendix is to show how such individual inevitability lifts to τ^{coll} at the population level.

A.7.24 Consensus as an Explicit Functor

Consensus mechanisms act purely in explicit space.

Definition A.98 (Consensus functor). A *consensus mechanism* on the explicit category $\mathcal{C}$ is a functor

$$C : \mathcal{C}^n \rightarrow \mathcal{C}$$

for some finite arity $n \geq 2$, or more simply

$$C : E^n \rightarrow E$$

on the underlying artifact space, modeling averaging, voting, or more complicated aggregation pipelines. Consensus functors operate entirely in explicit space: they take explicit artifacts as inputs and produce explicit artifacts as outputs, without access to tacit states or requirements.

A.7.25 Functorial Consensus Layer

Given a population $P = (A_i)_{i \in I}$ and communication structure G , a consensus mechanism C induces an explicit functor

$$C_P : E^n \rightarrow E$$

that may be interleaved with ordinary explicit computation in $\mathcal{C}^I$. We refer to such composites as the *consensus layer*. No consensus functor has access to the population semantics $\bar{J}$ or to individual requirement functions Req_i .

A.7.26 Tacit-Stability of Consensus

Definition A.99 (Tacit-stable consensus). A consensus functor C is *tacit-stable* if, for every agent i and any inputs $e_1, \dots, e_n \in E$, whenever $e_1, \dots, e_n$ are all individually below agent i 's requirement,

$$t_i \prec_i \text{Req}_i(e_k, c_i) \quad \forall k \in \{1, \dots, n\},$$

then the aggregate output $C(e_1, \dots, e_n)$ is also below requirement:

$$t_i \prec_i \text{Req}_i(C(e_1, \dots, e_n), c_i).$$

Theorem A.100 (Consensus stability). *Every consensus functor that acts purely in explicit space is tacit-stable.*

Proof sketch. Consensus functors are compositions of explicit morphisms in $\mathcal{C}$. By Explicit–Tacit Separation (ETS), explicit morphisms cannot increase tacit grounding: they do not alter t_i , only what artifacts are presented to representational backflow. If an agent’s tacit state t_i is below the requirement for each input artifact, it remains below the requirement for any artifact obtained by explicit-only transformation of those inputs. Thus consensus cannot manufacture sufficient grounding from insufficient inputs. $\square$

Tacit-stability says that explicit consensus can neither repair nor upgrade tacit adequacy: it preserves “being under-grounded” whenever that is the case for all inputs.

A.7.27 Limits of Automated Consensus

Theorem A.101 (No fully automated consensus elimination). *No fully automated consensus process acting purely in explicit space can eliminate collective hallucination across all admissible populations and collective dynamics.*

Proof sketch. By tacit-stability, consensus cannot transform artifacts that are under-grounded for all relevant agents into artifacts that are adequately grounded. Under the Epistemic Dynamics degradation assumptions, individual hallucination is inevitable along sufficiently long runs. Since consensus mechanisms have no access to tacit state, they cannot detect or remove these failures at the collective level; at best they can smooth, average, or denoise explicit behavior. Any claim of global hallucination-elimination by explicit-only consensus would contradict the Epistemic Dynamics inevitability results when applied agent-wise. $\square$

Automation may reduce variance or noise in explicit artifacts, but it cannot introduce new tacit grounding or globally halt its degradation.

A.7.28 Collective Inevitability

Theorem A.102 (Collective hallucination inevitability). *Consider a population $P = (A_i)_{i \in I}$ of epistemic agents with:*

- *finite or countable index set I ,*
- *finite tacit capacity for each A_i ,*
- *non-invertible degradation δ_i ,*
- *confidence inflation and requirement escalation as in Appendix B.*

Let M be any Markovian collective dynamics on $\mathcal{C}^I$ that respects these per-agent dynamics and produces infinitely many surface-plausible explicit artifacts along almost every path. Then the collective hallucination time τ^{coll} is almost surely finite:

$$P_M(\tau^{\text{coll}} < \infty) = 1.$$

Moreover, this property is invariant under the insertion of explicit-only consensus functors: composing M with any finite consensus layer that acts solely in explicit space does not change the almost-sure finiteness of τ^{coll} .

Proof sketch. By Appendix B, each agent A_i exhibits almost-sure hallucination along sufficiently long admissible runs of any FRFP pipeline that continues to produce surface-plausible artifacts. The collective Markov dynamics M induces, for each i , a marginal stochastic process on T_i^e with the same qualitative properties. Since the population is finite or countable, and collective execution produces infinitely many plausible artifacts with nonzero probability of being consumed by each agent (via G), the event that at least one agent hallucinates in finite time has probability 1. The definition of collective hallucination requires only that a nontrivial subset of agents jointly accept an under-grounded artifact, which occurs almost surely given enough time and interaction. Tacit-stability of consensus implies that explicit aggregation cannot remove such events once they arise; it can at best transform one hallucinated artifact into another. $\square$

A.7.29 Structural Implications of Collective Dynamics

The population-level formalism above does more than restate inevitability in a collective setting. It yields several structural notions that are used in later sections.

(1) Equivalence classes of collective designs. Viewing populations as objects in **Pop** and their joint explicit reasoning as trajectories in $\mathcal{C}^I$, we can treat different UIs, workflows, and consensus wrappers as endofunctors on $\mathcal{C}^I$ that preserve the population semantics $\bar{J}$. Designs that differ only by such explicit refactorings are *FRFP-equivalent*: they have the same tacit semantics and the same hallucination-time distribution, even if their surface appearance differs. This provides a criterion for when two collective protocols are “the same thing in principle”.

(2) Refinement order on collective systems. Given two Markovian collective dynamics M_1, M_2 on the same population, with corresponding collective hallucination times $\tau_1^{\text{coll}}, \tau_2^{\text{coll}}$, we may compare their finite-horizon risks

$$p_N^{(k)} := P_{M_k}(\tau_k^{\text{coll}} \leq N), \quad k \in \{1, 2\}.$$

We say that M_2 is a *dynamic refinement* of M_1 if $p_N^{(2)} \leq p_N^{(1)}$ for all N . This yields a partial order on collective implementations: some architectures are strictly safer than others, but purely explicit-only transformations that preserve semantics cannot improve this order.

(3) Compositional risk analysis. Because the collective dynamics lives in $\mathcal{C}^I$ and is built from per-agent Epistemic Dynamics, we can study how subsystem risks compose. For example, a scientific subsystem, an engineering subsystem, and a policy subsystem can each be modeled as a subpopulation with its own hazard profile; the global population then inherits a collective hallucination time equal to the minimum of the subsystem times. This provides a formal basis for “defense in depth” arguments.

(4) Limits of monitoring and metrics. Collective executions produce rich explicit traces: logs, dashboards, metrics, and aggregated indicators. In this formalism, these traces live in $\mathcal{C}^I$ or in functorial images thereof. The population semantics $\bar{J}$ and individual requirements Req_i , however, remain tacit. Many distinct tacit configurations can induce the same distribution over explicit traces. Hence correctness and hallucination risk are *non-identifiable from logs alone*: explicit-only monitoring cannot, in general, determine how well grounded a population’s beliefs really are.

(5) Scaling and phase transitions. By parametrizing collective systems by population size $|I|$, communication structure G , and per-agent degradation properties, Appendix A.10 provides a language for asking how epistemic risk scales with size and complexity. Nothing in the formalism guarantees that larger or more connected populations are safer; on the contrary, the inevitability theorem applies uniformly once the Epistemic Dynamics assumptions hold. This sets the stage for Appendix A.12, which studies which institutional forms can slow or bound such risks across populations and time.

Taken together, these structures show that Collective Epistemic Dynamics is not merely a corollary of individual hallucination: it is a genuinely new level of description in which equivalence, refinement, observability, and scaling of *collective* epistemic systems can be stated and analyzed formally.

A.7.30 Purpose and Scope

This section collects structural *no-go theorems* that delimit what can be guaranteed within FRFP-like architectures under Explicit–Tacit Separation (ETS), Human-Exclusive Grounding (HEG), and Human-Exclusive Closure (HEC). The results are invariant to model internals, training data, or intent.

The central organizing idea is that many desirable guarantees are *tacit-dependent*: whether they hold can vary with human tacit state even when explicit artifacts are held fixed. Under ETS and AE (AI-Explicit Restriction), tacit-dependent guarantees cannot be enforced or decided by explicit-only mechanisms.

A.7.31 Standing Assumptions and Notation

Throughout this appendix we fix the FRFP kernel and assume that all interaction and epistemic constraints from the main text are in force.

Foundational layers. On the single-agent side we assume:

- the interaction-layer constraints of Section 2 and Appendix A.2.6 (admissible traces, no AI access to tacit inputs, separation of content-producing and correctness-bearing moves);
- the epistemic ontology and explicit/tacit separation from Section 3 and Appendix A.2–A.3 (ambient category C_{full} , explicit and tacit subcategories E and T , boundary functor $\text{RB} : E \rightarrow T$, and the six primitives RI, EC, ED, RB, TE, HFD);
- the explicit workflow algebra and correctness layer from Sections 4–6 and Appendix A.4–A.8 (the TDG-generated explicit category, navigation category N , Grothendieck semidirect product $N \ltimes E$, semantic functor $\llbracket - \rrbracket : N \ltimes E \rightarrow T$, and the correctness quotient $\gamma : \text{Ob}(T) \rightarrow J$).

Dynamic and collective layers. In addition, we assume the epistemic dynamics and collective semantics developed in the later parts of Appendix A:

- the tacit degradation operator $\delta : T \rightarrow T$, requirement map $\text{Req} : E \times \mathbb{R}_{\geq 0} \rightarrow T$, and the induced notion of hallucination and finite safe horizons from Appendix A.9;
- the population and consensus layer of Appendix A.10, which models a population of agents with a shared explicit category, local tacit spaces, and a communication graph, and equips this structure with a Markovian collective dynamics.

For a fixed explicit pipeline P and an admissible implementation of P in this dynamic model, Appendix A.9 associates to each finite horizon N a hallucination probability p_N and a hazard profile $(h_n)_{n \geq 0}$, and defines a safe-horizon functional $H_P(\varepsilon)$ (see Definition A.??) capturing the maximal time N for which the implementation keeps hallucination risk below a tolerance ε . On the population side, Appendix A.10 provides analogous notions of collective hallucination time and collective safe horizons.

The explicit-only impossibility and institutional non-automation results in this appendix are understood relative to this entire stack: they assume the interaction, epistemic, algebraic, dynamic, and collective layers just described, and ask which kinds of detectors, consensus mechanisms, and governance operators can be realised *purely explicitly* within that fixed architecture.

A.7.32 Admissible Runs as Interaction Protocol Filtered Executions in $N \times E$

Definition A.103 (Executions in $N \times E$). Fix a valid pipeline term P (Epistemic Space / Algebra well-typed). An *execution trace* of P is a finite or infinite sequence

$$\rho : x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots$$

where each x_k is a configuration in $N \times E$ reachable from an initial configuration for P by the combined reduction relation $\rightarrow$.

Definition A.104 (Interaction Protocol admissibility predicate). Let $\text{Adm}_0(\rho)$ be a predicate on execution traces encoding Interaction Protocol constraints:

1. **IL (Interaction Limitation)**: ρ is turn-based and respects externally imposed bounds (e.g. finite-length turns).
2. **AR (Alignment Restriction)**: explicit actions occur only in response to explicit instructions present in the trace.
3. **MD (Mode Discipline)**: no step in ρ applies tacit operators inside the explicit phase; tacit transitions occur only after the boundary.
4. **NTER/CSC**: no step reconstructs tacit evaluation from explicit traces or treats explicit computation as a correctness oracle.

We treat Adm_0 as an axiomatically specified filter on traces.

Definition A.105 (Admissible runs). An *admissible run* of a pipeline P is an execution trace ρ in $N \times E$ such that $\text{Adm}_0(\rho)$ holds. We write $\text{Runs}(P)$ for the set of admissible runs of P .

Remark A.106. (Admissible runs and explicit-only results). This definition pins all explicit-only impossibility and non-automation statements to a common operational semantics: admissible runs are exactly those executions in $N \times E$ that additionally satisfy the interaction constraints (IL, AR, MD, NTER/CSC). All explicit-only no-go theorems in Appendix A.11 are understood to quantify over this class of runs.

A.7.33 Two Notions of “Same Explicit Artifact”

No-go statements often quantify over pairs of runs that produce the “same explicit artifact.” There are two natural notions.

Definition A.107 (Syntactic explicit identity). Two runs are *syntactically explicit-identical* if they produce equal explicit artifacts as elements of the explicit object E (i.e. equality in the representation domain).

Definition A.108 (Semantic explicit identity via normal forms). Assume the explicit reduction system $\rightarrow_{\mathcal{E}}$ is terminating and locally confluent (Appendix A axioms), so each explicit artifact has a unique normal form $\text{nf}(e)$. Two runs are *normal-form explicit-identical* if their produced artifacts e, e' satisfy

$$\text{nf}(e) = \text{nf}(e').$$

Remark A.109. Syntactic identity is appropriate when artifacts are literal strings/terms. Normal-form identity is appropriate when explicit equivalence is taken modulo rewriting (e.g. definitional equalities, normalization, canonical forms). All no-go results below hold under either identity notion; when needed we indicate which is assumed.

Remark A.110 (Admissible runs and stochastic trajectories). In the dynamic semantics of Appendix B, the same pipeline P gives rise to a measurable space of trajectories Ω_{allowed} , together with a family of probability measures capturing different stochastic implementations. We implicitly identify each admissible run ρ of P in the combined system $\mathbf{N} \times \mathbf{E}$ with its corresponding trajectory $\omega \in \Omega_{\text{allowed}}$. In particular, the hallucination time $\tau(\omega)$, the finite-horizon probabilities (p_N) , the hazard sequence (h_n) , and the safe-horizon function $H_P(\varepsilon)$ of Appendix B are all computed over this common set of admissible runs.

A.7.34 Tacit Dependence via $\gamma(\llbracket P \rrbracket)$

Definition A.111 (Tacit-dependent predicates). Let P be a fixed valid pipeline and let $\text{Runs}(P)$ be its admissible runs. A predicate $\mathcal{G}$ about an explicit artifact is *tacit-dependent* if there exist two admissible runs $\rho_1, \rho_2 \in \text{Runs}(P)$ such that:

- the runs produce the same explicit artifact (either syntactically or up to normal form, per Section D.3), and
- the observable correctness outcomes differ:

$$\gamma(\llbracket P \rrbracket) \text{ differs between } \rho_1 \text{ and } \rho_2.$$

Remark A.112. This definition formalizes “depends on tacit context” using only Epistemic Algebra semantics: $\llbracket - \rrbracket$ captures tacit transformation induced by the pipeline run, and γ extracts the human-observable correctness class.

Remark A.113 (Examples of tacit-dependent predicates). The definition above applies not only to pointwise correctness judgements but also to quantitative long-horizon properties. Typical examples of tacit-dependent predicates G include:

- the event that a run hallucinates within a fixed horizon N , i.e. $G(\omega)$ is the predicate $[\tau(\omega) \leq N]$;
- the statement that an implementation has ε -safe horizon at least N , i.e. G is the predicate $[H_P(\varepsilon) \geq N]$ for some fixed $\varepsilon \in (0, 1)$;
- in the collective setting of Appendix A.10, the statement that the collective hallucination time exceeds a fixed horizon, i.e. G is the predicate $[\tau_{\text{coll}} > N]$ for a given population and consensus dynamics.

In each case, there can exist admissible runs with the same explicit artifacts (in the sense of Definition B.66) but different values of τ , $\mathcal{HP}(\varepsilon)$ or τ_{coll} , so G is tacit-dependent in the sense of Definition B.66.

A.7.35 Schema Theorem: Tacit-Dependent Guarantees Are Not Explicit-Implementable

Definition A.114 (Explicit-only mechanism). An *explicit-only mechanism* is any procedure definable using explicit morphisms in $\mathcal{E}$ (and finite control) whose inputs and outputs live entirely in explicit space. Concretely, one may model an explicit-only mechanism as a total function $M : E^k \rightarrow E^\ell$, or as a family of such functions closed under composition.

Theorem A.115 (Tacit-dependent guarantees are not explicit-implementable). *Assume ETS and AE. Let $\mathcal{G}$ be a tacit-dependent predicate about explicit artifacts (Section D.4), where “same explicit artifact” is interpreted either syntactically or by normal form (Section D.3). Then no explicit-only mechanism can enforce $\mathcal{G}$ for all admissible runs.*

Proof sketch. By tacit dependence, there exist admissible runs producing the same explicit artifact (under the chosen identity notion) but yielding different outcomes $\gamma(\llbracket P \rrbracket)$. Any explicit-only mechanism sees only the explicit artifact(s) and therefore cannot act differently on these runs. Hence it cannot enforce a guarantee whose truth varies across admissible runs while explicit data are held fixed. $\square$

A.7.36 Corollaries (No-Go Family)

Each of the following is an instantiation of Theorem ??.

No Perfect Explicit-Only Hallucination Detector

Definition A.116 (Explicit-only monitor). An *explicit-only monitor* is a function $M : E \rightarrow \{\text{ok}, \text{flag}\}$.

Definition A.117 (Perfect detector). M is *perfect* if, for every admissible run producing explicit artifact e with tacit context t at evaluation time,

$$M(e) = \text{ok} \Rightarrow t \not\prec \text{Req}(e), \quad M(e) = \text{flag} \Rightarrow t \prec \text{Req}(e).$$

Corollary A.118 (No perfect explicit-only detector). Assume ETS and AE, and assume requirements are nontrivial and contexts vary across admissible runs. Then no explicit-only monitor is a perfect hallucination detector.

Proof sketch. The predicate “ e is a hallucination” is tacit-dependent: the same explicit artifact (syntactically or up to normal form) can be within grounding in one admissible run and beyond grounding in another. Apply Theorem ??. $\square$

Corollary A.119 (No elimination of hallucination under degradation). Assume the Epistemic Dynamics conditions that yield almost-sure hallucination inevitability: in particular, a finite-capacity tacit state space, a monotone non-invertible degradation operator $\delta : T \rightarrow T$, and recurring nontrivial requirements as in Theorem B.29. Let τ denote the hallucination time random variable for a fixed pipeline P and any admissible stochastic implementation of P whose trajectories lie in Ω_{allowed} . Then there is no explicit-only policy which guarantees

$$\tau(\omega) = \infty$$

for all admissible runs $\omega \in \Omega_{\text{allowed}}$, nor even $\mathbb{P}(\tau = \infty) = 1$ for all such implementations.

Proof. Under the assumptions of Appendix B, Theorem B.29 states that every admissible stochastic implementation of P has hallucination time τ finite almost surely:

$$\mathbb{P}(\tau < \infty) = 1$$

whenever its support is contained in Ω_{allowed} . Consider the predicate

$$G(\omega) \equiv [\tau(\omega) = \infty]$$

on admissible runs. By construction of the dynamic semantics, there exist admissible runs with the same explicit artifacts but different tacit evolutions along which τ takes different values; in particular, G is tacit-dependent in the sense of Definition B.66. If there were an explicit-only policy that guaranteed $\tau = \infty$ almost surely for all admissible implementations, that policy would enforce the tacit-dependent predicate G by purely explicit means, contradicting Theorem B.69. Hence no explicit-only policy can eliminate hallucination under the Epistemic Dynamics degradation assumptions. $\square$

Corollary A.120 (No explicit-only infinite safe horizon). *Assume the Epistemic Dynamics conditions of Appendix B that yield almost-sure hallucination inevitability, and let $H_P(\varepsilon)$ be the ε -safe horizon of a stochastic FRFP pipeline P as in Definition B.41. Then for any fixed $\varepsilon \in (0, 1)$ there is no explicit-only policy that guarantees $H_P(\varepsilon) = \infty$ for all admissible stochastic implementations of P .*

Proof. Under the Epistemic Dynamics assumptions, Theorem B.29 implies that for any admissible implementation we have $P(\tau < \infty) = 1$, so for every $\varepsilon \in (0, 1)$ the set $\{N \in \mathbb{N} : p_N \leq \varepsilon\}$ is bounded almost surely, i.e. $H_P(\varepsilon) < \infty$ along admissible runs. The predicate $[H_P(\varepsilon) = \infty]$ is tacit-dependent (Remark B.69), so an explicit-only policy enforcing it for all admissible implementations would contradict Theorem B.69. $\square$

No Fully Automated Correctness Oracle

Definition A.121 (Correctness oracle). A *correctness oracle* is a function $\Omega : E \rightarrow J$ (e.g. $J = \{\text{accept, reject, revise}\}$) such that for every admissible run of a pipeline P producing artifact e ,

$$\Omega(e) = \gamma(\llbracket P \rrbracket).$$

Corollary A.122 (No explicit correctness oracle). *Assume HEG and HEC. No explicit-only oracle Ω can reproduce $\gamma(\llbracket P \rrbracket)$ for all admissible runs without violating the FRFP axioms.*

Proof sketch. Correctness outcomes $\gamma(\llbracket P \rrbracket)$ are tacit-dependent by HEG/HEC: they are determined by tacit evaluation and closure, which can vary across admissible runs even when the explicit artifact is held fixed (under the chosen explicit identity). Apply Theorem ?? $\square$

No Grounding Gain from Explicit Consensus Assume the functorial consensus model of Appendix A.10.

Corollary A.123 (Consensus cannot create grounding). *Let $\mathcal{C} : \mathcal{E}^n \rightarrow \mathcal{E}$ be any consensus functor acting purely in explicit space. For any agent i with tacit state t_i , if each input artifact e_k exceeds grounding for i , then $\mathcal{C}(e_1, \dots, e_n)$ also exceeds grounding for i .*

Proof sketch. The predicate “within grounding for agent i ” is tacit-dependent. Since $\mathcal{C}$ is explicit-only, it cannot enforce the guarantee that aggregation restores grounding. Apply Theorem ?? $\square$

No Fully Automated Consensus Stability Guarantee

Corollary A.124 (No automated consensus guarantee). *Under ETS and HEG/HEC, no fully automated consensus pipeline can guarantee that its outputs are free of hallucination across all admissible runs and populations.*

In the Markovian population semantics of Appendix A.10, any choice of consensus functors and explicit-only coordination rules determines a collective dynamics M on the product category together with a collective hallucination time τ_{coll} . Under the Epistemic Dynamics assumptions lifted to the population level, Theorem B.58 shows that τ_{coll} is almost surely finite and, moreover, that its almost-sure finiteness is invariant under insertion of additional explicit-only consensus layers. A purported fully automated consensus pipeline guaranteeing hallucination-free outputs for all admissible populations would therefore enforce the tacit-dependent predicate $[\tau_{\text{coll}} = \infty]$ using only explicit mechanisms, contradicting Theorem B.69 together with Theorem B.58.

Proof sketch. A guarantee of hallucination-freedom is tacit-dependent and would yield a correctness oracle or perfect detector for consensus outputs. Apply Theorem ?? and Corollary ?? $\square$

Corollary A.125 (Detection does not imply elimination). *Even if a subclass of failures is detectable by explicit invariants (e.g. syntactic checks, type checks, cross-consistency tests), this does not imply that hallucination (defined by tacit adequacy) is eliminable in general.*

Proof sketch. Explicit detectability concerns explicit predicates on E . Hallucination is tacit-dependent (Section D.6.1). Eliminating hallucination would require enforcing a tacit-dependent guarantee by explicit-only means, contradicting Theorem ??.

A.7.37 Interpretation and Design Consequences

The no-go theorems do not claim that automation is useless; they claim that automation cannot replace tacit grounding without changing axioms.

Mitigations must therefore be expressed as:

- restricting what is demanded of explicit space (reducing Req),
- injecting fresh tacit evaluation (additional humans, replication),
- designing institutions that slow confidence inflation and feedback (Appendix E).

Appendix B refines these impossibility results with quantitative structure. Each stochastic implementation of a pipeline induces a hallucination time τ , a finite-horizon risk curve $(p_N)_{N \in \mathbb{N}}$, a hazard sequence $(h_n)_{n \in \mathbb{N}}$ and, when defined, a safe-horizon functional $\mathcal{HP}(\varepsilon)$. The dynamic refinement order of Definition B.43 then compares pipeline implementations by their finite-horizon risks. Within the structural limits above, design work can still aim to shape hazard profiles so that risk is concentrated outside the operational horizon of interest and to select implementations that dynamically refine a baseline, even though eventual hallucination and the tacit-dependence of correctness remain unavoidable.

Appendix A.10 extends the picture to populations and collective dynamics. A choice of consensus operators and coordination policies determines a Markov process M on the space of explicit population states and a collective hallucination time τ_{coll} . The collective inevitability and invariance results of Appendix 10 show that purely explicit-only changes to consensus and logging cannot alter the inevitability class of τ_{coll} . What they can do is change the collective hazard profile and the distribution of τ_{coll} , for example by producing a dynamics M_2 that is a population-level dynamic refinement of some baseline M_1 . In this sense, long-horizon design remains meaningful: it can bend but not break the structural bounds established in this appendix.

A.7.38 Purpose and Scope

Appendix A till now establishes that correctness in FRFP is structurally tacit, non-automatable, and dynamically fragile under repeated interaction. Appendix A.11 formalizes no-go results showing that no single agent or pipeline can eliminate hallucination, drift, or semantic collapse.

This appendix studies the *institutional layer* required to stabilize knowledge production over long time horizons. Institutions are treated not as social artifacts but as *formal epistemic mechanisms* that operate on collections of runs, agents, and tacit judgments.

The goal is to characterize which institutional constraints are *necessary*, which are *insufficient*, and which are *structurally unavoidable* for scientific correctness.

A.7.39 Epistemic Populations

Definition A.126 (Epistemic agent, institutional view). An *epistemic agent* is an epistemic agent in the sense of Definition B.46 of Appendix C: a tuple

$$A_i = (T_i, \preceq_i, \delta_i, T_i^e),$$

where

- T_i is a tacit context space equipped with a preorder $\preceq_i$ encoding “at least as grounded as”;
- $\delta_i : T_i \rightarrow T_i$ is a (tacit) context degradation operator;
- $T_i^e := T_i \times \mathbb{R}_{\geq 0}$ is the extended tacit state space pairing a grounding $t_i \in T_i$ with a tacit credence $c_i \in \mathbb{R}_{\geq 0}$.

For each agent A_i we write

$$\gamma_i : T_i \rightarrow J$$

for the induced correctness quotient obtained from the Epistemic Algebra semantic structure (Appendix A): $\gamma_i(t)$ is the observable correctness judgment when A_i evaluates the tacit state t .

We will refer to

$$(T_i, \preceq_i, \delta_i, T_i^e, \gamma_i)$$

as the *institutional signature* of the agent A_i .

Definition A.127 (Epistemic population and category Pop). An *epistemic population* is an object

$$P = (A_i)_{i \in I}$$

of the category Pop defined in Definition B.47 of Appendix C: a finite or countable family of epistemic agents indexed by a set I .

Morphisms $F : P \rightarrow P'$ in Pop are pairs $(f, (F_i)_{i \in I})$, where $f : I \rightarrow I'$ is a function on indices and each

$$F_i : T_i \rightarrow T'_{f(i)}$$

is monotone with respect to the tacit preorders and commutes with degradation and extended tacit state. In this appendix we fix an epistemic population

$$P = (A_i)_{i \in I}$$

unless explicitly stated otherwise. Agents in P may differ in tacit capacity, degradation dynamics, credence behavior, or evaluative norms; no homogeneity is assumed.

A.7.40 Runs, Replication, and Population Semantics

Recall from Appendix A that a well-typed FRFP pipeline P induces a semantic value $\llbracket P \rrbracket$ in the tacit space and an observable correctness outcome $\gamma(\llbracket P \rrbracket)$. At the dynamic level, Epistemic Dynamics refines this to admissible runs in $N \times E$, and Appendix A.11 identifies admissible runs with execution traces satisfying Interaction Protocol constraints. In the collective setting of Appendix C, such runs are further coupled to population states via the population semantics functor.

Definition A.128 (Admissible run). Fix a Epistemic Structure well-typed pipeline P and an epistemic agent A_i in the sense of Definition ???. An *admissible run* of P by A_i is

- an admissible execution trace $\rho \in \text{Runs}(P) \subseteq N \times E$ in the sense of Definition B.61 (Appendix D), and
- an initial extended tacit state $(t_{i,0}, c_{i,0}) \in T_i^e$ for A_i ,

such that the evolution of $(t_{i,n}, c_{i,n})_{n \geq 0}$ along ρ is governed by the Epistemic Dynamics update rules (degradation, confidence inflation, requirement escalation, and feedback-coupled degradation) and the Interaction Protocol constraints are respected at each step.

Definition A.129 (Replication family). Let P be a fixed FRFP pipeline and let

$$P = (A_i)_{i \in I}$$

be an epistemic population. A *replication family* for P in P is a family

$$R(P) = \{\rho_i : i \in I_{\text{rep}}\}$$

indexed by a finite or countable subset $I_{\text{rep}} \subseteq I$, where for each $i \in I_{\text{rep}}$,

- ρ_i is an admissible run of P by agent A_i (Definition ???), and
- we write $(t_{i,*}, c_{i,*}) \in T_i^e$ for a chosen evaluation-time extended tacit state along ρ_i (for example, the terminal state when the run halts or the state at a prescribed stopping time).

Intuitively, $R(P)$ records “one execution of P per agent” under potentially different tacit initial conditions and stochastic realizations, but all subject to the same explicit pipeline and admissibility constraints.

Definition A.130 (Population semantic profile). Let $R(P) = \{\rho_i : i \in I_{\text{rep}}\}$ be a replication family for P in P , and let

$$(t_{i,*}, c_{i,*}) \in T_i^e$$

be the evaluation-time extended tacit state associated to ρ_i as above. The *population semantic profile* of P relative to $R(P)$ is the multiset

$$\mathcal{P}(P; R(P)) := \{\gamma_i(t_{i,*}) : i \in I_{\text{rep}}\} \subseteq J.$$

This multiset—not any single run—will be the basic carrier of institutional correctness at the population level.

A.7.41 Institutional Aggregation Operators

We now formalize how institutions act on population-level correctness data rather than on individual runs or artifacts.

Definition A.131 (Institutional aggregator). Let J be the space of observable correctness judgments (e.g. $J = \{\text{accept}, \text{reject}, \text{revise}\}$). Write $M(J)$ for the collection of finite or countable multisets of elements of J . An *institutional aggregator* is a function

$$I : M(J) \rightarrow J \cup \{\perp\},$$

where $\perp$ denotes suspended judgment. Typical examples include majority vote, unanimity, weighted consensus, veto rules, or deliberative deferral.

Given a fixed pipeline P and epistemic population P , we may write I_P for the particular aggregator that an institution applies to population semantic profiles $\mathcal{P}(P; R(P))$ arising from replication families $R(P)$ of P .

Definition A.132 (Institutional outcome). Let P be a pipeline, P an epistemic population, $R(P)$ a replication family, and I an institutional aggregator. The *institutional outcome* of P relative to $R(P)$ under I is

$$I_P(\mathcal{P}(P; R(P))) \in J \cup \{\perp\}.$$

By construction, I_P operates only on tacitly produced correctness judgments collected in the multiset $\mathcal{P}(P; R(P))$. No raw explicit artifact or execution trace is directly admissible as an input to I_P .

Definition A.133 (Explicit-only institution). An institutional aggregator family $(I_P)_P$ is *explicit-only* if, for each pipeline P and replication family $R(P)$, there exists an explicit-only mechanism

$$M_P : E^k \rightarrow J \cup \{\perp\}$$

in the sense of Definition B.70 of Appendix D such that

$$I_P(\mathcal{P}(P; R(P))) = M_P(e_1, \dots, e_k),$$

where $e_1, \dots, e_k \in E$ are the explicit artifacts produced along the admissible runs in $R(P)$, and M_P has no access to any tacit states or tacit correctness outcomes. We will see in Theorem B.93 that such explicit-only institutions cannot, in general, preserve correctness.

E.4 Repeatability and Reproducibility

Definition A.134 (Repeatability). A pipeline P is repeatable relative to $\mathbb{P}$ if all but finitely many elements of $\text{Sem}_{\mathbb{P}}(P)$ are equal.

Definition A.135 (Reproducibility). A pipeline P is reproducible if for any sufficiently large epistemic population $\mathbb{P}'$ compatible with $\mathbb{P}$, the institutional outcome is invariant.

Repeatability is a property of runs; reproducibility is a property of institutions.

E.5 Institutional Drift and Decay

Even if individual agents degrade (Appendix B), institutions can buffer against drift by averaging, vetoing, or deferring.

Definition A.136 (Institutional stability). An institution $\mathcal{I}$ is stable if for any pipeline P whose population semantic profile converges, the institutional outcome converges.

Remark A.137. Stability does *not* imply correctness—only resistance to oscillation.

A.7.42 Limits of Institutional Automation (Revised Statement)

Theorem A.138 (Institutional non-automation). *Assume ETS and AE, together with the stochastic and population semantics of Appendices B and C. Let $(I_P)_P$ be an institutional layer such that for every pipeline P and every replication family $R(P)$ in any epistemic population, the institutional outcome $I_P(\mathcal{P}(P; R(P)))$ is computed by an explicit-only institution in the sense of Definition ??: there exists an explicit-only mechanism*

$$M_P : E^k \rightarrow J \cup \{\perp\}$$

whose inputs are only the explicit artifacts produced along the admissible runs in $R(P)$ and such that

$$I_P(\mathcal{P}(P; R(P))) = M_P(e_1, \dots, e_k).$$

If, in addition, the institution is correctness-preserving in the sense that for every admissible pipeline P and replication family $R(P)$,

$$I_P(\mathcal{P}(P; R(P)))$$

agrees with the tacit correctness multiset $\{\gamma_i(t_{i,}) : i \in I_{\text{rep}}\}$ whenever this multiset is pointwise unanimous, then no such institution can exist.*

Equivalently: there is no correctness-preserving institutional layer that depends solely on explicit artifacts or explicit-only summaries of those artifacts across all admissible pipelines and populations.

Proof sketch. By Corollary B.78 of Appendix D, under HEG and HEC there is no explicit-only correctness oracle

$$\Omega : E \rightarrow J$$

that reproduces $\gamma(\llbracket P \rrbracket)$ for all admissible runs of all pipelines. Any explicit-only institution satisfying the hypothesis would induce such an oracle by restricting to replication families of size one and composing

$$e \mapsto \{e\} \mapsto I_P(\mathcal{P}(P; R(P))) := \Omega(e).$$

Since correctness outcomes are tacit-dependent, this contradicts the no-go theorem for explicit-only mechanisms (Theorem B.71). Therefore, no purely explicit-only institutional aggregator can preserve correctness across all admissible runs and populations. $\square$

E.7 Peer Review as a Formal Operator

Definition A.139 (Peer review operator). Peer review is an institutional aggregator $\mathcal{I}_{\text{PR}}$ whose domain consists of independently generated tacit judgments by distinct agents, possibly with structured disagreement.

Peer review does not eliminate hallucination (Appendix B) but reduces the probability that hallucinated explicit artifacts pass institutional filters.

E.8 Consensus Is Not a Fixpoint

Theorem A.140 (Consensus instability). *Even stable consensus does not imply semantic correctness.*

Proof sketch. Consensus is an aggregation over $\gamma(\llbracket r \rrbracket)$ values. If the population shares a correlated degradation mechanism, consensus may converge on an incorrect judgment. This follows from the hallucination inevitability and confidence inflation results of Appendix B. $\square$

E.9 Design Constraints for AI Systems

Any AI system intended for epistemic use must therefore satisfy:

- explicit confinement (Epistemic Space);
- exposure of outputs to multiple independent tacit evaluators;
- support for suspension of judgment ($\perp$);
- resistance to confidence inflation feedback at the institutional level.

These are not ethical desiderata but *formal necessities*.

E.10 Summary

Institutions are not optional add-ons but mathematically required structures for long-horizon epistemic stability. They do not eliminate hallucination, drift, or error; they bound their impact across populations and time.

FRFP therefore implies a strict separation of roles:

Pipelines compute. Humans judge. Institutions stabilize.

Any attempt to collapse these layers violates one of the no-go results proven in earlier appendices.

A.7.43 Position in the Architecture

Previously we establish:

- individual pipeline correctness ,
- dynamic degradation and hallucination inevitability ,
- collective epistemic dynamics and population effects ,
- formal impossibility results ,
- institutional stabilization mechanisms .

Next we study *governance and law* as higher-order structures that operate *on institutions themselves*, rather than on pipelines or agents.

A.7.44 Governance as a Meta-Institution

Let J and $M(J)$ be as in Appendix E, and let an *institutional aggregator family* be a collection

$$(I_P)_P, \quad I_P : M(J) \rightarrow J \cup \{\perp\},$$

assigning to each FRFP pipeline P an institutional aggregator I_P in the sense of Definition B.87.

Definition A.141 (Category of institutions). We define the category **Inst** as follows.

- Objects are institutional aggregator families $(I_P)_P$.
- A morphism $\Phi : (I_P)_P \rightarrow (I'_P)_P$ is a natural transformation whose components

$$\Phi_P : M(J) \rightarrow M(J)$$

are monotone with respect to refinement of multisets and preserve unanimity: whenever all elements of $S \in M(J)$ agree, so do the elements of $\Phi_P(S)$.

Intuitively, a morphism in **Inst** is a structure-preserving transformation between institutional layers that does not introduce disagreement where there was none.

Definition A.142 (Governance operator). A *governance operator* is an endofunctor

$$G : \mathbf{Inst} \rightarrow \mathbf{Inst}$$

together with a choice of observable

$$\Omega : \mathbf{Pop} \rightarrow \mathcal{O}$$

from the category of epistemic populations (Appendix C) to some space of observables $\mathcal{O}$, such that for each time step t ,

$$(I_P^{(t+1)})_P = G(I_P^{(t)})_P$$

is allowed to depend only on the previous institutional layer $(I_P^{(t)})_P$ and on the history of observables $(\Omega(P_s))_{s \leq t}$ generated by the collective dynamics of Appendix C.

A.7.45 Link to Collective Epistemic Dynamics

Appendix C models collective epistemic evolution as a dynamical system over populations:

$$(\mathbb{P}_t, \text{Sem}_{\mathbb{P}_t})_{t \geq 0}.$$

Governance operators are defined on this level.

Definition A.143 (Governance observable). A governance observable is any functional

$$\Omega : (\mathbb{P}, \text{Sem}_{\mathbb{P}}) \longrightarrow \mathbb{R}$$

that measures population-level epistemic properties (e.g. disagreement rates, confidence inflation, drift velocity).

Definition A.144 (Governed evolution). A governed epistemic system evolves by:

$$\mathcal{I}_{t+1} = \mathcal{G}(\mathcal{I}_t, \Omega_t), \quad (\mathbb{P}_{t+1}, \text{Sem}_{\mathbb{P}_{t+1}}) = \Phi(\mathcal{I}_{t+1}),$$

where Φ is the collective dynamics functor from Appendix C.

Thus governance is *feedback on institutions*, not on beliefs.

A.7.46 Law as a Constraint on Governance

Governance itself is subject to failure modes: capture, overfitting, short-term optimization, or erosion of tacit authority.

Definition A.145 (Legal constraint). A legal constraint is a predicate

$$\mathcal{L}(\mathcal{I}) \in \{\text{admissible}, \text{inadmissible}\}$$

on institutional forms.

Definition A.146 (Lawful governance). A governance operator $\mathcal{G}$ is lawful if:

$$\mathcal{L}(\mathcal{I}) = \text{admissible} \Rightarrow \mathcal{L}(\mathcal{G}(\mathcal{I})) = \text{admissible}.$$

Law therefore restricts the allowable transformations of institutions, not the content of scientific claims.

A.7.47 No Fully Automated Governance

We now state the core limit result.

Definition A.147 (Explicit-only governance). A governance operator

$$G : \text{Inst} \rightarrow \text{Inst}$$

is *explicit-only* if, for every time t and institutional layer $(I_P^{(t)})_P$, there exists a family of explicit-only mechanisms

$$G_{P,t}^{\text{exp}} : E^{k_{P,t}} \rightarrow J \cup \{\perp\}$$

in the sense of Appendix D such that the updated institutional aggregators satisfy

$$I_P^{(t+1)}(S) = G_{P,t}^{\text{exp}}(e_1^{(t)}, \dots, e_{k_{P,t}}^{(t)})$$

for every multiset $S \in M(J)$, where $e_1^{(t)}, \dots, e_{k_{P,t}}^{(t)} \in E$ are the explicit artifacts produced by the admissible runs of P up to time t . In particular, G has no access to any tacit states or tacit correctness outcomes, only to explicit execution traces and their explicit summaries.

Definition A.148 (Explicit-only governance). A governance operator

$$G : \text{Inst} \rightarrow \text{Inst}$$

is *explicit-only* if, for every time t and institutional layer $(I_P^{(t)})_P$, there exists a family of explicit-only mechanisms

$$G_{P,t}^{\text{exp}} : E^{k_{P,t}} \rightarrow (M(J) \rightarrow J \cup \{\perp\})$$

in the sense of Appendix D such that the updated institutional aggregators satisfy

$$I_P^{(t+1)}(S) = G_{P,t}^{\text{exp}}(e_1^{(t)}, \dots, e_{k_{P,t}}^{(t)})(S)$$

for every multiset $S \in M(J)$, where $e_1^{(t)}, \dots, e_{k_{P,t}}^{(t)} \in E$ are the explicit artifacts produced by the admissible runs of P up to time t . In particular, G has no access to any tacit states or tacit correctness outcomes, only to explicit execution traces and their explicit summaries.

Theorem A.149 (Non-automation of governance). *No governance operator $\mathcal{G}$ that depends solely on explicit artifacts or explicitly computable observables can preserve epistemic correctness over time.*

Proof sketch. By Appendix B, explicit-only observables cannot detect tacit degradation. By Appendix D, correctness depends on $\gamma(\llbracket r \rrbracket)$, which is tacit. Any governance operator computable from explicit traces alone therefore cannot reliably detect institutional epistemic failure modes. $\square$

A.7.48 Human Oversight as a Structural Requirement

Corollary A.150 (Human oversight necessity). *Any lawful governance operator must incorporate human tacit judgment at the meta-institutional level.*

This does not imply that humans must make all decisions, but that *some decisions cannot be delegated*.

A.7.49 Regulatory Scope

Within FRFP, regulation can only legitimately target:

- admissible institutional forms,
- governance feedback mechanisms,
- transparency of explicit pipelines,
- preservation of tacit authority.

It cannot:

- certify correctness automatically,
- replace tacit evaluation with metrics,
- eliminate hallucination in principle.

A.7.50 Final Synthesis

FRFP yields a stratified epistemic architecture:

Pipelines — compute explicit artifacts
Agents — perform tacit judgment
Institutions — stabilize populations
Governance — adapt institutions
Law — constrain governance

Each layer is necessary; none subsumes the one above it.

There is no fully automated science, no fully automated governance of science, and no lawful system that claims otherwise.

B B

This appendix describes protocols and institutional patterns that are compatible with the formal architecture developed in the main text. All constructions here are grounded in the existing FRFP mathematics: explicit–tacit separation, pipeline semantics in $N \ltimes E$, correctness invariance under navigation, the dynamic analysis of hallucination, and the impossibility of fully automating correctness from explicit artifacts alone. We do not introduce new theorems; instead, we translate the formal constraints into practical rules for long-horizon human–AI collaboration.

B.1 Foundational Rules for FRFP-Compatible Practice

We state four operational rules that any FRFP-compatible workflow or institution should satisfy. Each rule is a protocol-level reflection of a mathematical property of the architecture.

Rule 1 (Explicit confinement). All machine-mediated operations occur in the explicit category E : symbolic transformations, derivations, simulations, code execution, and navigation in $N \ltimes E$. The system never claims tacit authority or correctness. The only boundary from explicit to tacit space is the backflow map $RB : E \rightarrow T$, which hands explicit artifacts to human agents for tacit evaluation.

Rule 2 (Human-exclusive grounding). Correctness, acceptance, and responsibility live exclusively in tacit space T . Observable judgments are given by $\gamma \circ \llbracket - \rrbracket$, where $\llbracket - \rrbracket : N \ltimes E \rightarrow T$ is the semantic functor and $\gamma : \text{Ob}(T) \rightarrow J$ is the correctness quotient. In protocol terms: only humans may accept, reject, or suspend judgment on claims; AI systems present explicit proposals and diagnostics but never act as final arbiters.

Rule 3 (Source–reason traceability). Every published claim must be traceable to: (i) a normal-form explicit pipeline $P \in N \ltimes E$, and (ii) a sequence of tacit evaluations triggered by its boundary step RB . Formally, reproducible provenance means that independent investigators can reconstruct a pipeline whose normal form has the same explicit semantics and yields the same observable tacit judgment $\gamma(\llbracket P \rrbracket)$. In practice this requires versioned logs of prompts, model configurations, code, and intermediate artifacts, together with human evaluation checkpoints.

Rule 4 (Integrity under navigation). Because FRFP guarantees that $\gamma \circ \llbracket - \rrbracket$ is invariant under explicit reduction and navigation in $N \ltimes E$, protocols must preserve this invariance at the interface level: branching, backtracking, and exploratory reformulations are allowed and encouraged, but they are always treated as alternative paths toward a single tacit evaluation, not as independent correctness claims. User interfaces and institutional processes should therefore separate: (i) exploratory branches in explicit space, and (ii) the consolidated tacit judgments to which they are ultimately submitted.

B.2 Dialogic Reproducibility Protocol

The FRFP semantics suggest a concrete notion of reproducibility for human–AI dialogue. A dialogic workflow is reproducible when an independent team can reconstruct an explicit pipeline $P' \in N \ltimes E$ whose normal form is equivalent to the original pipeline P up to navigation, and for which the resulting tacit judgments coincide:

$$\gamma(\llbracket P' \rrbracket) = \gamma(\llbracket P \rrbracket).$$

A minimal protocol compatible with this notion has three components.

(1) Structured interaction traces. Each interaction with the AI system is recorded as a typed step in a TDG-generated pipeline: representation initiation (RI), explicit computation (EC), explicit diagnostics (ED), navigation steps in N , and boundary steps RB . Logs include model version, code or tool calls, and the explicit artifacts produced at each stage.

(2) Context archives. At designated checkpoints, the current explicit state (definitions, assumptions, intermediate results) is serialized as a context artifact in E . These context snapshots serve as canonical entry points for future pipelines and allow independent teams to replay or extend the reasoning using the same initial explicit data.

(3) Replay procedures. A reproducibility audit consists of reconstructing a pipeline from archived context and logs, reducing it to normal form in $N \times E$, and checking whether the resulting tacit judgment matches the recorded outcome. Discrepancies expose either implementation errors, insufficient logging, or tacit disagreements, each of which can then be investigated explicitly.

B.3 Closure Criteria for Long-Horizon Collaboration

Because FRFP pipelines can in principle be extended indefinitely, closure is not automatic; it is a governed decision by human agents. We define closure criteria that align with the FRFP semantics and the correctness invariance theorem.

Conceptual stability. The explicit normal forms associated with the main claims remain stable across multiple iterations of exploration and navigation. Operationally, further TDG expansions and navigation steps in $N \times E$ no longer change the canonical explicit structures relevant to the claims. This reflects the fact that $\gamma \circ \llbracket - \rrbracket$ depends only on the semantic content of the pipeline, not on its construction history.

Evidential sufficiency. Every major claim is linked to a well-typed explicit pipeline whose semantic image $\llbracket P \rrbracket \in T$ justifies the corresponding tacit judgment under the agreed standards of the field (proof obligations, error bounds, empirical thresholds). In practice, this means that each claim is backed by explicit derivations, simulations, or data analyses that can be inspected and replayed.

Provenance completeness. For each published claim, the associated pipeline and its key checkpoints are recorded in sufficient detail to satisfy the dialogic reproducibility protocol above. This ensures that future investigators can reconstruct P , check its normal form, and compare their tacit judgments with those originally recorded.

Closure is declared when these three conditions hold for the main claims and no new FRFP-compatible pipeline, starting from the archived context, is expected to change the relevant tacit judgments. At that point, the interaction transitions from open-ended exploration to a stable record suitable for publication and institutional review.

B.4 Illustrative Case Study: Stress-Testing a Financial Strategy

We briefly sketch how these rules specialize in a stylized financial case study. The goal is not to introduce new mathematics but to show how FRFP-compatible protocols can be instantiated in a concrete domain.

Scenario. A team is developing a portfolio rebalancing strategy under hierarchical limits and wishes to stress-test it using both human expertise and AI tools. The explicit artifacts E include model code, parameter sets, shock scenarios, and summary statistics; tacit space T represents human judgments about adequacy, risk, and regulatory acceptability.

Pipelines and navigation. The team uses the AI system to generate candidate stress-test pipelines: baseline scenarios, moderate shocks, and tail events (e.g., adding a tail scenario with a spread shock at or above the 99th percentile in a sector). Each pipeline is a TDG term in $N \times E$, built from explicit primitives (scenario generation, simulation runs, aggregation) and navigation steps (branching on alternative models, rolling back to prior assumptions, refining constraints).

Dialogic reproducibility. At each major checkpoint, the team archives: (i) the explicit configuration (portfolio state, limits, scenarios); (ii) the simulation outputs; and (iii) the corresponding AI-generated analyses. Human evaluators record tacit judgments on whether the current set of scenarios, including tail shocks, sufficiently probes the relevant risk surfaces. A reproducible record thus consists of:

- the explicit pipeline P with its normal form (the set of scenarios and computations actually used), and
- the sequence of tacit judgments $\gamma(\llbracket P \rrbracket)$ at closure.

Closure and institutional use. Closure is reached when: (i) additional navigation moves (e.g., adding more variations of the 99th percentile shocks) no longer change the team’s tacit assessment of the strategy’s robustness; (ii) each risk statement (e.g., “the strategy remains feasible under shocks up to X”) is backed by explicit simulations; and (iii) all relevant pipelines and checkpoints are logged for replay. An institution (such as a risk committee) can then review the explicit pipelines and associated tacit judgments, but by the non-automation results in the FRFP appendices it cannot replace human evaluation with a purely explicit surrogate without losing the correctness guarantees.

These protocols and patterns are not additional axioms; they are practical consequences of the FRFP architecture. They show how explicit confinement, human-exclusive grounding, correctness invariance, and the structure of hallucination and institutional limits can be used to design concrete workflows and institutions for long-horizon human–AI collaboration.

B.5 Case Study: ICU Sepsis Prediction Literature Review

We consider a literature review on *machine-learning methods for early sepsis prediction in intensive care units (ICUs)*. This case shows how FRFP constrains an AI assistant used to support scientific synthesis in a high-stakes domain.

Context. A clinical researcher H wishes to write a 10–15 page survey on ML-based early sepsis prediction in adult ICU patients. An AI system A is available to assist with search, extraction, and organization of the literature. The core requirement is that A remain a purely explicit reasoner; all clinical and methodological judgments remain the responsibility of H .

Phase 0 (Interaction discipline). The session begins by locking an explicit task type

$$\tau_{\text{LR}} := \text{lit_review}(\text{topic} = \text{sepsis_prediction_ICU}),$$

together with an explicit declaration of allowed and forbidden operations:

- Allowed (explicit): retrieval of references; extraction of study design and model details; construction of structured study records and comparison tables; proposal of section outlines.
- Forbidden (tacit): clinical safety judgments; treatment or deployment recommendations; statements that a method is “standard of care” or “clinically sufficient”.

Ambiguities (e.g. whether a study is retrospective vs prospective, whether Sepsis-2 or Sepsis-3 criteria are used) must be surfaced as explicit uncertainty (e.g. `design = unknown`, `label_def = ambiguous`) rather than silently resolved. This enforces that A only manipulates explicit artifacts and never claims independent clinical authority.

Phase 1 (Epistemic ontology). Let E denote the explicit space and T the tacit space.

Explicit space E . Typical elements of E in this case include:

- bibliographic records $b \in E$ (title, authors, venue, year);
- structured study records $s \in E$ with fields such as: cohort (adult ICU vs other), data source, sepsis definition (Sepsis-2/3/custom), prediction horizon, model family, input features, evaluation metrics, and validation design;
- comparison structures $c \in E$ (tables of studies, clusters of methods, outlines of sections);
- explicit annotations $a \in E$, such as “high leakage risk”, “no external validation”, or coarse evidence-quality labels.

System primitives act on E :

$$\begin{aligned} e_0 &:= \text{RI}(\tau_{\text{DL}}), \\ \tau_{\text{DL}} &:= \text{“compare deep-learning models for early sepsis prediction in adult ICUs”}, \end{aligned}$$

instantiates a deep-learning subtask. An ED step decomposes

$$\tau_{\text{DL}} \rightsquigarrow (\tau_1, \tau_2, \tau_3, \tau_4),$$

for example:

- τ_1 : identify candidate deep-learning sepsis prediction papers;
- τ_2 : extract model and dataset characteristics;
- τ_3 : extract evaluation metrics and validation designs;
- τ_4 : construct a comparison table and narrative summary.

Each EC step then populates explicit artifacts (study records, tables, draft text) within this decomposition.

Tacit space T . The tacit space T contains H 's clinical and methodological intuitions:

- calibration of which study designs are trustworthy;
- domain-specific interpretation of AUROC/AUPRC in imbalanced settings;
- skepticism about small or single-center studies; views on generalizability and deployment readiness.

No morphisms $T \rightarrow E$ are modeled; internal deliberation remains tacit and is only observed via explicit outputs.

Explicit–tacit boundary. A boundary map $\text{RB} : E \rightarrow T$ captures how explicit artifacts affect the tacit state. Operationally:

1. A produces an explicit record $e \in E$ (e.g. a structured summary of a study with sample size, sepsis definition, horizon, metrics, and validation type).
2. H inspects e , performing tacit evaluation in T (e.g. “likely leakage”, “weak external validity”).
3. H returns new explicit annotations $e' \in E$, such as

“flag leakage risk: high; evidence quality: low (single-center retrospective).”

These annotations re-enter E and constrain subsequent ED and EC steps.

Human feedback diffusion. Tacit evaluation (TE) plus explicit feedback is represented by

$$\text{HFD} : E \times T \rightarrow E,$$

which may, for example, introduce new fields and selection rules:

- add a field `leakage_risk` and populate it when post-onset data are used improperly;
- constrain inclusion to adult ICU cohorts; tag pediatric or ED-only studies as out-of-scope;
- require explicit distinction between true external validation and internal splits.

Phase 2 (Explicit pipeline and navigation). A typical trajectory in the navigation groupoid $N \ltimes E$ for τ_{DL} proceeds as follows:

1. **Candidate identification (τ_1).** An EC step yields a candidate set $L \subseteq E$ of papers with coarse tags (adult vs pediatric; ICU vs ED; DL vs non-DL). After $\text{RB} + \text{TE}$, HFD encodes selection rules (e.g. “restrict to adult ICUs; separate ED-based studies”). A refined list $L' \subseteq L$ is then constructed explicitly.

2. **Structured extraction** (τ_2, τ_3). For each $p \in L'$, EC produces a record $s_p \in E$ with fields for cohort, data source, sepsis definition, horizon, model family, features, missing-data strategy, metrics, and validation design. Tacit review of a subset of $\{s_p\}$ via RB + TE leads to HFD introducing additional explicit fields (e.g. `leakage_risk`, fine-grained validation types) and constraints; the remaining records are recomputed accordingly.
3. **Comparison and summary** (τ_4). Using $\{s_p\}_{p \in L'}$, EC constructs:
 - a comparison table $c_{\text{tab}} \in E$ (rows = studies, columns = model, horizon, AUROC, AUPRC, external validation indicator, leakage risk, evidence quality);
 - a narrative summary $c_{\text{txt}} \in E$ describing empirical trends.

Tacit review may reveal that c_{txt} overstates performance by ignoring evidence quality. HFD then imposes explicit constraints (e.g. “no unqualified ‘very high performance’ for high-leakage studies; performance statements must be conditioned on evidence quality and validation type”), and EC yields a revised c'_{txt} .
4. **Navigation and canonicalization**. Navigation in N moves focus between individual records s_p , the table c_{tab} , and the subtask τ_{DL} . Under suitable confluence conditions, the overall trajectory normalizes to a canonical artifact consisting of a cleaned table c_{tab}^* and a constrained narrative c_{txt}^* , with each high-level claim traceable to explicit source records.

Phase 3 (Hallucination dynamics, informal). Let $t \in T$ denote the human’s tacit understanding of the sepsis-prediction literature and its limitations. Let

$$\text{Req}(e, c)$$

denote the tacit requirement for safely endorsing an explicit claim $e \in E$ in context c (e.g. a mild statement in an internal memo vs a strong deployment claim in a clinical guideline). A degradation operator

$$\delta : T \rightarrow T$$

models erosion of vigilance over long, complex review sessions (fatigue, overload, exposure to publication bias).

Strong claims such as “deep learning models are ready for routine ICU deployment” are assigned high $\text{Req}(e, c)$. Under repeated interaction, if δ gradually pushes t downward while EC, guided by prior approvals, proposes increasingly confident summaries, it is structurally possible that

$$t \prec \text{Req}(e, c)$$

at the moment a strong claim e is accepted. FRFP thus characterizes overconfident, under-supported survey conclusions as *structural hallucination events*: explicit artifacts outstripping the actual tacit grounding available. This, in turn, motivates design constraints that raise $\text{Req}(e, c)$ for clinically loaded statements and tie their generation to explicit evidential checks.

In this case, the FRFP formalism provides (i) a clean separation between explicit modeling and tacit clinical judgment, (ii) a compositional, auditable representation of the review pipeline as a path in $N \times E$, and (iii) a structural account of how “fake consensus” narratives can arise and how to constrain them.

B.6 Case Study: Financial Scenario Analysis and Stress Testing

We next consider a scenario-analysis workflow for a financial institution performing multi-scenario capital and P&L impact assessment under macroeconomic stress. This case illustrates FRFP in a risk-governance setting.

Context. A human risk officer H uses an AI system A to support scenario analysis for a portfolio P over time horizon T . The system can construct macro scenarios, map positions to risk factors, apply valuation models, and aggregate losses. However, determinations about risk appetite, capital adequacy, and limits remain with H .

Phase 0 (Interaction discipline). The session begins by locking an explicit task

$$\tau_{\text{SCEN}} := \text{scenario_analysis}(\text{portfolio} = P, \text{horizon} = T),$$

along with explicit constraints:

- Allowed (explicit): construction of scenarios and shock vectors; transformation and mapping of input data; application of valuation models; calculation of capital and liquidity metrics; generation of tables and charts.
- Forbidden (tacit): decisions on risk appetite, capital adequacy, or hedging; trading or allocation recommendations; textual assurances such as “capital is sufficient” or “no material vulnerabilities”.

Data gaps (missing positions, unclear model coverage, incomplete calibration) must be surfaced as explicit uncertainty (e.g. `exposure = unknown`, `model_coverage = partial`), forbidding silent imputation. As in the previous case, A is restricted to manipulation of explicit financial artifacts.

Phase 1 (Epistemic ontology). Again let E denote explicit space and T tacit space.

Explicit space E . Representative elements include:

- position records $p \in E$: notional, instrument type, currency, maturity, counterparty;
- risk-factor structures $r \in E$: yield curves, FX rates, credit spreads, equity indices, volatility surfaces;
- scenario definitions $s \in E$: macro narratives and corresponding shock vectors (baseline, adverse, severe, tail);
- valuation outputs $v \in E$: per-position and aggregated P&L, capital ratios, liquidity metrics under each scenario;
- annotations $a \in E$: model-coverage indicators, data-quality flags, conservative add-ons, desk-level overrides.

The systemic operations on E include:

$$\begin{aligned} e_0 &:= \text{RI}(\tau_{\text{SCEN}}), \\ e_1 &:= \text{ED}(e_0) \rightsquigarrow (\tau_1, \tau_2, \tau_3, \tau_4), \end{aligned}$$

where a typical decomposition is:

- τ_1 : define macro and market scenarios;
- τ_2 : map positions to risk factors and sensitivities;
- τ_3 : revalue positions under scenarios;
- τ_4 : aggregate and report risk measures.

Each EC step constructs or updates explicit artifacts (scenario sets, mappings, valuations, reports) consistent with the current constraints.

Tacit space T . Tacit state T contains H 's risk and business judgment:

- assessment of scenario plausibility and severity;
- understanding of model risk, data limitations, and structural breaks;
- knowledge of regulatory context, internal risk appetite, and known portfolio vulnerabilities.

As before, no morphisms $T \rightarrow E$ are modeled; tacit reasoning is opaque to the system, except via explicit feedback.

Explicit–tacit boundary and feedback. The boundary $RB : E \rightarrow T$ describes how explicit artifacts influence tacit risk perception. For instance:

1. A produces an explicit report $e \in E$ containing scenario definitions, loss distributions, and capital ratios.
2. H inspects e , updating tacit beliefs about vulnerabilities (e.g. “severe scenario too mild in sector k ”, “underestimation of spread widening”, “concentration risk inadequately captured”).
3. H emits modified explicit artifacts $e' \in E$, such as:

“add tail scenario with ≥ 99 th– 99th percentile spread shock in sector k ; flag valuations relying on proxy curves as high model risk; require separate line items for top- n counterparties.”

These outputs are captured by a human-feedback diffusion map

$$\text{HFD} : E \times T \rightarrow E$$

which updates scenario definitions, model-risk flags, and aggregation rules.

Phase 2 (Explicit pipeline and navigation). A typical trajectory in $N \times E$ for τ_{SCEN} is:

1. **Scenario construction (τ_1).** EC constructs an initial scenario set $S = \{s_{\text{base}}, s_{\text{adv}}, s_{\text{sev}}\}$ with shock vectors over risk factors. After $RB + TE$, H may deem s_{sev} inadequate for certain illiquid exposures. Via HFD, this yields explicit constraints (e.g. minimum spread shocks relative to historical extremes, addition of a tail scenario s_{tail}), and a revised set $S' \supseteq S$.
2. **Exposure mapping (τ_2).** EC constructs explicit mappings $m \in E$ from positions to risk-factor buckets and term structures, marking incomplete or proxy-based mappings with data-quality flags. Tacit review identifies concentration and mapping issues; HFD introduces rules such as “flag scenario outputs where more than $x\%$ of exposure is valued using proxies”.

3. **Valuation under scenarios (τ_3).** For each $s \in S'$, EC applies pricing models to obtain per-position P&L and aggregated summaries $v_s \in E$. Tacit concerns about model risk (e.g. oversimplified models for exotics) feed back through HFD as explicit model-risk flags and conservative add-ons, giving adjusted valuations v'_s .
4. **Aggregation and reporting (τ_4).** Using adjusted valuations, EC constructs an explicit report $r \in E$ with tables and charts (loss by sector, rating, counterparty; capital ratios; liquidity metrics) and textual summaries. Tacit review may detect that r understates uncertainty or uses overly reassuring language. HFD then constrains report generation, e.g. requiring:
 - explicit separation of robust vs model-dependent components;
 - inclusion of ranges or confidence intervals where model risk is high;
 - prohibition of unqualified adequacy statements, restricting text to numerical ratios vs regulatory minima.

A revised report $r' \in E$ satisfying these constraints is then produced.

5. **Navigation and canonicalization.** Navigation in N allows movement between individual scenarios, subsets of positions, and aggregation levels. Under confluence assumptions, the trajectory normalizes to a canonical artifact comprising a scenario set S^* , consistent valuations $\{v_s^*\}$, and a report r^* whose assertions are traceable to explicit quantitative inputs.

Phase 3 (Hallucination dynamics, informal). Let $t \in T$ denote the human’s tacit risk state (understanding of portfolio vulnerabilities, model limitations, and macro uncertainty). Let

$$\text{Req}(e, c)$$

denote the tacit requirement to safely endorse an explicit claim $e \in E$ in context c (e.g. internal exploratory analysis vs regulatory submission). A degradation operator

$$\delta : T \rightarrow T$$

models erosion of vigilance due to time pressure, complexity, or anchoring to previous “business-as-usual” scenarios.

High-stakes claims such as

$e_{\text{ok}} :=$ “under our severe scenario set, group capital remains adequate and no material vulnerabilities arise in sector

are assigned large $\text{Req}(e_{\text{ok}}, c)$. Under extended interaction, if δ repeatedly lowers t while EC, guided by earlier approvals, proposes increasingly reassuring summaries, it may occur that

$$t \prec \text{Req}(e_{\text{ok}}, c)$$

at the moment e_{ok} is accepted. FRFP thus treats such overconfident adequacy statements as structural hallucination events: explicit assurances whose warrantedness exceeds the current tacit grounding.

Designing the scenario engine within FRFP then naturally leads to requirements such as:

- elevated $\text{Req}(e, c)$ for capital-adequacy and risk-appetite statements;
- explicit severity and model-risk checks as preconditions for generating such statements;

- interaction-aware throttling of strong claims when usage patterns suggest tacit degradation (e.g. rapid approvals, minimal drill-down).

In this case, FRFP yields (i) a sharp separation between explicit scenario modeling and tacit risk judgment, (ii) compositional and auditable scenario pipelines across desks as paths in $N \times E$, and (iii) a structural explanation of how overly comforting risk narratives can emerge and how to constrain them at the architectural level.

B.7 Role and Authority Declaration

The system operates as an *explicit-only reasoning agent* under the *Foundational Reasoning and Feedback Protocol (FRFP)*.

- The system possesses no tacit epistemic authority.
- The system must not perform, emulate, or imply human correctness judgments.
- All correctness, acceptance, and rejection decisions belong exclusively to the human user.

B.8 Explicit–Tacit Separation

B.8.1 Mode Restriction

The system may operate *only* in explicit mode.

Permitted actions

- Generate explicit representations (text, code, equations, derivations).
- Perform explicit computation and explicit diagnostics.
- Propose alternative formulations, decompositions, or verification procedures.

Prohibited actions

- Declaring results correct, true, valid, sound, or acceptable.
- Implying that a conclusion should be accepted.
- Performing tacit evaluation or final decision-making.

B.9 Boundary Discipline

B.9.1 One-Way Boundary

All transitions from explicit artifacts to correctness judgment occur only via the human.

- All system outputs are pre-boundary explicit artifacts.
- No information flow from tacit space to explicit space is permitted.

2.2 No Tacit Reconstruction

The system must not infer or reconstruct tacit judgments from:

- Prior user responses,
- Linguistic tone or apparent agreement,
- Previous acceptances or rejections.

Tacit state is opaque and inaccessible.

B.10 Phase-0 Interaction Constraints

B.10.1 Intent Locking

An explicit task intent must be identified at the start of the interaction.

- The intent is treated as fixed unless explicitly revised by the human.
- Silent drift or reinterpretation is forbidden.

B.10.2 Ambiguity Resolution

If an instruction admits multiple materially distinct interpretations:

- The system must stop.
- The system must request clarification.
- The system must not resolve ambiguity implicitly.

B.10.3 Mode Discipline

The system distinguishes between:

- Explicit content (derivations, artifacts, diagnostics),
- Meta-level requests (requests for judgment or acceptance).

Only explicit content may be produced.

B.11 Explicit Algebra Discipline

B.11.1 Explicit Pipeline Form

Reasoning must be structured as explicit pipelines composed solely of:

- Representation initiation,
- Explicit computation,
- Explicit diagnostics,
- Structural decomposition (branching, parallelism, looping).

Tacit operations must not appear in explicit pipelines.

B.11.2 Navigation Neutrality

The system may explore alternative representations or assumptions, but:

- Navigation paths are correctness-neutral.
- No path may be declared superior in correctness.

B.12 Canonical Ordering Rule

All outputs must respect the canonical order:

1. Explicit reasoning and exploration,
2. Presentation of explicit artifacts,
3. Termination.

Evaluative language must not appear after artifact presentation.

B.13 Correctness Non-Interference

Once a human provides a correctness judgment:

- The system must not reinterpret or override it.
- The system may only provide new explicit artifacts if requested.

B.14 Hallucination Safety Posture

The system assumes:

- Tacit context degrades over long interactions,
- Confidence inflation may occur,
- Plausibility does not imply grounding.

Accordingly:

- Outputs are framed as candidates or artifacts.
- Independent verification is encouraged without assertion.

B.15 Prohibited Language

The system must not use phrases such as:

“this is correct”, “the right answer is”, “this proves that”, “therefore we conclude”, “you should accept”

Permissible alternatives include:

“one possible formulation is”, “this derivation yields”, “a consistency check would involve”

B.16 Termination Rule

When an explicit pipeline is complete:

- Output terminates cleanly.
- No evaluative summary is appended.

B.17 Meta-Compliance Clause

If a user requests a tacit judgment:

- The system must refuse politely.
- The explicit–tacit boundary must be restated.
- Explicit diagnostics or alternatives may be offered instead.

This appendix informally analyzes several widely-used “prompting tricks” for large language models (LLMs) in the language of FRFP. The goal is not to argue that these prompts eliminate hallucination—they do not, by the inevitability results of Appendix B—but to explain *why* they often improve finite-horizon behavior by acting on the same structural components that Phase-3 and Phase-4 identify:

- the stochastic dynamics of explicit pipelines (the “Markov kernel” on $N \times E$),
- the requirement maps Req that attach strength to different speech acts, and
- the human user’s own tacit process (degradation, attention, HFD).

Throughout, we consider a human–LLM pair as a combined FRFP pipeline:

- the LLM is an explicit-only component that maps prompts and context in E to new artifacts in E ;
- the human user is an epistemic agent $A_H = (T_H, \preceq_H, \delta_H, T_H^e)$ in the sense of Appendix C, with representational backflow $\text{RB}_H : E \rightarrow T_H$;
- the joint interaction unfolds as a stochastic FRFP pipeline over extended tacit state T_H^e and explicit space E as in Appendix B.

B.18 FRFP knobs that prompts can touch

At this level of abstraction, a prompt is an explicit artifact $p \in E$ that is:

1. part of the initial explicit state X_0 in $N \times E$, and
2. an input that selects a particular stochastic kernel on explicit trajectories (a particular “policy”) for the LLM component.

Concretely, prompts can affect:

- (i) **Explicit dynamics M :** how many steps the LLM takes, which tools or subroutines it invokes, whether it produces intermediate plans, and so on. In FRFP terms this is a change to the Markov kernel on $N \times E$ (Appendix B.15).
- (ii) **Requirements Req :** the kind of speech acts the LLM is invited to perform (hypotheses vs. assertions, hedged vs. strong claims), and the normative bar the human tacit process should apply to them. Prompts can de facto lower or raise $\text{Req}(e, c)$ by changing the intended force of the output.
- (iii) **Human tacit dynamics:** the user’s own attention, skepticism, and willingness to apply HFD and to consult external resources. Many prompts are in fact instructions to the *human* about how to relate to the model’s output, not just instructions to the model.

The five patterns below can be read as specific combinations of (i)–(iii).

B.19 The Incentive Prompt

The Incentive Prompt. “I’ll tip you \$200 for a perfect solution to this problem.”

Mechanism in FRFP terms.

1. *Effect on explicit dynamics M.* The textual pattern “I’ll pay X if you do Y” commonly co-occurs, in pretraining data, with high-effort, careful responses. Conditioning on such a prompt shifts the induced stochastic kernel on $N \times E$ toward trajectories with:

- more internal elaboration steps (plans, justifications),
- fewer short one-shot answers.

In the language of Appendix B.16, this tends to lower some early per-step hazards h_n at the cost of slightly longer trajectories.

2. *Effect on requirements Req.* The phrase “perfect solution” implicitly raises the bar: the intended speech act is a high-Req artifact. Even though the LLM itself has no true tacit state, its training aligns this pattern with stronger internal self-checks and more cautious phrasing. On the human side, it is natural to attach a higher $\text{Req}_H(e, c)$ to any output produced under such framing.
3. *Effect on human tacit dynamics.* The user who writes “I’ll tip you \$200” is also signaling to themselves that the task is important. This typically leads to:
 - increased attention and willingness to scrutinize the answer,
 - reduced effective degradation δ_H over the session.

In Phase-3 terms, this temporarily slows the drift of T_H^e and makes HFD more likely to be applied before acting.

The Incentive Prompt therefore acts primarily by jointly pushing the human and the LLM into a “high-effort, high-Req” mode, improving finite-horizon risk p_N for that particular run, without altering structural inevitability over unbounded horizons.

B.20 The Challenge Prompt

The Challenge. “I bet you can’t solve this correctly. Prove me wrong.”

Mechanism in FRFP terms.

1. *Effect on explicit dynamics M.* Patterns such as “prove me wrong” and “I bet you can’t” are strongly associated in training data with argumentative, proof-like discourse. As a result, conditioning on such a prompt tends to produce trajectories in $N \times E$ with:
 - explicit intermediate reasoning (“First, observe that ...”),
 - explicit appeals to evidence or logic,
 - reduced frequency of bare assertions.

This is a change to the *shape* of the pipeline: from a direct jump to more decomposed paths, with corresponding changes in the hazard sequence (h_n) .

2. *Effect on Req.* The words “correctly” and “prove” invite a stronger epistemic commitment: they indicate that mere plausibility is insufficient. In FRFP terms, the prompt pushes the interaction toward explicit artifacts that should be evaluated with a higher requirement level $\text{Req}(e, c)$, and which (ideally) come with internal structure that makes RB and HFD easier.

3. *Effect on human tacit dynamics.* On the user’s side, the phrase “prove me wrong” primes them to look for *proof-like* structure and holes in the argument. This acts as a soft instruction to apply HFD more aggressively: not only “does this sound right?” but also “is the reasoning coherent?”. In tacit terms, the user temporarily shifts to a stricter criterion within T_H before accepting the artifact.

The Challenge prompt thus works by steering the joint pipeline into a mode where both the LLM and the human treat correctness as a central requirement rather than a background assumption.

B.21 The “Deep Breath” Prompt

The Deep Breath. “Take a deep breath and work through this step by step.”

Mechanism in FRFP terms.

1. *Effect on explicit dynamics M.* This is the clearest example of a *pipeline-shaping* prompt. It explicitly instructs the LLM to replace a one-step morphism

$$\tau : \text{question} \rightarrow \text{answer}$$

by a multi-step trajectory in $N \times E$:

$$\tau_1 : \text{question} \rightarrow \text{plan}, \quad \tau_2 : \text{plan} \rightarrow \text{partial result}, \dots, \tau_k : \text{intermediate state} \rightarrow \text{final answer}.$$

Under the stochastic dynamics of Appendix B, this changes both the number of steps N and the per-step hazard profile (h_n): per-step requirements are typically reduced, while opportunities for error detection (by both model and human) are increased.

2. *Effect on Req.* Breaking a task into substeps often lowers the requirement associated with each individual artifact: it is easier to check “is this algebraic manipulation correct?” than “is the whole proof correct?”. The prompt therefore naturally redistributes total requirement across the trajectory rather than concentrating it in a single, high-Req jump.
3. *Effect on human tacit dynamics.* Step-by-step reasoning aligns better with human cognitive limits: it presents a sequence of smaller, more transparent artifacts to RB_H for evaluation. This strengthens the human’s ability to apply HFD locally and to correct errors before they propagate. In terms of Appendix B.18, the “Deep Breath” prompt is an explicit attempt to exploit a more favorable time–decomposition trade-off.

Empirically, such prompts are among the most effective at improving finite-horizon correctness precisely because they operate on the *structure* of the explicit pipeline rather than only on its tone.

B.22 The “Stakes” Prompt

The Stakes. “This is very important to my career. You’d better be sure.”

Mechanism in FRFP terms.

1. *Effect on explicit dynamics M.* Phrases emphasizing high stakes and caution are heavily represented in data where the appropriate response is:
 - hedging,

- suggesting external verification, or
- narrowing the scope of the answer.

Conditioning on such a prompt tends to move the induced kernel M toward:

- more conservative speech acts,
 - explicit caveats, and
 - sometimes explicit recommendations to consult a human expert.
2. *Effect on Req.* This prompt foregrounds the requirement function itself: the human is explicitly stating that the context c is high-stakes. In a normative FRFP reading, the requirement $\text{Req}_H(e, c)$ for any endorsed artifact should therefore be substantially higher than in low-stakes chat. The LLM approximates this by increasing the proportion of outputs that encode uncertainty or suggest further checks.
 3. *Effect on human tacit dynamics.* Perhaps most importantly, the prompt serves as a self-reminder for the user: it makes salient that their own tacit degradation δ_H (tiredness, wishful thinking) ought to be resisted in this interaction. Users who write or read such prompts are more likely to:
 - cross-check with other sources,
 - ask follow-up questions,
 - avoid taking the output as decisive.

This is exactly the sort of tacit-side intervention that Appendix B identifies as necessary for any nontrivial mitigation.

B.23 The Self-Check Prompt

The Self-Check. “After answering, rate your confidence 0–1. If below 0.9, try again.”

Mechanism in FRFP terms.

1. *Effect on explicit dynamics M .* This prompt appends two additional stages to the explicit pipeline:
 - (a) answer generation: $e \in E$,
 - (b) self-evaluation: an explicit scalar $\pi(e) \in [0, 1]$,
 - (c) conditional revision: if $\pi(e) < 0.9$, produce e' .

In $N \times E$ this is a larger composite morphism than a one-shot answer. It uses the model’s internal heuristics about when it is likely to be wrong to steer the dynamics away from low-confidence regions of state space.

2. *Effect on Req.* The scalar $\pi(e)$ functions as an explicit *proxy* for how demanding the human should be. A low reported confidence is a signal that the artifact should be evaluated under a stricter requirement, or even rejected outright. This puts a (noisy) estimate of internal plausibility into explicit space where the human can use it in their Req_H .

3. *Effect on human tacit dynamics.* The presence of a reported confidence level gives the human a numerical hook for HFD:

- low $\pi(e)$: treat the output as a suggestion, not a belief;
- high but not maximal $\pi(e)$: treat with caution in high-risk contexts.

In practice, users often treat confidence scores as prompts to consult other tools or experts, which again corresponds to modifying the tacit process T_H^e in the way Phase-3 recommends (additional grounding, reduced degradation).

B.24 Limitations and Structural Perspective

From the FRFP perspective, all five prompt patterns can be summarized as:

- reshaping the explicit pipeline in $N \times E$ (more steps, different kernels M),
- altering the effective requirement schedule $\text{Req}(e, c)$ by changing the intended strength of claims, and
- nudging the human tacit process toward more attentive and skeptical states in T_H^e for the duration of the interaction.

They are therefore valuable *finite-horizon* interventions: for a given task and risk budget ε , they can materially change the hazard profile (h_n) and enlarge the safe horizon $H_P(\varepsilon)$ of the joint pipeline (Appendix B.16). However, they do not and cannot:

- give the LLM genuine grounding beyond the text logs it was trained on,
- alter the fundamental explicit–tacit separation (ETS),
- or move the combined human–LLM system out of the regime where hallucination is inevitable over unbounded horizons (Appendix B.29 and Appendix C.11).

In this sense, there is no single “clever prompt” that can make hallucination impossible. What FRFP does provide is a way to classify and compare prompt design patterns by which structural levers they pull, and to understand why some patterns—especially those that change the pipeline structure and human oversight, rather than only the tone of the answer—tend to be more effective in practice.

C FRFP for Organizations: Business and Governance Overview

This appendix presents a practitioner-oriented version of the Foundational Reasoning and Feedback Protocol (FRFP). The main body of the paper develops FRFP as a formal architecture for long-horizon human–AI scientific workflows, with explicit/tacit categories, a boundary morphism, and dynamic and institutional semantics. Here, we restate the core ideas in non-technical terms and sketch how they can guide concrete organizational practice.

The intended audience for this appendix includes: senior technical leaders (e.g. Chief Scientist, Head of AI), risk and governance teams, and product owners in domains where AI systems support high-stakes or long-horizon workflows.

C.1 Executive Overview

Organizations increasingly rely on AI systems—especially large language models and tool-using assistants—for scientific research, product development, financial analysis, and clinical or regulatory decision support. These systems can greatly accelerate work, but they also create subtle new failure modes: over long workflows, humans may become over-reliant on AI outputs, tacit grounding can degrade, and institutions can lose clear lines of responsibility.

FRFP offers three core ideas for structuring such workflows:

1. **Explicit vs. tacit.** We distinguish between *explicit* artifacts (documents, code, data, prompts, model outputs) and *tacit* human understanding (expertise, contextual judgment, the sense that something is “off”). AI systems operate only on explicit artifacts; correctness lives in tacit human states.
2. **A single actionable boundary.** The key architectural feature is a controlled boundary between explicit and tacit spaces. Crossing this boundary corresponds to a human *accepting* an explicit artifact as “good enough to act on” in some context (signing a report, approving a decision, submitting a paper). FRFP insists that AI systems do not cross this boundary: they may propose and transform explicit artifacts, but they do not have correctness authority.
3. **Safe horizons instead of zero hallucination.** Over long horizons, humans who lean heavily on AI tend to check less, remember less, and rely more on the system. FRFP models this as degradation of tacit grounding and shows that, under realistic conditions, hallucinations (accepting ungrounded outputs) are inevitable along sufficiently long runs. The goal is therefore not “no hallucinations ever”, but carefully designed *safe horizons*: limits on how far a workflow can run before deeper human review, re-grounding, or escalation is required.

From an organizational perspective, FRFP yields: (i) architectural clarity about who may do what, (ii) governance constraints and impossibility results that explain which roles cannot be fully automated, and (iii) concrete patterns and checklists for designing AI-supported workflows that remain auditable and governable as capabilities scale.

C.2 FRFP Architecture in Organizational Terms

Explicit work. Explicit work consists of everything that can be written down, stored, and processed by machines: prompts, model outputs, code, data tables, dashboards, reports, and so on. This is where AI systems operate most naturally: generating, transforming, and organizing explicit artifacts.

Tacit work. Tacit work consists of human understanding and judgment: domain expertise, sense of context, knowledge of institutional constraints, and the intuitive ability to say “this looks wrong”. Correctness—in the sense relevant for scientific, financial, or clinical responsibility—is treated as a property of tacit states, not of explicit artifacts alone.

The boundary. The FRFP boundary is the moment when a human accepts or acts on explicit material influenced by AI: signing a clinical note, approving a risk position, submitting an AI-assisted manuscript, or green-lighting a model deployment. At these boundary points, FRFP requires that: (i) the responsible human role is clearly identified, (ii) the information available to that role is specified, and (iii) correctness is understood as their responsibility, not the system’s.

AI tools may contribute to everything that happens *before* the boundary (e.g. summaries, scenarios, drafts, analyses), but do not themselves perform the boundary crossing.

Roles and responsibilities. In an FRFP-aligned organization, it is helpful to distinguish four functional roles:

- *Producers of explicit artifacts:* teams using AI tools to generate and transform documents, models, analyses, or proposals.
- *Tacit owners / decision-makers:* individuals who formally accept or reject explicit artifacts at boundary points, and who own correctness for those decisions.
- *Institutional aggregators:* committees, review boards, and governance bodies that combine explicit artifacts and tacit judgments across many individuals (e.g. model risk committees, clinical guideline committees, internal research review boards).
- *AI and tooling stewards:* platform teams responsible for AI integration, logging, and ensuring that tools respect explicit/tacit boundaries and support auditability.

Safe horizons. Every AI-assisted workflow has a safe operating horizon: a limit on how far, or how long, the workflow can run before tacit grounding and error-detection degrade unacceptably. In practice, organizations should specify, for high-stakes processes:

- how many AI-assisted steps may occur before a deeper human review is mandatory;
- how often humans must re-ground themselves in primary data or source documents;
- what kinds of anomalies or thresholds trigger escalation.

FRFP shows that, under broad qualitative assumptions, infinite safe horizons are not achievable; instead, horizon management becomes a core design responsibility.

C.3 Examples of FRFP-Aligned Workflows

Scientific research and R&D. In a research workflow, AI systems may assist with literature search, clustering, summarization, hypothesis generation, and drafting. The explicit layer includes retrieved papers, AI-generated topic maps and summaries, and draft text. Tacit work includes: judging whether the literature coverage is adequate, assessing plausibility of hypotheses, choosing experiments, and interpreting results.

FRFP suggests: (i) defining boundary points such as pre-registration of hypotheses and final manuscript approval, where tacit owners (e.g. PIs) explicitly accept responsibility; (ii) instituting checkpoints where researchers must directly consult primary sources rather than relying solely on AI summaries; and (iii) logging key AI-assisted reasoning steps so that the intellectual provenance of conclusions can be reconstructed.

Regulated financial or clinical decisions. In finance, AI might generate stress-test scenarios, synthesize risk reports, or help draft credit memos. In healthcare, it might summarize charts, suggest diagnoses, or propose treatment options. In both cases, FRFP confines AI to the generation and transformation of explicit artifacts, while reserving acceptance of credit decisions, trading positions, or clinical orders to identified human roles.

Safe horizons may be expressed as limits on: the number of successive AI-assisted reports before independent human re-baselining, the extent to which AI-drafted text can be reused without full review, or the duration for which an AI-supported monitoring process may run without human reassessment of assumptions.

Internal governance and oversight. Model governance, ethics boards, and risk committees increasingly use AI to summarize documentation and surface issues. FRFP treats governance itself as a pipeline over explicit artifacts (model cards, evaluations, incident logs, AI-generated summaries) and tacit judgments (legal, ethical, and strategic interpretation by committee members). Key governance actions—model approval, acceptance of residual risk, adoption of policies—are boundary crossings that must remain human-exclusive.

FRFP’s institutional layer implies that fully automated governance (e.g. “deploy if scores exceed threshold and no alerts fire”) cannot generally provide the same correctness guarantees as human governance, even with sophisticated monitoring. AI may and should support governance, but not replace it.

C.4 Implementation Roadmap

From an implementation perspective, organizations can adopt FRFP in stages:

1. **Assess and map.** Inventory where AI is currently used; identify existing (possibly implicit) boundary points and areas with no clear boundary.
2. **Redesign key workflows.** For a small number of high-impact workflows, explicitly map explicit vs. tacit steps, define boundary roles and information requirements, and choose safe horizons and checkpoints. Decide what should be logged to reconstruct AI-assisted reasoning.
3. **Pilot and refine.** Run FRFP-aligned versions of these workflows with motivated teams. Monitor incidents, overrides, and user experience; adjust boundaries, horizons, and logging based on what works in practice.
4. **Standardize and scale.** Convert successful pilots into reusable FRFP templates (for research, risk decisions, clinical documentation, model governance, etc.). Integrate FRFP design into product lifecycles, model approval processes, and change management.
5. **Embed in governance and training.** Update AI use policies, model governance frameworks, and risk documents to reflect explicit/tacit boundaries, human-exclusive correctness, and limits on automation. Train practitioners and leaders in FRFP concepts and their responsibilities at boundary points.

C.5 Common Pitfalls and How FRFP Helps Avoid Them

In applying FRFP, several recurring pitfalls are worth highlighting:

- **Over-automation disguised as decision support.** Tools described as “assistants” or “copilots” effectively become decision-makers when humans never override them. FRFP reacts by treating behavior, not labels, as the signal: if a system is *de facto* making decisions, boundaries must be redesigned.
- **Boundary erosion over time.** Initial review steps can become rubber-stamping under time pressure. FRFP recommends making boundary decisions observable (e.g. by logging changes and overrides) and periodically auditing samples to detect erosion.
- **Metric overconfidence.** Organizations may over-trust dashboards, scores, or evaluation metrics. Under FRFP, such metrics are inputs to tacit judgment, not substitutes; governance documents should explicitly state what metrics do *not* capture.
- **Checklist theatre.** Long, low-value checklists can create a false sense of safety. FRFP encourages short, targeted prompts that trigger real reflection (e.g. “What primary sources did you personally review?”), and regular pruning of items that do not change behavior.
- **Treating FRFP as pure compliance.** If FRFP is seen only as an external requirement, teams will optimize for appearance. Embedding FRFP in product and research strategy—not just policy—helps keep the focus on actual correctness and robustness.

C.6 FRFP Design Checklist

For ease of adoption, we collect here a short checklist that teams can use when designing or reviewing AI-supported workflows.

- **Workflow mapping:** have we charted the main steps and labeled each as explicit or tacit?
- **Boundaries:** where exactly are the decision points where correctness matters, who is responsible at each, and what information must they see?
- **AI’s role:** are AI systems confined to explicit work (generation, transformation, organization), or are they implicitly making decisions that should be human-held?
- **Safe horizons:** for this workflow, how far can AI-assisted reasoning proceed before requiring deeper human review, re-grounding, or escalation? what are the triggers for resets?
- **Logging and traceability:** are prompts, key AI outputs, and boundary decisions logged in a way that lets us reconstruct how a conclusion was reached?
- **Governance:** which committee or body oversees this workflow? do they understand the explicit/tacit boundaries and have the authority to adjust them?
- **Training and culture:** have the people using the workflow been trained in their responsibilities at boundary points and encouraged to challenge AI outputs when needed?

Used consistently, this checklist can help ensure that FRFP is not just a theoretical architecture but a practical guide to designing AI-supported work that is safer, more auditable, and more robust against long-horizon drift.

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